

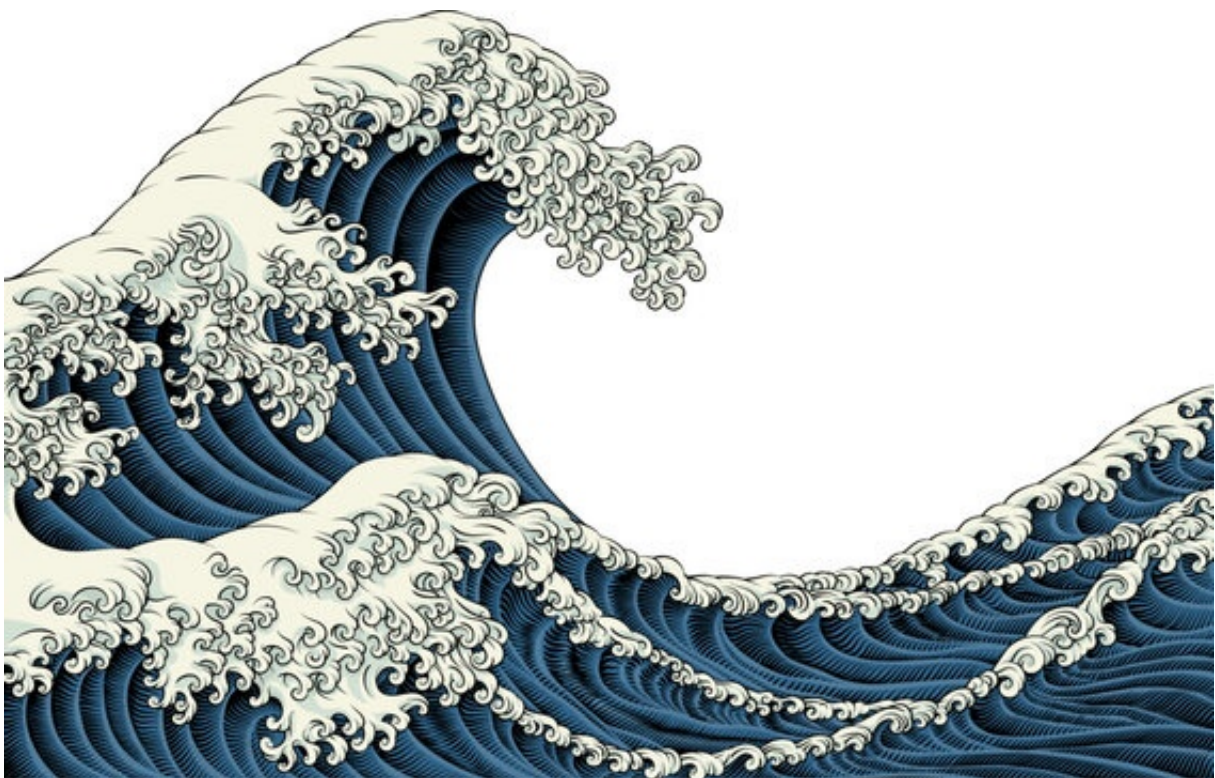
pyCoastal: a Python Tool for Coastal Engineering

Stefano Biondi

University of Florida

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Preface

pyCoastal began as a teaching and research package for numerical modeling, fluid dynamics and hydraulics. Numerical models follow recurrent structural patterns across modeling frameworks. Configured and validated, they reproduce measured behavior and allow the analysis of conditions that cannot be observed directly.

The first edition of this manual documented the numerical framework. The package now also implements a design chain: a wave record yields a design condition, the design condition sizes a structure, and the structure is issued as a dimensioned drawing with quantities and specification notes. Each relation records its source and its range of validity, and reports application outside that range. PyCoaTools provides the design modules in a browser and presents each result together with the supporting literature held in PyCoaPedia, the knowledge base of coastal and ocean engineering publications distributed with the package.

This manual covers both. Part I describes installation and conventions. Part II documents the standalone engineering formulae. Part III documents the design applications, one chapter per module, in the order of the design chain, each with formulation, interface, a worked example reproducing the program output, and stated limitations. Part IV covers drawings and deliverables. Part V documents PyCoaTools. Part VI documents PyCoaPedia, its construction, and its interfaces for browser, Python, command line and SQL access, together with the repository layout for AI agents. Part VII documents the numerical framework: grids, operators, time integration, boundary conditions, governing equations, and the simulation examples. Part VIII documents the test suite.

Stefano Biondi

Part I

Getting started

1 Introduction

Coastal zones concentrate population, infrastructure and ecosystems in energetic, time-varying environments. Waves, tides, surges and currents interact with the shoreline and drive erosion, flooding, sediment transport and loads on structures. Quantifying these processes supports navigation, coastal protection, habitat conservation and climate adaptation.

pyCoastal is a Python toolbox for coastal, port and ocean engineering. It contains two independent layers in one code base.

1. **A numerical framework:** uniform grids, finite-difference operators, explicit time integrators, boundary-condition classes, Poisson solvers, and governing-equation components (shallow water, incompressible Navier-Stokes, eddy-viscosity closures). Each numerical component is exposed and can be replaced.
2. **A design chain:** scenario-level applications that take an engineering design and return project quantities. These include the 100-year wave with its confidence band, armor stone size, crest level, seawall base width, scour depth over a tidal cycle, dredge level, fender size, flood extent, and the corresponding drawing sheet.

1.1 The design chain

pyCoastal is organized in the sequence a coastal scheme is designed (Table 1). Each step passes a structured object to the subsequent module. This maintains consistency between calculations and drawings.

Table 1: The design chain and the modules that implement each step.

Step	Module	Output
Design condition	<code>applications.extremes</code>	100-year wave or water level, with a confidence band
Bed and soil	<code>applications.sediment</code>	grain mobility, fall velocity, earth pressure
Nearshore	<code>applications.port</code>	phase-resolved diffraction into a harbor, berth agitation
Structure	<code>applications.structures</code> , <code>applications.seawall</code>	armor size, crest level, stability checks
Loads	<code>applications.piles</code> , <code>applications.berthing</code>	Morison loads through the wave cycle, berthing energy and fenders
Scour	<code>applications.scour</code> , <code>applications.river</code>	contraction, pier and abutment scour over the tidal cycle; the flow split that supplies it
Navigation	<code>applications.channel</code>	dredge level, channel width, dredge volume
Coastline	<code>applications.nourishment</code> , <code>applications.surge</code>	fill life and profile, water level budget and a connected flood map
Deliverable	<code>drafting</code> , <code>applications.sections</code>	dimensioned drawing sheet and DXF

1.2 Implementation rules

Three rules apply throughout the package.

Source attribution. Module docstrings list the sources implemented (Van der Meer 1988, EurOtop 2018, Goda 2010, PIANC 2014, HEC-18, among others). Each function records the equation it implements.

Range reporting. Where a relation has a documented range of validity, the function computes outside that range and reports the condition through a `warnings` list, a flag such as `within_range`, `impulsive` or `screening_only`, or an entry in the returned dictionary.

Assumption reporting. Unspecified channel-width components are assigned the least restrictive class and reported as assumed. The +1 safety term in Froehlich's abutment equation is returned separately as `safety_margin`. The steady-current floor applied to the Sumer and Fredsoe scour relation is flagged as `current_governs`.

1.3 Terminology

The following terms are used in Part VII. A *field* is a variable defined over the domain, such as water depth, velocity or pressure. A *grid* divides the domain into cells holding those variables; `pyCoastal` stores values at cell centers. A *stencil* is the pattern of neighboring cells used to approximate a derivative by finite differences; these approximations form the spatial *operators*. A *time integrator* advances the discretized solution using a right-hand side containing fluxes, advection, diffusion, pressure gradients and sources. *Boundary conditions* define the solution at the domain edges.

In Parts III and IV, a *design condition* is the wave and water level used to size a structure, a *design object* (for example `SeawallDesign` or `ChannelDesign`) is the dimensioned result returned by a design function, and a *sheet* is a drawing at a stated scale with border, title block and specification notes.

1.4 Manual organization

- **Part I:** installation, repository layout, conventions, quick start.
- **Part II:** the standalone engineering formulae in `pyCoastal.tools`, used by the design modules.
- **Part III:** one chapter per design application, in the order of the design chain.
- **Part IV:** the drafting layer, sections, sheets and DXF export.
- **Part V:** `PyCoaTools`, the browser implementation of the design modules.
- **Part VI:** `PyCoaPedia`, its interfaces, and the repository layout for AI agents.
- **Part VII:** the numerical framework, the governing equations, and the simulation examples. The design modules have no dependency on this layer.
- **Part VIII:** the test suite and the verification of the browser engine against the Python implementation.
- **Appendices:** list of symbols and engineering references.

2 Installation and first steps

2.1 Requirements

pyCoastal needs Python 3.10 or newer. The runtime dependencies are kept to two, so the design modules install anywhere:

- `numpy` ≥ 1.21 for arrays, gradients, and linear algebra;
- `PyYAML` ≥ 5.4 for YAML case files.

Everything else is optional and imported lazily, only by the functions that need it (Table 2).

Table 2: Optional dependency groups declared in `pyproject.toml`.

Extra	Installs	Needed for
<code>sparse</code>	<code>scipy</code> ≥ 1.7	sparse direct solver in <code>physics.poisson</code>
<code>vtk</code>	<code>vtk</code> ≥ 9.0	<code>io.write_vtk</code>
<code>plots</code>	<code>matplotlib</code> ≥ 3.4	drafting, applications, sections, plotting, and the examples
<code>dev</code>	<code>pytest</code> , <code>scipy</code> , <code>matplotlib</code>	running the test suite
<code>docs</code>	<code>sphinx</code> , <code>sphinx-rtd-theme</code>	the Sphinx API pages in <code>docs/</code>

2.2 Installing

From PyPI:

```
pip install pyCoastal
pip install "pyCoastal[plots]"      # with drawings and plotting
```

From a clone, in editable mode, for development and for running the examples:

```
git clone https://github.com/stebiondi/pyCoastal.git
cd pyCoastal
python -m venv .venv
source .venv/bin/activate          # on Windows: .venv\Scripts\activate
pip install -e ".[dev]"
```

Check the installation and run the tests:

```
python -c "import pyCoastal; print(pyCoastal.__version__)"
python -m pytest tests/
```

Continuous integration runs the same suite on Ubuntu and Windows under Python 3.10, 3.11, and 3.13 on every push to `main` and on every pull request (`.github/workflows/tests.yml`). A separate workflow builds and publishes the distribution to PyPI (`.github/workflows/publish.yml`).

2.3 Repository layout

```
pyCoastal/
├── pyCoastal/          the package
│   ├── numerics/       grids, operators, time integrators, BCs, solver
│   ├── physics/        shallow water, Navier-Stokes, Poisson, turbulence
│   ├── tools/          standalone formulae: waves, sediment, structures,
│                       one-line shoreline, morphodynamics
│   └── applications/   design modules (extremes, structures, seawall,
```

	port, piles, scour, river, channel, berthing, nourishment, surge, sediment, sections)
— drafting.py	drawing primitives, sheets, title blocks, DXF
— plotting.py	color maps and helpers for the examples
— config.py, io.py	case-file reading and VTK output
— examples/	worked examples and their YAML configs
— tests/	the pytest suite
— pyCoastal/pedia/	PyCoaPedia reader and its SQLite database
— pedia/	PyCoaPedia as Markdown, its schema, and the builder
— webapp/	PyCoaTools and the PyCoaPedia explorer (browser)
— media/	figures, animations, drawing sheets, DXF files
— docs/	this manual (docs/manual), the API reference (docs/reference) and the example index
— AGENTS.md, llms.txt	the repository map for AI agents
— paper/	JOSS-style paper
— pyCoastal manual.pdf	this manual

2.4 Conventions

The following conventions apply in every module. Departures are recorded in the docstring of the module concerned.

Coordinates. x and y span the horizontal plane and z is elevation, positive upward. In 2D arrays x is axis 0 and y is axis 1, matching `UniformGrid`, which builds its coordinates with `indexing="ij"` and flattens as $i \cdot ny + j$. In cross-shore work x is the chainage, and in alongshore work (the one-line model) x is alongshore and y is the cross-shore shoreline offset, positive seaward.

The design applications and the operators follow that axis order. Two older pieces of the numerical framework do not, and this manual says so where they appear: `numerics.domain.Mesh2D` builds its coordinate arrays with NumPy's default `indexing="xy"`, so they are shaped (ny, nx) , and the solvers in `physics.poisson` take their right-hand side in that same (ny, nx) form.

Units. SI throughout: meters, seconds, kilograms, newtons. Wall forces and wall weights are kN per meter run. Overtopping discharge is liters per second per meter ($l/s/m$), the unit of the tolerability tables. Ship speeds are knots in the functions whose source formula is written in knots. Grain sizes are meters. Geotechnical and navigational angles are degrees; wave-direction and slope-angle arguments of the low-level tools are radians. Each function states the unit it takes.

Levels. Levels are meters relative to chart datum (m CD) and increase upward. Depths and draughts are positive. Pier scour drawings use the initial bed as datum; dimensions on those drawings are depths below it.

Wave parameters. Design wave heights are spectral significant heights H_{m0} at the structure toe. `EurOtop` relations take the spectral period $T_{m-1,0}$. `DesignConditions.from_peak_period` converts a peak period with $T_{m-1,0} = T_p/1.1$. Goda's pressure method is formulated on the significant period; the seawall module passes T_p to it.

Slopes. Slope input: $\cot \alpha$, defined as horizontal run per unit rise. This convention is used in the generated drawings.

2.5 Quick start

A seawall from a design condition to a drawing sheet:

```
from pyCoastal.applications.structures import DesignConditions
from pyCoastal.applications.seawall import design_seawall
```

```

from pyCoastal.applications.sections import seawall_sheet

conditions = DesignConditions.from_peak_period(Hm0=2.8, Tp=9.5, depth=8.5)
wall = design_seawall(conditions, still_water_level=2.9, seabed_level=-5.6,
                      tolerable_use="trained_staff")
print(wall.summary())
sheet = seawall_sheet(wall, project="Bayfront promenade protection", size="A3")
sheet.save("seawall.png")      # 600 dpi, at a true scale stated on the sheet
sheet.to_dxf("seawall.dxf")    # the geometry on named layers, for CAD

```

The printed summary is the design report: crest level, founding level, base dimensions, wave force, safety factors and bearing pressures. The full worked example is in [Section 8](#).

Every example in `examples/` runs from the repository root:

```

python examples/seawall_section.py
python examples/breakwater_design.py
python examples/design_wave.py

```

and writes its figures to `media/`. `docs/examples.md` lists them all.

2.6 Reading case files

The numerical examples read their parameters from YAML files in `examples/configs/`. Two helpers parse case files into dictionaries:

- `pyCoastal.config.load_config(path)` reads `.yaml/.yml` (PyYAML), `.json`, and `.ini/.cfg` files;
- `pyCoastal.io.read_data(path)` does the same for any path-like object, and `pyCoastal.io.write_vtk(grid, filename)` writes a VTK grid object to a legacy `.vtk` file (needs the `vtk` extra).

These are used by the simulation examples of [Section 22](#). `pyCoastal.numerics.domain.Domain(cfg)` builds a `Mesh1D` or `Mesh2D` from a domain: block with dimension, `x0`, `x1`, `nx` (and `y0`, `y1`, `ny` in 2D).

Part II

Engineering tools

3 Standalone formulae: `pyCoastal.tools`

`pyCoastal.tools` provides standalone coastal engineering formulae. Every parameter is an explicit argument, so defaults can be set or overridden from a case file. The design applications of Part III build on the same relations and add validity checks and reporting. Where a tool and an application implement the same relation, the application is the design interface.

3.1 Waves: `tools.wave`

3.1.1 Linear dispersion and derived numbers

Under linear (small-amplitude) theory, frequency, wavenumber and depth are related by the dispersion relation, which defines the phase and group speeds:

$$\omega^2 = gk \tanh(kh), \quad c = \frac{\omega}{k}, \quad c_g = \frac{c}{2} \left(1 + \frac{2kh}{\sinh 2kh} \right). \quad (1)$$

`dispersion(T, h, g=9.81, tol=1e-12)` solves it for the wavelength L by Newton iteration and raises on a non-positive period or depth; `wave_number(T, h)` returns $k = 2\pi/L$. The deep-water wavelength is $L_0 = gT^2/(2\pi)$.

Two nondimensional numbers classify the waves:

- **Iribarren (surf similarity)**, $\xi_0 = \tan \beta / \sqrt{H_0/L_0}$: `surf_similarity(alpha, H, T)` returns it and `breaker_type(alpha, H, T)` names the breaker, spilling for $\xi < 0.4$, plunging for $0.4 \leq \xi \leq 2$, and collapsing or surging above 2.
- **Ursell**, $U = HL^2/h^3$: `ursell_number(H, T, h)` returns the value and an interpretation, linear theory for $U < 32$ and a nonlinear theory (Stokes, Boussinesq) above.

`wave_setup(Hb, gamma=0.8)` gives the shoreline setup $\bar{\eta} = \frac{5}{16}\gamma H_b$.

3.1.2 Spectral synthesis

A wave spectrum gives the distribution of sea-state energy over frequency. For irregular-sea forcing, a surface elevation record is synthesized by random-phase superposition consistent with a target spectrum (Equation 2):

$$\eta(t) = \sum_{i=1}^N \sqrt{2S(f_i)\Delta f} \cos(2\pi f_i t + \varphi_i), \quad \varphi_i \sim U[0, 2\pi]. \quad (2)$$

The spectral moments $m_n = \int_0^\infty f^n S(f) df$ give $H_s \approx 4\sqrt{m_0}$ and $T_z \approx \sqrt{m_0/m_2}$. `generate_irregular_wave(Hs, Tp, duration, dt, spectrum="pm", gamma=3.3)` returns `(t, eta)` for a Pierson-Moskowitz (fully developed) or JONSWAP (fetch-limited, peak enhancement `gamma`) spectrum scaled to the requested H_s .

3.2 Sediment transport: `tools.sediment_transport`

The Shields parameter gauges bed mobility under a bed shear stress $\tau_b = \rho u_*^2$:

$$\theta = \frac{\tau_b}{(\rho_s - \rho) g d}, \quad (3)$$

with a critical value $\theta_c \approx 0.03$ to 0.06 for sand (the design modules compute it from the grain size, Section 5). The transport formulae in Table 3 convert mobility into rates.

Table 3: Transport relations in `pyCoastal.tools.sediment_transport`.

Function	Relation
<code>shields_parameter(tau_b, rho_s, rho, d)</code>	Equation 3
<code>van_rijn_bedload(Ue, h, d50, rho_s, rho)</code>	$q_b = 0.015 \rho_s U_e h (d_{50}/h)^{1.2} M_e^{1.5}$, $M_e = (U_e - U_{cr}) / \sqrt{(s-1)gd_{50}}$ (Van Rijn 1984)
<code>van_rijn_suspended(Ue, h, d50, rho_s, rho)</code>	$q_s = 0.008 \rho_s U_e d_{50} M_e^{2.4} D_*^{-0.6}$
<code>bijker_bedload(tau_wave, tau_current, rho_s, rho, d50)</code>	$\propto \sqrt{\tau_{\text{total}}}$, waves and current combined (Bijker 1971)
<code>cerc_transport(Eb, angle, K=0.39)</code>	$Q = K E_b / (\rho g) \sin \varphi_b \cos \varphi_b$ (CERC)
<code>bagnold_sediment(H, c, rho_s)</code>	$(\rho_s / \rho) H^2 c$, energetics estimate
<code>izbash_current(rho_s, rho, d)</code>	$u_c = 1.7 \sqrt{\Delta g d}$, stone stability in current
<code>einstein_bedload(tau_star, d, s)</code>	$q_b = 8 \sqrt{g(s-1)d^3} \tau_*^{1.5}$

The critical velocity in the Van Rijn functions is $U_{cr} = 0.19 d_{50}^{0.1} \log(12h/3d_{50})$ and $s = \rho_s / \rho$.

3.3 Structures and runup: `tools.structural`

- `hudson_dn50(Hs, Delta, theta, Kd=3.0)`: $H_s / (\Delta D_{n50}) = (K_D \cot \theta)^{1/3} / 1.27$ (SPM 1984).
- `vandermeer_dn50(Hs, Delta, P, N, alpha, xi_m, damage=2.0, safety=1.0)`: Van der Meer (1988), plunging and surging branches (Equation 13 and Equation 14 in Section 7), with the wave count saturating at 7500. The surf similarity is a required argument; it depends on the wave period, which this function does not receive.
- `hunt_runup(beta, H, L)`: $R \approx H \tan \beta / \sqrt{H/L}$ (Hunt 1959).
- `stockdon_runup(H, L, beta)`: the Stockdon et al. (2006) 2% runup.
- `goda_wave_force(H, T, h, beta)`: a screening estimate of the Goda horizontal force on a vertical face, in N/m, taking $H_{\text{design}} = 1.8H$. The full Goda distribution with uplift and stability checks is `applications.seawall.goda_pressures` (Section 8).
- `iribarren_stability(H, alpha, rho_s, rho_w, mu, N)`: the Iribarren stone weight, $W \geq N \rho_s g H^3 / [\Delta^3 (\mu \cos \alpha - \sin \alpha)^3]$.

3.4 One-line shoreline model: `tools.shoreline`

The one-line model takes the shoreline position $y(x, t)$ as the only morphological variable and conserves sediment alongshore:

$$\frac{\partial y}{\partial t} = -\frac{1}{(1-p)D} \frac{\partial Q_{ls}}{\partial x}, \quad Q_{ls} = K_{\text{cerc}} H_b^{5/2} \sin(2\alpha_b), \quad \alpha_b = \alpha_0 - \frac{\partial y}{\partial x}, \quad (4)$$

with $H_b = K_d K_t H_{\text{free}}$, where K_d is a diffraction shadow factor and K_t a transmission factor. The explicit scheme is stable when Equation 5 holds,

$$\Delta t \leq 0.45 \frac{\Delta x^2}{\max G}, \quad G = \frac{2K_{\text{cerc}} H_b^{5/2}}{(1-p)D}. \quad (5)$$

`OneLineParams(D=8, p=0.4, Kcerc=0.4, Hfree=1.0, alpha0=10 deg, Kt=1.0, beta0=-30 deg, morfac=1.0)` holds the parameters; any of `Hfree`, `alpha0`, `Kt` can be arrays over x . The building blocks are `make_grid(Lx, dx)`, `breakwaters_geometry(x, tips, y_tip, crest_freeboard)` for any number of detached breakwaters, `kd_diffraction_field(x, geom, beta0)` for a simple shadow factor $K_d = \frac{1}{2}(1 + \cos \theta)$ from each tip, `compute_flux_Qls, rhs_y_t, suggest_dt`, and `apply_bcs(y, bc="fixed_ends" | "zero_slope")`. `run_one_line_model()` runs a default two-breakwater case. The nourishment application (Section 14) drives this solver with a fill planform.

3.5 Morphodynamics helpers: `tools.morphodynamics`

- `bruuns_rule(S, beta, L=None, h=None, B=None)`: shoreline retreat under sea-level rise S , $R = SL/(h + B)$ when the profile length, closure depth, and berm height are given, and $R = S/\tan \beta$ otherwise.
- `exner_change(qs_dx, porosity=0.64)`: bed change rate from the transport divergence, $\partial \eta / \partial t = -q_{s,x}/(1 - n)$.
- `linear_slope(grid, north_level, south_level)`: a plane beach on a grid.
- `circle_through_points, two_point_arc`, and `build_tombolo_arcs(Lshore, Gb, Lb1, Lb2, yb, ...)`: the three-arc equilibrium planform behind a pair of detached breakwaters used by `examples/equilibrium_shoreline.py`, with retreat $Y_e = 1.204Y_i - 0.07G_b$ below the breakwater line; `accretion_metrics(x_eq, y_eq, geom)` returns the retreat and the accretion percentage.

Part III

Design applications

4 Design conditions from a record

Module: `pyCoastal.applications.extremes`. *Example:* `examples/design_wave.py`.
Browser: Return values.

The module derives a design condition from a measured or hindcast record and returns a return value with a confidence band. The design conditions used by the structural modules of this part originate here.

4.1 Methods

Peaks over threshold (POT). Independent storm peaks above a threshold are retained, a generalized Pareto distribution (GPD) is fitted to the excesses, and the fit is combined with the arrival rate of the peaks. The result depends on the threshold, which is selected with the diagnostics described below.

Block maxima. The largest value in each year is retained and a generalized extreme value (GEV) distribution is fitted. The method uses less of the record and provides an independent check on a POT result.

Both are fitted by **L-moments** (Hosking and Wallis 1997). L-moment estimators are close to unbiased at the sample sizes of a coastal record, are stable on the flat likelihood surfaces of short records, and require no optimizer. Shape parameters follow the extreme-value convention: a positive shape gives a heavy tail with no upper bound, a negative shape a finite upper limit. Hosking's $k = -\text{shape}$ is converted internally and is not exposed in the interface.

4.2 Formulation

Declustering. `decluster(values, threshold, separation)` scans the record and retains the largest value in each run above the threshold. Two runs separated by fewer than `separation` samples below the threshold are treated as one storm. For a 3-hourly record, 24 samples corresponds to a three-day separation, the standard value for storm waves.

GPD fit. With survivor function $1 - F(x) = (1 + \xi x/\sigma)^{-1/\xi}$ (exponential as $\xi \rightarrow 0$), the L-moment estimators from the mean λ_1 and L-CV $\tau = \lambda_2/\lambda_1$ of the excesses are given by Equation 6,

$$\xi = 2 - \frac{1}{\tau}, \quad \sigma = \lambda_1(1 - \xi). \quad (6)$$

A shape at or above one implies an undefined fitted mean. The fit is returned with a warning and is not clipped.

POT return value. With λ peaks per year, the T -year value is Equation 7,

$$x_T = u + \frac{\sigma}{\xi} [(\lambda T)^\xi - 1], \quad (7)$$

with the logarithmic form in the exponential limit. Return periods shorter than $1/\lambda$ are outside the model.

GEV fit. With $F(x) = \exp\{-[1 + \xi(x - \mu)/\sigma]^{-1/\xi}\}$, the parameters follow Hosking's L-moment estimators. The polynomial approximation for k from the L-skewness has an accuracy better than 10^{-4} for $-0.5 < k < 0.5$.

Plotting positions. Empirical return periods use the Gringorten position $(i - a)/(n + 1 - 2a)$ with $a = 0.44$, which is close to unbiased for the Gumbel and GEV families.

Threshold diagnostics. Above a threshold at which the GPD applies, the mean excess is linear in the threshold (`mean_residual_life`), and the fitted shape and the *modified* scale $\sigma^* = \sigma - \xi u$ are constant (`threshold_stability`). The raw scale is not threshold-invariant and is not used for threshold selection.

Confidence band. `ExtremeFit.confidence` applies a percentile bootstrap over the fitted peaks. It quantifies sampling error in the parameters. It excludes measurement error, threshold selection, and non-stationarity of the wave climate.

4.3 Interface

```
from pyCoastal.applications.extremes import (
    fit_pot, fit_block_maxima, mean_residual_life, threshold_stability,
)

mrl = mean_residual_life(record, thresholds)          # threshold diagnostics
stab = threshold_stability(record, thresholds, separation=24)
fit = fit_pot(record, threshold=2.5, separation=24, samples_per_year=2920)

fit.return_value(100)                                # 100-year Hm0
fit.confidence(100, level=0.90)                      # confidence band
fit.extrapolation_note(100)                          # extrapolation beyond the record
print(fit.summary())

check = fit_block_maxima(record, block=2920)         # annual maxima cross-check
```

`fit_pot` and `fit_block_maxima` return an `ExtremeFit` with `kind` ("pot" or "gev"), `parameters`, the peaks or maxima in data, their `positions`, the `rate` and record length `years`, the bootstrap `samples`, and a `warnings` list. A warning is issued for fewer than one or two peaks per year and for fewer than about thirty peaks in total.

4.4 Worked example

`examples/design_wave.py` generates a 40-year synthetic 3-hourly record of H_{m0} with known parameters, selects the threshold from the diagnostics, fits POT and annual maxima, and passes the 100-year wave to the seawall module. The record is synthetic, so the target value is known. The example repeats the analysis on 60 independent 40-year records, measures the sampling spread of the 100-year estimate, and compares the bootstrap band with that spread.

Output of `examples/design_wave.py`

```
Record: 40 years, 116800 samples, max 5.21 m, 240 storms (6.0/yr)
Threshold 2.50 m chosen from the diagnostics (the record was built with 2.50 m)

Peaks over threshold
Model           peaks over threshold
Record          40.0 years, 240 peaks
Threshold       2.50
Rate            6.00 peaks per year
```

Scale, shape 0.560, -0.071

Return period	value	90% band
10 yr	4.49	4.19 to 4.83
50 yr	5.12	4.61 to 5.78
100 yr	5.38	4.74 to 6.19

the record was built with scale 0.550, shape -0.080, rate 6.0/yr

Annual maxima, as a cross-check

Model	annual maxima
Record	40.0 years, 40 maxima
Location	3.489
Scale, shape	0.484, -0.058

Return period	value	90% band
10 yr	4.51	4.22 to 4.74
50 yr	5.18	4.63 to 5.58
100 yr	5.45	4.77 to 5.95

100-year Hm0 = 5.38 m (4.74 to 6.19 m, 90% bootstrap)

true value for this climate: 5.25 m

2.5 times the 40 year record. Treat the confidence band as a lower bound on the real

↪ uncertainty.

Repeating the study on 60 fresh 40-year records

100-year estimate: mean 5.23 m, sd 0.44 m

5th to 95th percentile: 4.61 to 5.94 m (width 1.32 m)

bootstrap band from the single record: width 1.45 m

this record sits at the 70th percentile of that spread

The bootstrap band matches the real sampling spread.

Design wave at the toe 1.93 m (depth limited)

Seawall crest +6.19 m CD, base 10.75 m, concrete 19.4 m3/m

Wrote media/design_wave.png

The bootstrap band from the single record has a width of 1.45 m against a sampling spread across records of 1.32 m, and contains the target value of 5.25 m. The 100-year return period is 2.5 times the record length; `extrapolation_note` reports this ratio, and the band is a lower bound on the total uncertainty.

4.5 Limitations

- Stationarity is assumed: no trend or climate change in the record.
- The confidence band quantifies sampling error only.
- Short records and thresholds retaining few peaks are fitted and flagged.

5 Sediment and soil

Module: `pyCoastal.applications.sediment`.

The module defines bed and backfill materials and the relations that use them. Material properties propagate into the other design modules: the grain size controls bed mobility, the friction angle controls the lateral pressure of a backfill, and the unit weight sets the split between effective stress and pore pressure. The seawall, breakwater, channel, pile, scour and nourishment modules accept a `Sediment` instance or a catalogue key.

The module implements two groups of relation. **Mobility** (grain size, fall velocity, Shields threshold) determines whether the bed is in motion and gates the scour relations. **Earth pres-**

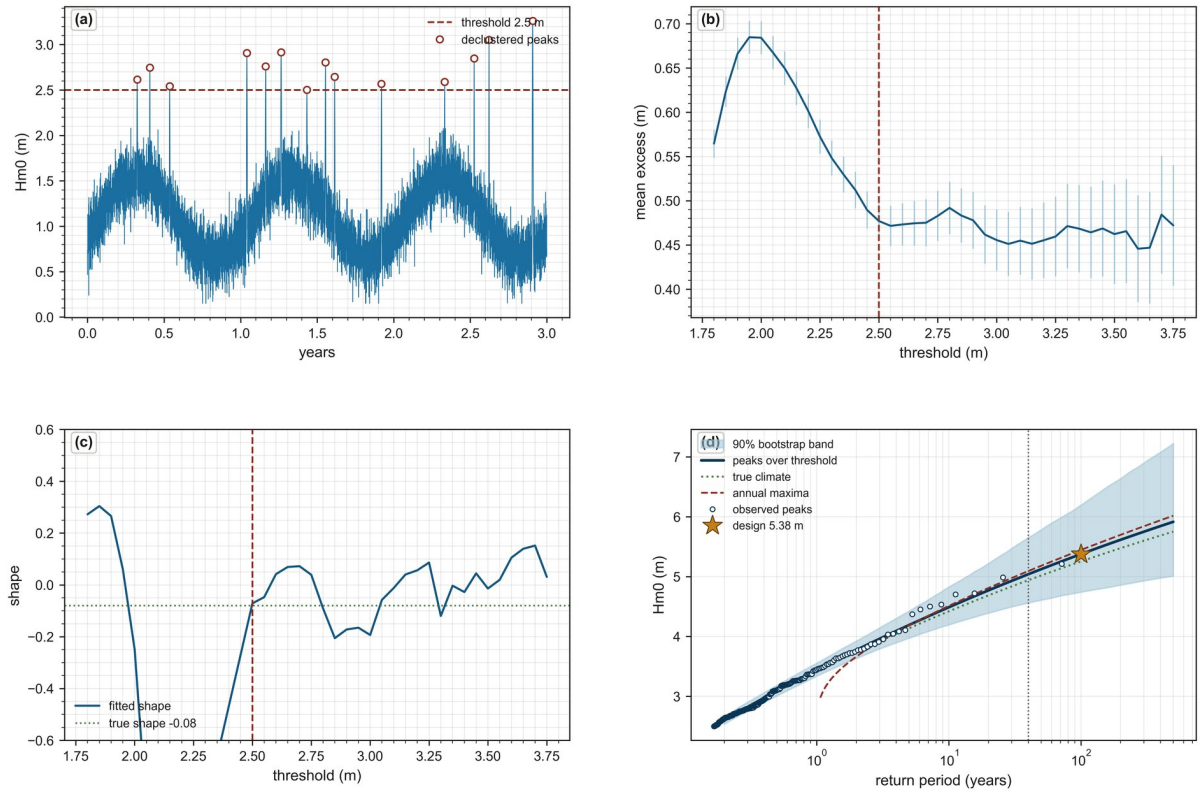


Figure 1: Extreme value analysis of a 40-year record. (a) The first three years of the record, with the threshold and the declustered peaks. (b) Mean residual life against threshold. (c) Fitted shape against threshold. (d) Return level plot with the 90% bootstrap band, the annual maxima fit and the design value.

sure (Rankine and Coulomb coefficients, and the pressure diagram including the water table and any surcharge) supplies the load on the back of a wall.

5.1 The material catalogue

`Sediment(name, d50, specific_gravity=2.65, porosity=0.4, friction_angle=32, cohesion=0, phi_sorting=0.6, d90=None, description="")` stores a material, with derived `relative_density` ($s - 1$), `dry_unit_weight`, `saturated_unit_weight`, `submerged_unit_weight`, `grading` (d_{90}/d_{50} , with d_{90} defaulting to $2.5 d_{50}$), and `cohesive`. `sediment(key)` returns a catalogue material (Table 4); an unknown key raises with the available keys listed.

Table 4: The built-in material catalogue.

Key	d_{50} (mm)	ϕ' (deg)	Description
<code>soft_clay</code>	cohesive	22	normally consolidated, undrained strength governs
<code>stiff_clay</code>	cohesive	26	overconsolidated
<code>silt</code>	0.03	28	mobile at almost any wave
<code>very_fine_sand</code>	0.09	29	suspends readily, high siltation
<code>fine_sand</code>	0.19	31	the usual beach and nearshore sand
<code>medium_sand</code>	0.38	33	typical dredged fill
<code>coarse_sand</code>	0.75	35	good drained backfill
<code>fine_gravel</code>	6	38	free draining
<code>coarse_gravel</code>	30	40	shingle beach
<code>rock_fill</code>	150	42	engineered granular backfill

5.2 Mobility

The dimensionless grain size, the critical Shields parameter (Soulsby and Whitehouse 1997), and the settling velocity (Soulsby 1997) are given by Equation 8:

$$\begin{aligned}
 D_* &= d_{50} \left[\frac{g(s-1)}{\nu^2} \right]^{1/3}, \\
 \theta_{cr} &= \frac{0.30}{1 + 1.2D_*} + 0.055 [1 - e^{-0.020D_*}], \\
 w_s &= \frac{\nu}{d_{50}} \left[\sqrt{10.36^2 + 1.049D_*^3} - 10.36 \right].
 \end{aligned} \tag{8}$$

The fall velocity relation applies from silt to gravel and covers the full grain-size range of the catalogue.

Under waves, `wave_orbital_velocity(Hs, T, depth)` gives the near-bed orbital amplitude from linear theory and `wave_shields(material, Hs, T, depth)` the Shields parameter, with the Swart friction factor for a rough turbulent bed and a Nikuradse roughness $k_s = 2.5 d_{50}$ (Equation 9):

$$f_w = \exp \left[5.213 \left(\frac{A}{k_s} \right)^{-0.194} - 5.977 \right], \quad f_w \leq 0.3. \tag{9}$$

`bed_mobility(material, Hs, T, depth)` returns the mobility state used by the scour functions: whether the bed is live, the regime (live bed, clear water or cohesive), and the ratio of the Shields parameter to its critical value. The scour relations implemented in this package are live-bed results and are gated on this state.

5.3 Earth pressure

`earth_pressure_coefficient(friction_angle, kind="active", wall_friction=0, backslope=0)` returns the Rankine coefficient for "active", "at_rest" ($K_0 = 1 - \sin \phi'$) or "passive". A non-zero wall friction selects the Coulomb active coefficient, which applies to a rough wall face and returns a smaller force.

Selection between the coefficients is governed by wall movement. A wall free to move by the displacement that mobilizes the active state carries active pressure. A wall restrained against that displacement carries at-rest pressure, which is approximately 1.5 times the active value for the materials in the catalogue.

`lateral_earth_pressure(material, height, water_table=0, surcharge=0, kind="active")` returns the pressure diagram with effective stress and pore pressure as separate arrays. `lateral_earth_force(...)` integrates the diagram into `soil`, `water` and `total` forces and the lever arm of the total. Below the water table the soil contributes its submerged unit weight and the pore water contributes the full hydrostatic gradient at $K = 1$. The total therefore exceeds the dry-soil value. The soil component depends on the backfill material; the water component is removed only by drainage.

```
from pyCoastal.applications.sediment import (
    sediment, critical_shields, fall_velocity, wave_shields, lateral_earth_force,
)

sand = sediment("medium_sand")
critical_shields(sand), fall_velocity(sand)          # about 0.035 and 0.05 m/s
wave_shields(sand, Hs=2.0, T=9.0, depth=8.0)["mobile"]
lateral_earth_force(sand, height=10.0, water_table=0.0, surcharge=10.0,
                    kind="at_rest")["total"]         # kN/m
```

6 Harbor agitation and port layout

Module: `pyCoastal.applications.port`. *Examples:* `examples/port_diffraction.py`, `examples/port_layout_comparison.py`.

The module computes phase-resolved wave propagation into a harbor and reports the wave energy at the berths. Breakwaters are rasterized onto the grid as reflecting walls with a configurable absorption. Waves are introduced at a soft source line. The open boundaries carry sponge layers that damp outgoing and reflected energy.

6.1 The solver

The model integrates the second-order wave equation in flux form (Equation 10),

$$\frac{\partial^2 \eta}{\partial t^2} = \nabla \cdot (c^2 \nabla \eta), \quad (10)$$

with zero flux on every face adjacent to a breakwater cell, using an explicit leapfrog step at a Courant number of 0.35. The celerity is taken from the linear dispersion relation (Equation 1) at the local depth and the run period, which gives the correct wavelength in intermediate water.

The solution applies to the single frequency of the run. The model excludes spectral content, breaking, wave-current interaction, refraction over a varying bed and nonlinear transfers.

Harbor performance is the disturbance coefficient (Equation 11)

$$K_d = \frac{H_{\text{local}}}{H_{\text{reference}}}, \quad (11)$$

where the reference height is measured at a probe in open water outside the harbor. This normalization removes the dependence on source calibration. For a semi-infinite breakwater the solver returns K_d near 0.5 on the geometric shadow boundary and Fresnel fringes in the illuminated field.

6.2 Geometry

`Breakwater(points, width=20.0, name="breakwater", absorption=0.0)` defines a polyline of given thickness; cells within `width/2` of the centerline are set to land. `absorption` applies a damping collar in the water adjacent to the structure. It is a model parameter; the resulting reflection coefficient is obtained from `measure_reflection`. `measure_reflection(absorption, period, depth)` propagates a normally incident wave onto a full-width wall and evaluates $K_r = (H_{\text{max}} - H_{\text{min}})/(H_{\text{max}} + H_{\text{min}})$ from the standing-wave envelope (Table 5). The mapping is non-monotonic: above about 0.75 the collar acts as an impedance step and reflection increases. Values of 0.3 to 0.4 correspond to a rubble mound (K_r near 0.4); 0 corresponds to a vertical caisson.

Table 5: Reflection coefficient against the absorption setting.

absorption	0.00	0.25	0.50	0.75	1.00
K_r	0.95	0.48	0.22	0.19	0.25

`PortLayout(Lx=1200, Ly=900, dx=4.0, depth=10.0, breakwaters=[])` holds the domain, grid spacing, constant depth, and structures. Five layouts are built in (`LAYOUTS`):

- `harbour_layout ("two_arm")`: two shore-normal arms with a gap entrance, and optionally a back wall (drop "east" from the sponge sides so the quay reflects);
- `offset_entrance_layout ("offset_entrance")`: overlapping arms forming a dog-leg entrance, which requires two diffractions to reach the basin;
- `hooked_breakwater_layout ("hooked")`: a main breakwater with a shore-parallel hook, with the entrance oriented along the coast;
- `detached_breakwater_layout ("detached_screen")`: two arms with a detached screen off the gap;
- `marina_layout ("marina")`: an outer harbor protecting an inner basin through an offset second opening.

Wave direction. `IncidentWave(height=1.5, period=8.0, direction=0.0)` accepts an oblique direction. A phased source line illuminates only a parallelogram of the domain (`illuminated_mask`), and `simulate_port` issues a warning when a structure lies outside that region. Oblique cases are therefore run by rotating the layout with `rotate_layout(layout, degrees)` and driving it shore-normal; the solution depends on the relative angle only.

Absorption setting. With a fully reflecting detached screen, the screen and the arms form a resonant pocket and basin agitation increases. For a 150 m gap with 280 m arms at $T = 9$ s,

mean basin K_d is 0.35 without the screen, 0.51 with a reflecting screen, and 0.04 with the same screen and arms at an absorption of 0.35.

6.3 Running a simulation

```
import numpy as np
from pyCoastal.applications.port import IncidentWave, harbour_layout, simulate_port

layout = harbour_layout(gap=130, arm_length=300, back_wall=True, absorption=0.35)
wave = IncidentWave(height=1.5, period=9.0, direction=np.deg2rad(0))

result = simulate_port(layout, wave, sponge_sides=("west", "north", "south"))

result.disturbance_coefficient      # Kd over the whole basin
result.probe((1120, 450))          # Kd at one point
result.berth_report({"quay": (1120, 450)})
result.operable_fraction({"quay": (1120, 450)}, limit=0.5)

simulate_port(layout, wave, duration=None, cfl=0.35, sponge_thickness=
None, sponge_strength=2.0, sponge_sides=..., source_x=None, ramp_periods=3.0,
analysis_periods=6.0, n_snapshots=120, store_from=None, reference_point=None) re-
turns a PortResult with the coordinate axes, stored surface snapshots, the land mask, the
analysis-window wave_height and the reference_height. The default duration covers two
domain crossings plus the ramp and analysis windows. The default sponge thickness is 1.5
wavelengths.
```

6.4 Worked examples

examples/port_diffraction.py propagates a 9 s swell at 25 degrees into a two-arm harbor by rotating the layout, writes an animation of the instantaneous surface, and reports K_d at four berths:

Output of examples/port_diffraction.py

```
Wavelength      : 81.7 m   (20 cells per wave)
Celerity         : 9.08 m/s
Entrance gap    : 130 m = 1.59 wavelengths
Approach angle  : 25 deg off shore-normal
Incident height : 1.50 m
```

berth	Kd	Hs (m)
entrance	0.12	0.18
outer basin	0.33	0.49
north berth	0.55	0.82
south berth	0.13	0.20

```
Berths within a 0.5 m operational limit:
entrance      yes
outer basin   yes
north berth   no
south berth   yes
```

```
Wrote media/port_disturbance.png
Wrote media/port_diffraction.gif
```

examples/port_layout_comparison.py runs all five layouts under the same design wave and ranks them by basin agitation:

Output of examples/port_layout_comparison.py

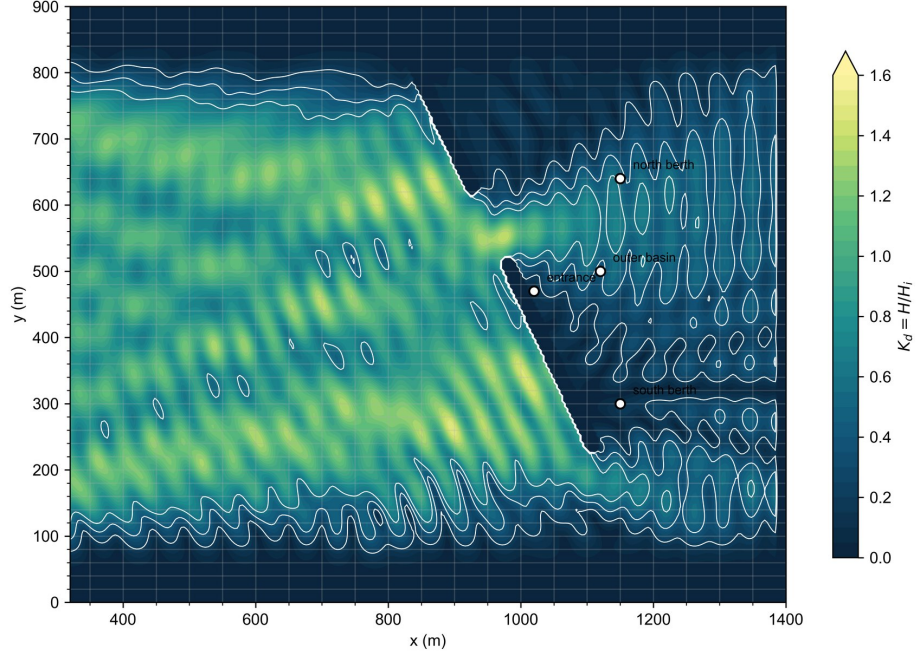


Figure 2: Disturbance coefficient in the harbor. The north berth lies in the diffracted field from the entrance and exceeds the 0.5 m operational limit.

```

ran two_arm
ran offset_entrance
ran hooked
ran detached_screen
ran marina

```

layout	mean K_d	95th pct K_d
Outer harbour and inner basin	0.007	0.022
Overlapping arms, dog-leg	0.008	0.029
Detached screen off the gap	0.039	0.067
Two arms, straight gap	0.146	0.370
Hooked main breakwater	0.180	0.768

Quietest basin: Outer harbour and inner basin (0.007 mean K_d)

Noisiest basin: Hooked main breakwater (0.180 mean K_d)

Wrote media/port_layouts.png

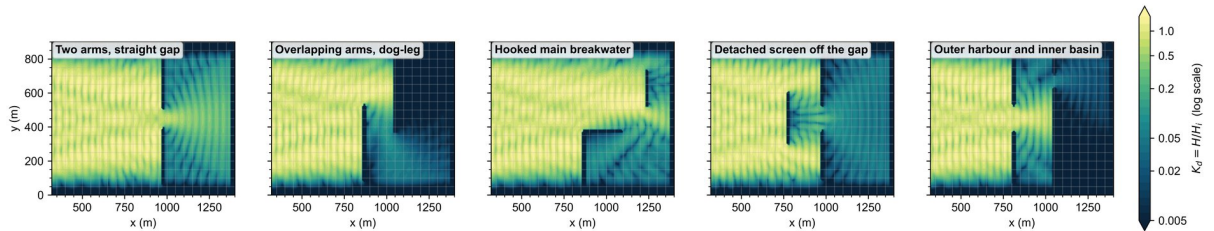


Figure 3: The five built-in layouts under the same design wave, each panel labeled with its layout and mean basin K_d .

7 Rubble-mound breakwaters

Module: `pyCoastal.applications.structures`. *Example:* `examples/breakwater_design.py`.
Browser: Breakwater.

The module implements armor sizing, wave overtopping and the crown wall. The sources are Van der Meer (1988) in the form of Van Gent et al. (2003) for rock, Van der Meer (1988) and the unit design numbers for concrete armor, Hudson (SPM 1984) as a screening check, the Rock Manual (2007) for layer geometry, EurOtop (2018) for overtopping and the tolerable discharge limits, and Pedersen (1996) for the crown wall. Each relation records its source and its validity range, computes outside that range, and reports the condition.

7.1 The design condition

`DesignConditions(Hm0, Tm10, depth=15.0, storm_duration=6*3600)` carries the wave and water level at the toe. `DesignConditions.from_peak_period(Hm0, Tp, ...)` converts with $T_{m-1,0} = T_p/1.1$ for a single-peaked spectrum. The mean period is $T_m = T_{m-1,0}/1.164$, from $T_p = 1.1 T_{m-1,0} = 1.28 T_m$ for a JONSWAP spectrum (Goda 2010). Derived properties are the wave count $N = t_{\text{storm}}/T_m$, capped at 7500, the deep and local wavelengths, and the surf similarity (Equation 12)

$$\xi_{m-1,0} = \frac{\tan \alpha}{\sqrt{H_{m0}/L_{m-1,0}}}, \quad L_{m-1,0} = \frac{gT_{m-1,0}^2}{2\pi}. \quad (12)$$

7.2 Armor stability

Van der Meer, Van Gent form. Van der Meer's (1988) coefficients 6.2 and 1.0 belong to the mean period T_m . The Rock Manual (2007) adopts the form of Van Gent et al. (2003), written in $T_{m-1,0}$ and the 2 % wave height, which is used here. Two regimes are selected by the surf similarity against a critical value (Equations 13, 14, 15):

$$\frac{H_s}{\Delta D_{n50}} = 8.4 P^{0.18} \left(\frac{S}{\sqrt{N}} \right)^{0.2} \frac{H_s}{H_{2\%}} \xi_{m-1,0}^{-0.5} \quad (\xi < \xi_{cr}, \text{ plunging}), \quad (13)$$

$$\frac{H_s}{\Delta D_{n50}} = 1.3 P^{-0.13} \left(\frac{S}{\sqrt{N}} \right)^{0.2} \frac{H_s}{H_{2\%}} \sqrt{\cot \alpha} \xi_{m-1,0}^P \quad (\xi \geq \xi_{cr}, \text{ surging}), \quad (14)$$

$$\xi_{cr} = \left(\frac{8.4}{1.3} P^{0.31} \sqrt{\tan \alpha} \right)^{1/(P+0.5)}. \quad (15)$$

$H_{2\%}/H_s$ is taken as 1.4, the Rayleigh value (`height_ratio`). On a shallow foreshore the ratio is lower (Battjes and Groenendijk 2000), so the default is conservative there.

$\Delta = \rho_s/\rho_w - 1$ (1.585 for 2650 kg/m³ rock in seawater), P is the notional permeability (0.1 impermeable core with a filter, 0.4 permeable core, 0.5 homogeneous, 0.6 very permeable), and S the damage level (`DAMAGE_LEVELS`: 2 start of damage, the design value; 8 and 12 to 17 failure). `rock_armour_vandermeer(conditions, cot_alpha, Delta=1.585, permeability=0.4, damage=2.0, safety_factor=1.0)` returns D_{n50} , M_{50} , the governing regime, ξ , and ξ_{cr} .

Depth at the toe. `depth_limit_warnings(conditions)` reports $H_{m0}/h > 0.6$, a wave the toe depth cannot carry, as an inconsistent input, and $H_{m0}/h > 0.2$ as a shallow toe where the Rayleigh $H_{2\%}$ overstates the load.

Concrete armor units. `concrete_armour(conditions, unit, cot_alpha, damage=0.5, density=2400)` sizes a unit by its own relation, with $\Delta = \rho_c/\rho_w - 1$, N_{od} the number of displaced units per strip one D_n wide (0.5 is the start of damage) and $s_{om} = H_s/L_{om}$ on the mean period:

- cubes, two layers (Van der Meer 1988): $H_s/(\Delta D_n) = (6.7N_{od}^{0.4}/N^{0.3} + 1.0) s_{om}^{-0.1}$;
- tetrapods, two layers (Van der Meer 1988): $H_s/(\Delta D_n) = (3.75N_{od}^{0.5}/N^{0.25} + 0.85) s_{om}^{-0.2}$;
- single-layer units: the design stability number $N_s = 2.7$ for Accropode and 2.8 for Core-Loc and Xbloc (CEM Table VI-5-37);
- dolos: Hudson with $K_D = 16$ on H_s .

`CONCRETE_UNITS` holds the layer count, layer coefficient, porosity and reference slope of each unit. `design_rubble_mound` selects the relation from the `armour` argument, reports the unit mass at the concrete density, and sets the underlayer at a tenth of the unit mass in rock.

Hudson (SPM 1984). $H_s/(\Delta D_{n50}) = (K_D \cot \alpha)^{1/3}/1.27$, where 1.27 converts the SPM's $H_{1/10}$ basis to H_s . `rock_armour_hudson(conditions, cot_alpha, Delta, Kd=4.0)` provides a screening estimate. The relation has no dependence on wave period, storm duration, permeability or damage level.

Layer geometry (Rock Manual). `armour_layer(Dn50, n_layers=2, layer_coefficient=1.0, porosity=0.37)` returns the layer thickness $t = nk_t D_{n50}$ and the number of stones per square meter.

7.3 Overtopping

For a sloping structure, EurOtop (2018) takes the lesser of the breaking and the maximum expressions (Equations 16, 17):

$$\frac{q}{\sqrt{gH_{m0}^3}} = \frac{0.023}{\sqrt{\tan \alpha}} \gamma_b \xi \exp \left[- \left(2.7 \frac{R_c}{\xi H_{m0} \gamma_b \gamma_f \gamma_\beta \gamma_v} \right)^{1.3} \right], \quad (16)$$

$$\frac{q}{\sqrt{gH_{m0}^3}} = 0.09 \exp \left[- \left(1.5 \frac{R_c}{H_{m0} \gamma_f \gamma_\beta} \right)^{1.3} \right]. \quad (17)$$

For a plain vertical wall in non-impulsive conditions (EurOtop eq. 7.1) (Equation 18):

$$\frac{q}{\sqrt{gH_{m0}^3}} = 0.047 \exp \left[- \left(2.35 \frac{R_c}{H_{m0} \gamma_\beta} \right)^{1.3} \right], \quad (18)$$

valid when the impulsiveness parameter $h_* = 1.35 (h/H_{m0}) (2\pi h/gT_{m-1,0}^2)$ exceeds about 0.23; `overtopping_vertical` reports `h_star` and `impulsive`.

The factors are roughness γ_f (`ROUGHNESS_FACTORS`: 1.0 for smooth concrete, 0.55 and 0.40 for impermeable and permeable two-layer rock, 0.38 to 0.49 for concrete units), obliquity γ_β (`obliquity_factor(beta_degrees)`), berm γ_b , and wave wall γ_v . EurOtop reports roughly a factor-of-three scatter about the mean discharge; `overtopping_with_uncertainty(result, factor=3.0)` returns the band, and `required_crest_freeboard(conditions, q_allowable, cot_alpha, ...)` inverts the relation by bisection. `assess_overtopping(q)` lists which uses a discharge is tolerable for (Table 6).

Table 6: Tolerable mean discharges q in l/s/m (TOLERABLE_
DISCHARGE, EurOtop 2018).

Key	q	Use
pedestrians_unaware	0.03	unaware pedestrians, narrow walkway, sea in view
pedestrians_aware	0.1	aware pedestrians, able to see and avoid the hazard
harbour_quay_equipment	0.4	equipment set back 5 to 10 m from the crest
trained_staff	1	trained staff, well shod and protected, wide walkway
buildings_structural	1	structural damage to buildings behind the defense
vehicles_low_speed	10	vehicles at low speed on a road behind the crest
vehicles_moderate_speed	50	driving at moderate speed, pulsating overtopping
embankment_seaward	50	damage to a maintained grass or armored embankment

7.4 Designing a section

`design_rubble_mound(conditions, cot_alpha=2.0, armour="rock_two_layer_permeable", damage=2.0, permeability=0.4, Delta=1.585, tolerable_use="trained_staff", safety_factor=1.0, scatter_factor=3.0)` sizes the stone with Van der Meer and sets the crest so that the **upper** confidence bound on overtopping meets the limit. The mean discharge is exceeded in approximately half of the realizations and is not used for the crest level. The function returns a `BreakwaterDesign` with `Dn50`, `M50`, `regime`, `crest_freeboard`, `q_mean`, `q_upper`, `layer`, `governing_limit`, `section`, and `summary()`.

The rest of the section is built from the design:

- **Toe scour.** `reflection_coefficient(cot_alpha, xi, permeable=True)` (Seelig and Ahrens 1981, $K_r = a\xi^2/(b + \xi^2)$ with $a = 0.6$, $b = 6.6$ for rubble) and `toe_scour(bed, Hs, T, depth, reflection)`, which scales Xie's standing-wave result by K_r : $S/H_s = 0.4 K_r / \sinh(kh)^{1.35}$. The scaling is an assumption with the correct limits at $K_r = 1$ and $K_r = 0$, and the result carries the flag `screening_only`. `breakwater_toe_scour(design, depth, bed)` applies it to a design, gated by `bed_mobility`.
- **Foundation.** `mound_foundation(design, depth, bed="medium_sand", bedding_material="coarse_sand", settlement_allowance=0.0)` returns the bedding blanket on a geotextile, carried past each toe by the larger of the scour apron and twice the scour depth, with a minimum of 3 m, and a toe berm of filter stone. The berm is sized as filter stone; the near-bed orbital velocities at its level are lower than at the waterline.
- **Crown wall.** `crown_wall(design, still_water_level, deck_width=7.5, parapet_width=2.0, berm_width=None, ...)` places a stepped concrete crown block behind an armor berm ($3D_{n50}$ by default) and loads it with Pedersen (1996), as given in CEM Table VI-5-61 (`pedersen_crown_loads`): $F_{h,0.1\%} = 0.21\sqrt{L_{om}/B}(1.6p_m y_{\text{eff}} + Ap_m h'/2)$, $M_{0.1\%} = 0.55(h' + y_{\text{eff}})F_{h,0.1\%}$ and $p_{b,0.1\%} = Ap_m$, with $p_m = \rho_w g(R_{u,0.1\%} - A_c)$, $R_{u,0.1\%} = 1.12H_s \xi_m$ ($\xi_m \leq 1.5$) or $1.34H_s \xi_m^{0.55}$, $y = (R_{u,0.1\%} - A_c) \sin 15^\circ / (\sin \alpha \cos(\alpha - 15^\circ))$ and $y_{\text{eff}} = \min(y/2, f_c)$. $A = \min(A_2/A_1, 1)$ is taken as 1, its upper bound, and the horizontal force and the uplift are combined, which is conservative since they do not peak together. The deck is widened until sliding reaches 1.2 and overturning about the rear heel 1.5. Parameters outside Pedersen's tested range are reported.
- **Roundhead.** A head receives wave attack from a wider range of directions, and armor on a convex surface has reduced interlock. Design practice applies a lower stability coefficient at the head. With a ratio $r = K_{D,\text{head}}/K_{D,\text{trunk}}$ from `roundhead_kd_ratio(armour, cot_`

alpha) (Table 7), $D_{n50,head} = D_{n50,trunk}/r^{1/3}$ and $M_{50,head} = M_{50,trunk}/r$. `roundhead(design, kd_ratio=None, raise_crest=0.0)` returns the head design.

Table 7: Head to trunk stability coefficient ratios (ROUNDHEAD_KD_RATIO).

Armor family	slope 1:1.5	1:2	1:3
rock	0.95	0.80	0.65
cubes	0.85	0.75	0.60
tetrapod	0.72	0.64	0.50
accropode	0.80	0.75	0.65
dolos	0.65	0.58	0.45

```
from pyCoastal.applications.structures import (
    DesignConditions, design_rubble_mound, roundhead, mound_foundation,
)

conditions = DesignConditions.from_peak_period(
    Hm0=4.0, Tp=11.0, depth=12.0, storm_duration=6 * 3600,
)
design = design_rubble_mound(conditions, cot_alpha=2.0, tolerable_use="trained_staff")
print(design.summary())
head = roundhead(design) # heavier armor at the head
base = mound_foundation(design, depth=12.0, bed="medium_sand")
```

7.5 Worked example

Output of examples/breakwater_design.py

```
Design condition  Hm0 = 4.00 m, Tm-1,0 = 10.00 s, depth = 12.0 m
Storm            6.0 h, N = 2513 waves
Section         trunk
Slope           1 : 2
Armour          rock_two_layer_permeable, Dn50 = 1.67 m, M50 = 12.3 t (plunging)
Layer           3.34 m thick, 0.45 stones/m2
Crest freeboard  Rc = 5.69 m (Rc/Hm0 = 1.42)
Overtopping     mean 0.333 l/s/m, upper bound 1 l/s/m
Governing limit  1 l/s/m, Trained staff, well shod and protected, wide walkway
```

Warnings

- Hm0/h = 0.33: the toe is in shallow water. H2% is taken as 1.4 Hs from the Rayleigh
- ↪ distribution, which overstates it here and errs on the heavy side; Battjes and Groenendijk
- ↪ (2000) with the foreshore slope gives the real ratio.

Tolerability of the mean discharge:

```
NO    0.03 l/s/m  pedestrians_unaware
NO    0.10 l/s/m  pedestrians_aware
ok    0.40 l/s/m  harbour_quay_equipment
ok    1.00 l/s/m  trained_staff
ok    1.00 l/s/m  buildings_structural
ok    10.00 l/s/m vehicles_low_speed
ok    50.00 l/s/m vehicles_moderate_speed
ok    50.00 l/s/m embankment_seaward
```

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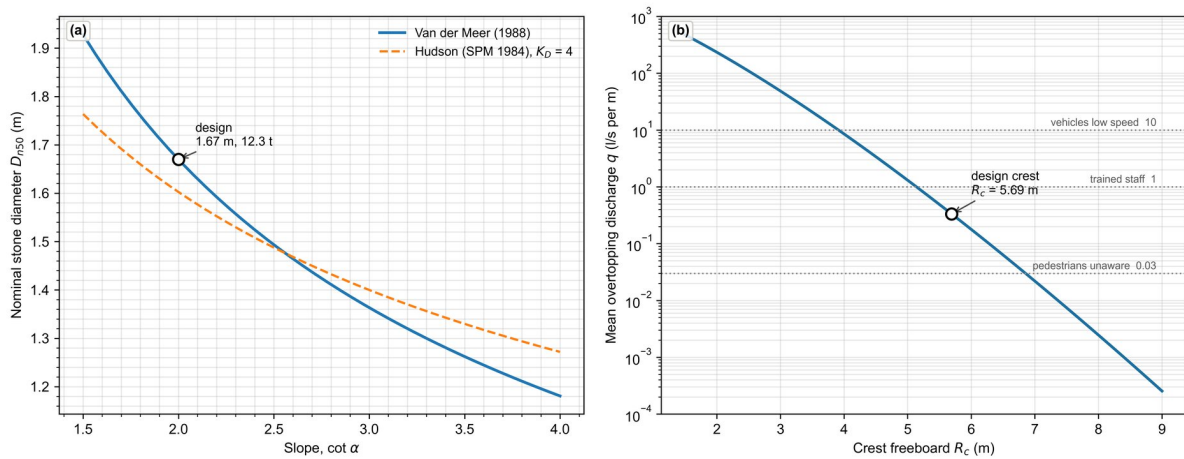


Figure 4: Breakwater design curves. (a) Nominal stone diameter against slope. (b) Mean overtopping discharge against crest freeboard, with the EurOtop scatter band.

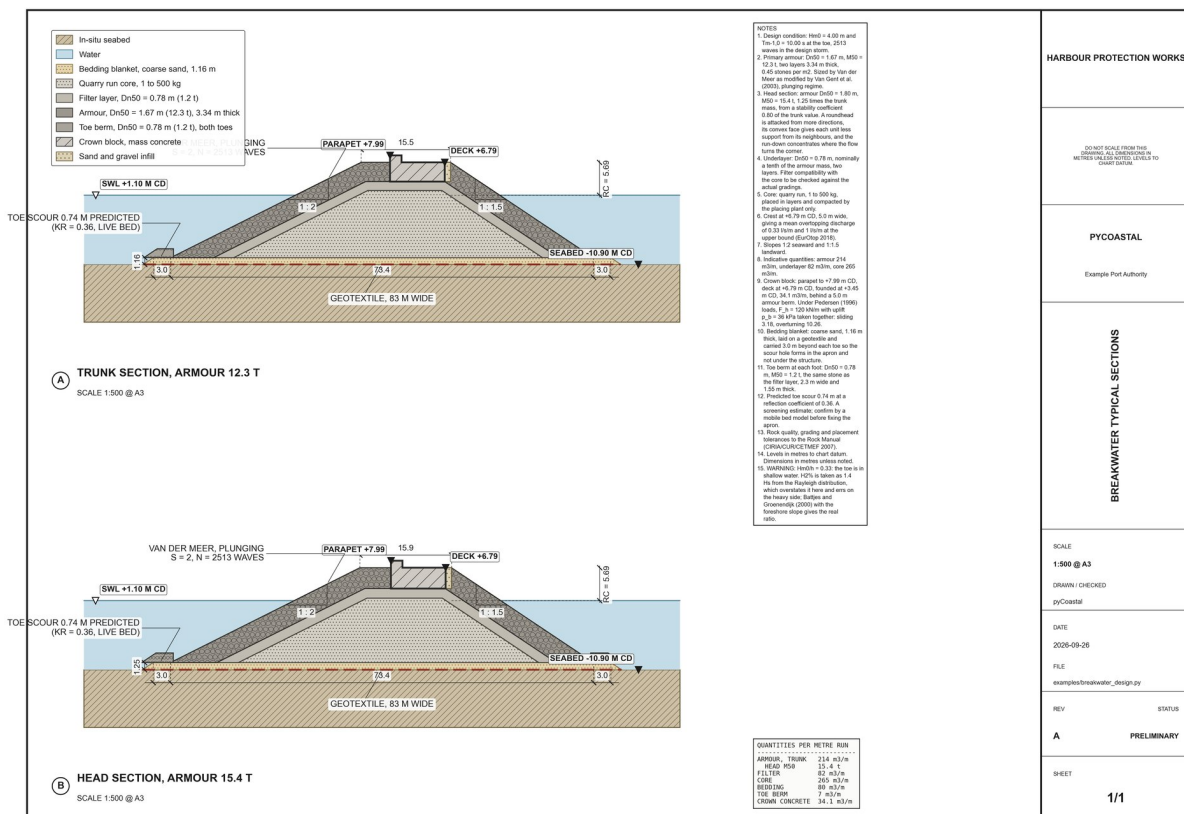


Figure 5: The same design issued as a drawing sheet: trunk and head sections with the armor drawn as individual stones at the computed D_{n50} , the bedding blanket, toe berms, crown wall, and generated specification notes.

8 Vertical seawalls

Module: `pyCoastal.applications.seawall`. *Example:* `examples/seawall_section.py`.
Browser: Seawall.

The module sizes an L-shaped gravity seawall: crest level from `EurOtop`, founding level from the scour allowance, toe stone from `Tanimoto`, Goda wave pressures on the wetted face, earth and water pressure from the backfill, stem and base thickness from the bending and shear at the foot of the stem, and a base width incremented until sliding, overturning, the middle third and the allowable bearing pressure are satisfied under each load case. `applications.sections` converts the result into a drawing and a bill of quantities (Section 16).

8.1 Goda pressures

Goda's method (1974, 2010) gives a trapezoidal pressure distribution on the face of a vertical wall. The design wave is $H_{\max} = 1.8H_{m0}$, capped at the breaking height of Goda's breaker index at the depth h_b five significant wave heights seaward (`depth_limit=True`, `goda_breaking_height(T, depth, slope)`):

$$H_b = 0.17L_0 \left\{ 1 - \exp \left[-1.5\pi \frac{h_b}{L_0} \left(1 + 15 \tan^{4/3} \theta \right) \right] \right\}, \quad L_0 = \frac{gT^2}{2\pi}. \quad (19)$$

The pressure at the still water level, the elevation of the distribution above it, and the pressure at the base of the upright are given by Equation 20:

$$p_1 = \frac{1}{2}(1 + \cos \beta)(\alpha_1 + \alpha_2 \cos^2 \beta)\rho g H_{\max}, \quad \eta^* = 0.75(1 + \cos \beta)H_{\max}, \quad p_3 = \alpha_3 p_1, \quad (20)$$

with the pressure at the crest p_4 interpolated where the crest is below η^* , and a triangular uplift $p_u = \frac{1}{2}(1 + \cos \beta)\alpha_1\alpha_3\rho g H_{\max}$ at the seaward edge. The coefficients in Equation 20 are given by Equation 21:

$$\begin{aligned} \alpha_1 &= 0.6 + \frac{1}{2} \left(\frac{4\pi h/L}{\sinh 4\pi h/L} \right)^2, \\ \alpha_2 &= \min \left[\frac{h_b - d}{3h_b} \left(\frac{H_{\max}}{d} \right)^2, \frac{2d}{H_{\max}} \right], \\ \alpha_3 &= 1 - \frac{h'}{h} \left[1 - \frac{1}{\cosh 2\pi h/L} \right], \end{aligned} \quad (21)$$

where d is the depth over the berm and h' the depth to the underside of the upright. `goda_pressures(Hm0, T, depth, wall_toe_depth, berm_depth=None, crest_freeboard=5.0, beta_degrees=0.0, slope=1/30, Hmax_factor=1.8, depth_limit=True)` returns the pressures, `eta_star`, the force F , its lever arm, and the uplift. The pressures are in excess of hydrostatic about the still water level; the still-water pressure is applied separately in the stability calculation. This is the standard non-impulsive distribution; Takahashi's impulsive coefficient is not applied, so a wall on a high mound needs a separate check.

8.2 Scour, toe, and stability relations

- `scour_depth_vertical_wall(Hm0, T, depth, coefficient=0.4)`: Xie (1981), $S/H = 0.4/\sinh(kh)^{1.35}$, the equilibrium depth a quarter wavelength from the wall, used as a scour allowance for the founding level.

- `toe_stone_tanimoto(Hs, T, berm_depth, berm_width, Delta=1.585, beta_degrees=0.0)`: Tanimoto et al. (1982) with Takahashi (2002), the berm in front of a vertical wall (Equation 22):

$$\frac{H_s}{\Delta D_{n50}} = \max \left\{ 1.8, 1.3 \frac{1-\kappa}{\kappa^{1/3}} \frac{h'}{H_s} + 1.8 \exp \left[-1.5 \frac{(1-\kappa)^2}{\kappa^{1/3}} \frac{h'}{H_s} \right] \right\}, \quad (22)$$

with $\kappa = \kappa_1 \kappa_2$, $\kappa_1 = (4\pi h'/L')/\sinh(4\pi h'/L')$ and $\kappa_2 = \max\{0.45 \sin^2 \beta \cos^2(2\pi B_M \cos \beta/L'), \cos^2 \beta \sin^2(2\pi B_M \cos \beta/L')\}$, h' the depth over the berm, B_M its width and L' the wavelength at h' . κ_2 carries the standing-wave velocity at the berm: a narrow berm close to the face sits near a velocity node and takes small stone. The design reports when the berm does not reach the quarter-wavelength scour zone.

- `toe_stone_size(Hm0, toe_depth, water_depth, Delta=1.585, damage=0.5)`: Van der Meer's toe formula for a sloping structure, kept for that use.
- `bearing_pressures(normal, base_width, net_moment)`: $p_{\max}$, $p_{\min}$, the eccentricity, and `middle_third`; outside the middle third the no-tension triangular distribution is used. The moment may be taken about either edge.
- `hydrostatic_force(depth)`: $\frac{1}{2}\gamma h^2$ at $h/3$.
- `stem_section(moment, shear)`: the thickness a reinforced concrete cantilever needs to EN 1992-1-1 at concept level, per meter width: bending $d_M = \sqrt{M_{Ed}/(0.9\rho f_{yd})}$ with $\rho = 1\%$, and shear without links $d_V = V_{Ed}/v_{Rd,c}$ with $v_{Rd,c} = \max[0.12k(100\rho f_{ck})^{1/3}, 0.035k^{1.5}f_{ck}^{1/2}]$, C35/45 and B500, plus 100 mm to the bar centroid.
- `sliding_safety` and `overturning_safety` remain as general relations for a single block under a single water level.

8.3 The free body

Stability is computed on the wall and the fill standing on its heel, with total unit weights (2400 kg/m³ concrete; the backfill's dry unit weight above its water table and saturated unit weight below it) and every water pressure applied explicitly:

- on the seaward face, hydrostatic from the sea level of the load case down to the founding level;
- on the virtual back plane through the rear of the heel, the backfill pore pressure from its water table, inside the earth pressure of `lateral_earth_force`;
- under the base, a linear uplift from the sea-level head at the seaward edge to the water-table head at the heel.

Buoyant weights are not used. They imply one water level on both sides of the wall, and a seawall with a saturated backfill has two. The backfill water table is never taken below the still water level, since the sea feeds it.

8.4 Designing a wall

`design_seawall(conditions, still_water_level, seabed_level, tolerable_use="pedestrians_aware", allowable_bearing=300.0, ...)` follows the chain a design office follows:

1. **Crest level** from EurOtop, set so the upper bound of the overtopping scatter meets the tolerable limit.
2. **Founding level** from the scour allowance, bounded by `minimum_embedment` and `maximum_embedment`.
3. **Toe stone** from Equation 22, iterated: the berm thickness changes the depth over the berm and the stone sets the thickness and width.
4. **Goda pressures** on the wetted face, from the still water level down to the seabed.
5. **Stem and base thickness** from the characteristic moment and shear at the foot of the stem, factored by 1.35. The stem is bent seaward by the at-rest backfill at the trough and landward by the wave and the still water less the active backfill. `stem_thickness` and `base_thickness` are minimums, and the base is never thinner than the stem.
6. **Base width** grown in `step` increments until both load cases pass sliding (1.2), overturning (1.5), the middle third and `allowable_bearing`:
 - *wave crest*: Goda pressure and uplift plus the still water in front, against the active backfill, overturning about the rear heel;
 - *drawdown*: the at-rest backfill and its pore water pushing seaward, the sea at the trough ($SWL - H_{m0}/2$, `drawdown_level`), overturning about the seaward toe.

Full Goda uplift is applied under the base including the embedded part, and passive resistance in front of the embedment is ignored; both are conservative. The default allowable bearing of 300 kPa is a presumptive value for a medium dense sand and is replaced by the geotechnical designer's figure. A retained height above 8 m is reported, since a cantilever L-wall is normally limited to about that height. The result is a `SeawallDesign` with every level and dimension, the pressures and forces, both load cases with their bearing, the `stem` demand, `governing_case`, the overtopping, `warnings`, `summary()`, and `quantities()`.

8.5 Worked example

Output of `examples/seawall_section.py`

```

Design condition      Hm0 = 2.00 m, Tm-1,0 = 7.27 s
Water level           SWL +2.50 m CD, depth at wall 3.50 m

Crest level           +6.36 m CD (Rc = 3.86 m, Rc/Hm0 = 1.93)
Promenade level       +5.36 m CD
Founding level        -3.00 m CD (embedment 2.00 m, scour 2.00 m)
Wall height           9.36 m
Base                  10.48 m wide x 1.00 m thick, stem 1.00 m
Sizing                59 iterations on base width

Goda Hmax             3.22 m (depth limited)
Pressures             p1 = 34.7 kPa, p3 = 30.9 kPa, pu = 27.9 kPa
Wave force            195 kN/m at 5.42 m above the base
Weight, uplift        1840 kN/m total, 731 kN/m hydrostatic + 146 kN/m wave

Backfill              Medium sand, phi' = 33 deg, retained 8.36 m
Water table           +5.36 m CD behind the wall
Earth pressure        548 kN/m at 2.88 m (K = 0.455, 196 soil + 351 water)

Load case 1, wave crest pushing landward
  sliding              inf
  overturning          1.90
  bearing              p_max = 107 kPa (allowable 300), e = -0.28 m
Load case 2, trough with the backfill pushing seaward
  front water          102 kN/m at +1.50 m CD
  net seaward          446 kN/m

```

sliding	1.56
overturning	1.74
bearing	$p_{\max} = 221 \text{ kPa}$ (allowable 300), $e = +1.74 \text{ m}$
Governing case	drawdown
Stem at the base	$M_{\text{Ed}} = 1403 \text{ kNm/m}$, $V_{\text{Ed}} = 501 \text{ kN/m}$, needs 1.00 m
Toe protection	$D_{n50} = 0.31 \text{ m}$, $M_{50} = 0.08 \text{ t}$, berm 2.0 m wide x 0.63 m thick (Tanimoto, $N_s = 4.02$)
Overtopping	mean 0.333 l/s/m, upper bound 1 l/s/m
Governing limit	1 l/s/m, trained staff, well shod and protected, wide walkway
Concrete	18.84 m ³ /m (45.2 t/m)
Backfill	69.78 m ³ /m
Toe rock	2.10 t/m
Excavation	21.00 m ³ /m

Warnings

- $h^* = 0.10$ is below 0.23, so the conditions are impulsive. The non-impulsive EurOtop formula
↳ used for the crest level understates the discharge, and Goda understates the load; check
↳ against the impulsive formulae before fixing the section.
- The 2.0 m toe berm sits close to the wall, where the standing wave moves the bed least, which
↳ is why Tanimoto allows 0.08 t stone. The scour hole forms about a quarter wavelength out, 11
↳ m from the face; widen the berm to reach it if the toe is also the scour protection, and the
↳ stone will grow.
- The wall retains 8.4 m. A cantilever L-wall is ordinary practice up to about 8 m; beyond that
↳ a counterforted wall, a caisson or an anchored sheet-pile wall is the usual form, and the
↳ stem here (1.00 m) shows why.
- Pore water is 64% of the pressure on the back of the wall. A drain holding the water table at
↳ still water level would cut the total from 548 to about 403 kN/m.
- The drawdown case governs: the wall is sized by the saturated backfill pushing it seaward at
↳ the trough, so the backfill grading and the drainage detail set the base width more than the
↳ design wave does.

Overtopping tolerability at the design discharge

no	0.03 l/s/m	pedestrians_unaware
no	0.1 l/s/m	pedestrians_aware
no	0.4 l/s/m	harbour_quay_equipment
ok	1 l/s/m	trained_staff
ok	1 l/s/m	buildings_structural
ok	10 l/s/m	vehicles_low_speed
ok	50 l/s/m	vehicles_moderate_speed
ok	50 l/s/m	embankment_seaward

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The drawdown case governs this wall: the base width is set by the saturated backfill acting seaward at the wave trough, with the resultant close to the edge of the middle third. Pore water accounts for 64% of the pressure on the back of the wall, and a drain holding the water table at the still water level would reduce the total from 548 to 403 kN/m. The retained height of 8.4 m is at the upper limit for a cantilever L-wall, which is reflected in the 1.0 m stem. The conditions are impulsive ($h^* = 0.10$), so the crest and the load both need the impulsive check before the section is fixed.

9 Wave loads on piles

Module: pyCoastal.applications.piles. *Example:* examples/pile_wave_loads.py.
Browser: Monopile.

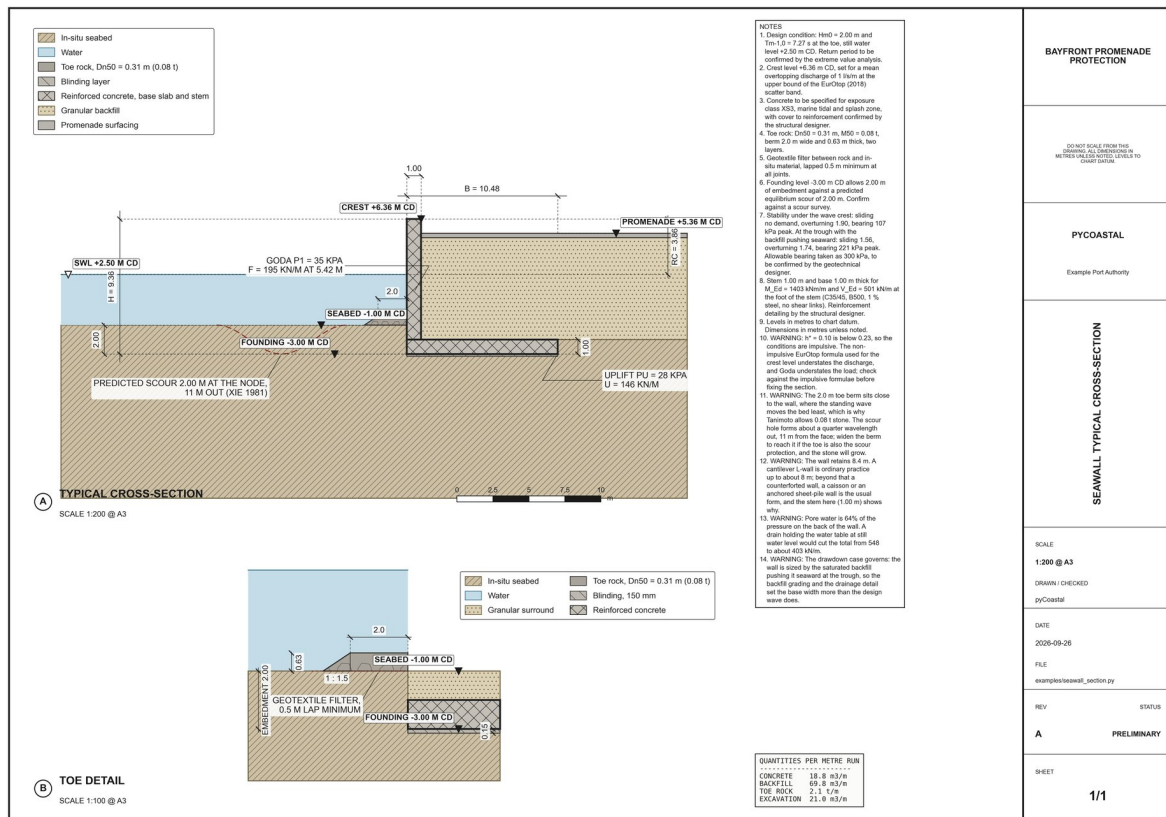


Figure 6: The seawall drawing sheet: typical section at a true stated scale, an enlarged toe detail, the design parameter block, and specification notes generated from the design.

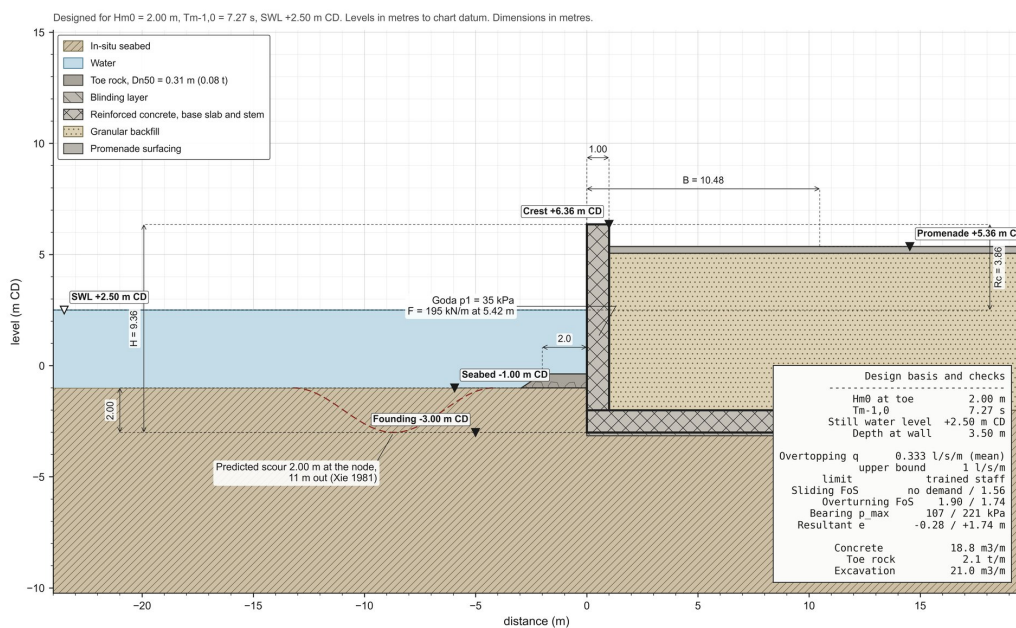


Figure 7: Standalone seawall section with plot axes, for a report.

The wave load on a pile is a distribution along the pile that varies through the wave cycle. The base shear and the mudline overturning moment reach their maxima at different phases. The module computes the load distribution, integrates it, and sweeps the wave phase to locate both maxima.

9.1 The Morison equation

The Morison equation (Morison et al. 1950) splits the load per unit length on a slender member into a drag term in phase with the velocity and an inertia term in phase with the acceleration:

$$f(z) = \frac{1}{2}\rho C_d D u|u| + \rho C_m \frac{\pi D^2}{4} \frac{\partial u}{\partial t}. \quad (23)$$

The slender-member condition is D/L below about 0.2. Above that value the wave diffracts around the member and the Morison equation does not apply; the functions report the condition through the returned `diffraction_ratio` and a warning.

Kinematics. A design wave in intermediate or shallow water is nonlinear: at $H/h = 0.4$ and an Ursell number of 17 the crest stands about $0.65H$ above the still water level, not $0.5H$, and the steep front carries the largest accelerations. The load is computed on a stream function wave (`theory="stream"`, the default), Fenton's (1988) Fourier solution of the full nonlinear free-surface problem with no Eulerian mean current. `StreamFunctionWave(H, T, depth, N=20)` solves for the wave number, celerity, surface and the N Fourier coefficients by Newton's method, with the height raised in six steps from a linear start, and its `kinematics(z, phase)` returns the horizontal velocity and local acceleration up to the instantaneous surface. A wave that does not converge is at breaking; the load then falls back to linear theory with a warning.

`wave_kinematics(H, T, depth, z, phase=0.0, stretching="wheeler")` returns linear-theory kinematics (`theory="linear"`). Linear theory is undefined between the still water level and the crest; Wheeler (1970) stretching maps the still-water profile onto the instantaneous water column and is the default for that case, with `"none"` and `"extrapolate"` for comparison. For the design wave above, linear theory understates the mudline moment by about 30 %.

Regime and coefficients. The Keulegan-Carpenter number $KC = u_{\max}T/D$ determines the dominant term: below about 3 the load is inertia dominated, above about 20 drag dominated. `drag_inertia_coefficients(KC, rough=True)` returns post-critical values from DNV-RP-C205: $C_d = 1.05$ for a rough cylinder, which applies to a pile carrying marine growth, and 0.65 for a clean cylinder. C_m decreases from the potential-flow value of 2.0 with increasing KC . The values are concept-design defaults; a detailed design takes them from the governing code for the actual roughness, KC and Reynolds number.

Integration and sweep. `morison_load_profile` evaluates Equation 23 up the pile, `integrate_load(z, load, mudline)` integrates the force and the mudline moment, and `morison_pile_load(diameter, H, T, depth, phase=0.0, ...)` returns a `PileLoad` at one phase. `phase_sweep(diameter, H, T, depth, phases=181)` sweeps the cycle and reports where each peaks. `design_monopile(diameter, H, T, depth, rough=True, phases=181, bed=None, theory="stream", Hs=None, current=0.0)` returns the worst-phase load, the scour, and the ratio `crest_underestimate` by which a crest-phase evaluation would understate the moment.

9.2 Scour at a pile

Scour is driven by the sea state and the current; the design wave governs the load only. The Keulegan-Carpenter number for scour is taken at the bed from the significant wave height (`bed_keulegan_carpenter(Hs, T, depth, diameter)`): $U_m = \pi H_s / (T \sinh kh)$ and $KC = U_m T / D$. It is several times smaller than the KC at the surface under the design wave, which sets the Morison coefficients. H_s defaults to $H/1.86$, the Rayleigh ratio for about 1000 waves.

Sumer and Fredsoe (2001), for combined waves and current (Equation 24):

$$\frac{S}{D} = 1.3 \{1 - \exp[-A(KC - B)]\} \quad (KC > B), \quad A = 0.03 + 0.75U_{cw}^{2.6}, \quad B = 6e^{-4.7U_{cw}}, \quad (24)$$

with $U_{cw} = U_c / (U_c + U_m)$ the current share of the near-bed velocity. $U_{cw} = 0$ recovers the waves-only relation of Sumer, Fredsoe and Christiansen (1992), with no scour below $KC = 6$, where the horseshoe vortex does not form. Under a steady current $S/D = 1.3$ with a standard deviation of 0.7, which is the `current_only` branch and the limit of the relation. `scour_depth_pile(diameter, KC, current_only=False, live_bed=True, bed=None, ..., current_ratio=0.0)` returns the equilibrium depth; pass a bed to gate it by mobility. `design_monopile` takes the depth-averaged current in m/s and reports when none is given, since a tidal current dominates the scour at most monopile sites.

```
from pyCoastal.applications.piles import design_monopile

result = design_monopile(diameter=8.0, H=12.0, T=13.0, depth=30.0, current=1.0)
result["load"].force / 1e3      # kN, at the worst phase
result["load"].moment / 1e6     # MNm about the mudline
result["load"].inertia_fraction # share carried by the inertia term
result["crest_underestimate"]   # what a crest-phase check would miss
```

9.3 Worked example

Output of examples/pile_wave_loads.py

```
Pile      8.00 m diameter in 30.0 m of water
Wave      H = 12.00 m, T = 13.0 s, stream function theory
KC, regime 6.3, mixed
Cd, Cm    1.05, 1.87
D / L     0.039
```

```
Worst phase 28 degrees (crest at 0), surface at +5.62 m
Base shear  5,271 kN
Mudline moment 111.5 MNm
Effective arm 21.16 m above the mudline
Inertia share 76% of the load
```

Warnings

- The worst moment is 53% larger than the moment at the crest phase. Designing on the crest alone would have understated it.

```
At the crest phase 2,180 kN, 52.3 MNm
At the worst phase 5,271 kN, 111.5 MNm
Designing on the crest would miss 53% of the moment
```

```
Max base shear at 36 degrees
Max mudline moment at 28 degrees
```

```
Scour, waves + current 1.47 m (0.18 D) at bed KC = 2.3, Ucw = 0.42
```

waves alone: 0.00 m
 steady current alone: 10.40 m (1.3 D, standard deviation 5.60 m)

How the load splits with pile diameter

D (m)	KC	regime	inertia share	moment (MNm)
1.0	50.7	drag dominated	3%	6.7
2.0	25.3	drag dominated	15%	14.0
4.0	12.7	mixed	42%	33.7
8.0	6.3	mixed	76%	111.5
12.0	4.2	mixed	86%	243.0

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The 8 m monopile is inertia dominated. On the stream function wave the largest moment, 111.5 MNm, occurs 28 degrees ahead of the crest, on the steep front of the wave where the acceleration is largest; evaluation at the crest phase alone understates it by 53 %. `phase_sweep` returns both phases and the ratio. The diameter table gives the transition from drag to inertia dominance with increasing diameter. For scour, the bed KC from the significant wave is 2.3, below the waves-only threshold, so waves alone give no scour; a 1.0 m/s tidal current ($U_{cw} = 0.42$) gives 1.47 m, and a steady current alone 10.40 m.

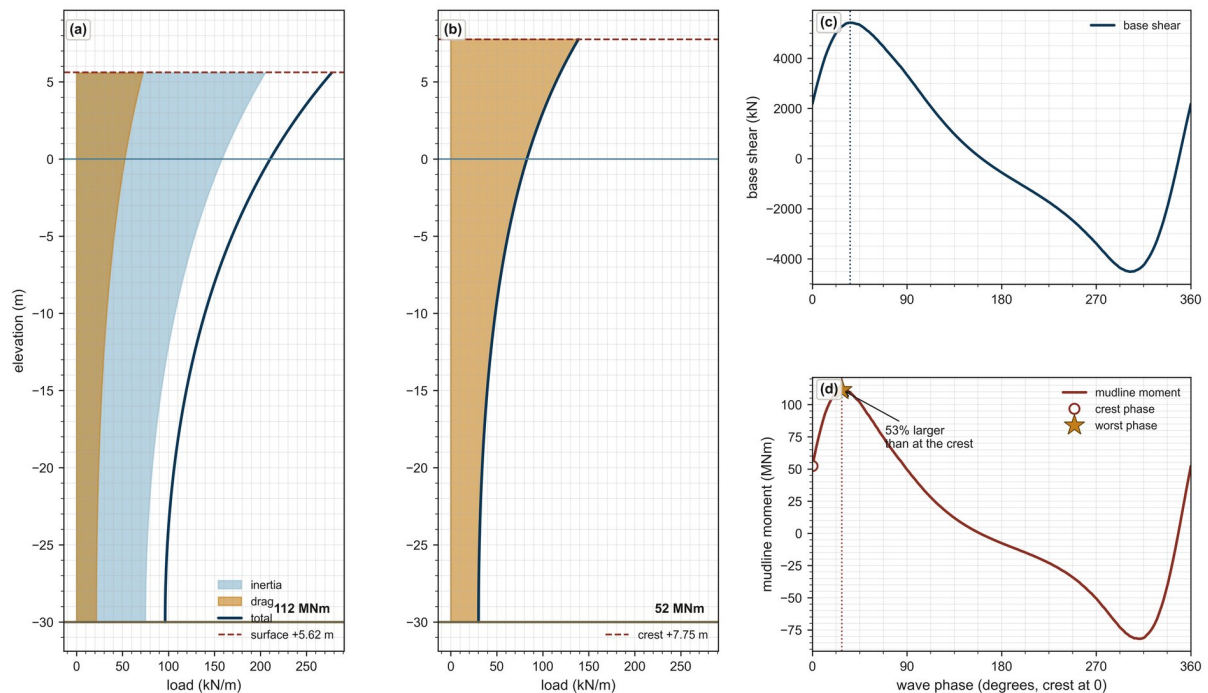


Figure 8: Morison loads on an 8 m monopile. (a) Load profile at the governing phase. (b) Load profile at the crest phase. (c) Base shear through the wave cycle. (d) Mudline moment through the wave cycle.

10 Open-channel hydraulics and backwater

Module: `pyCoastal.applications.river`. Example: `examples/backwater.py`.

The module provides the open-channel hydraulics of a river crossing. A bridge contracts the flow, which raises the upstream water level and produces a backwater profile extending upstream from the structure. The module also supplies one input to the scour chain: the fraction of the discharge passing through the bridge opening, computed from conveyance.

10.1 The channel

`Channel(width=30.0, side_slope=2.0, roughness="natural_clean")` is a trapezoidal prismatic section (zero width is triangular, zero side slope rectangular). Roughness is a Manning n or a key into MANNING: 0.013 and 0.017 for smooth and rough concrete, 0.022 and 0.030 for clean and weedy earth, 0.028 gravel, 0.035 cobbles, 0.030 and 0.045 for clean and weedy natural channels, and 0.035, 0.07, 0.10 for pasture, brush, and wooded floodplains. The section provides `area`, `perimeter`, `top_width`, `hydraulic_radius`, `velocity`, and the conveyance (Equation 25)

$$K = \frac{AR^{2/3}}{n}, \quad Q = K\sqrt{S}. \quad (25)$$

10.2 Two depths

Normal depth is where friction balances the bed slope, $Q = K(y_n)\sqrt{S_0}$ (`normal_depth`, which raises on a horizontal or adverse slope, where uniform flow does not exist). **Critical depth** is where the specific energy $E = y + V^2/2g$ is least and the Froude number is one, $Q^2T/(gA^3) = 1$ (`critical_depth`), with $Fr = V/\sqrt{gA/T}$ (`froude_number`), the form that is right for any section. `classify_slope(channel, discharge, slope)` compares them: mild, steep, critical, horizontal, or adverse. The classification belongs to the channel and the discharge together, so the same reach can be mild in flood and steep at low flow. `profile_type(depth, classification)` names the profile in Chow's notation: M1 behind a bridge on a mild reach, M2 a drawdown to a free overfall, S1 behind a structure on a steep reach.

10.3 Gradually varied flow

`gvf_profile(channel, discharge, slope, control_depth, steps=200, approach=0.99)` integrates the direct step method (Equation 26),

$$\Delta x = \frac{\Delta E}{S_0 - S_f}, \quad S_f = \left(\frac{Qn}{AR^{2/3}} \right)^2, \quad (26)$$

away from the control. Depths are prescribed and distances computed, which is exact for a prismatic channel and requires no iteration. The profile approaches normal depth asymptotically, so the integration terminates at a prescribed fraction of the approach to normal depth. `approach` sets that fraction, default 0.99; a value of 0.95 returns a shorter reach. The criterion used is recorded in the notes of the returned `BackwaterResult`, which also carries `distance`, `depth`, `water_surface`, `bed_level`, `profile`, `reach` and `depth_at(distance)`.

10.4 Afflux at bridge piers

Yarnell (1934) (Equation 27):

$$\Delta H = K(K + 5Fr^2 - 0.6)(a + 15a^4)Fr^2 y, \quad (27)$$

with a the fraction of the area blocked by the piers, Fr the downstream Froude number, and K the pier shape coefficient (PIER_SHAPE: 0.9 semicircular nose to 2.5 for a ten-pile trestle). The fourth power on the blockage gives a strongly nonlinear dependence on pier area. Yarnell's tests were in a rectangular flume at blockages up to about 0.4 with the flow class unchanged through the bridge. Where a bridge chokes the flow to critical, the energy or momentum methods of HEC-RAS apply. `yarnell_afflux(channel, discharge, downstream_depth, blockage, shape="semicircular_nose")` returns the afflux and the upstream depth.

10.5 Flow distribution

`flow_distribution(main, main_depth, slope, floodplains=[(Channel, depth), ...])` splits the discharge by conveyance (Equation 28),

$$\frac{Q_i}{Q} = \frac{K_i}{\sum_j K_j}. \quad (28)$$

The result is the `flow_fraction` input of `scour.BridgeOpening`. Conveyance depends on the roughness and the hydraulic radius, so a wooded floodplain occupying half the section width can carry a tenth of the discharge. An assumed flow fraction propagates directly into the contraction scour depth.

```
from pyCoastal.applications.river import (
    Channel, classify_slope, flow_distribution, gvf_profile, yarnell_afflux,
)

channel = Channel(width=30.0, side_slope=2.0, roughness="natural_clean")
plain = Channel(width=120.0, side_slope=3.0, roughness="floodplain_trees")

state = classify_slope(channel, 250.0, 0.0008)      # normal, critical, mild
afflux = yarnell_afflux(channel, 250.0, state["normal"], blockage=0.18)
profile = gvf_profile(channel, 250.0, 0.0008, afflux["upstream_depth"])
profile.profile, profile.reach, profile.depth_at(1500.0)

split = flow_distribution(channel, state["normal"], 0.0008, [(plain, 1.2)])
split["main_fraction"]
```

10.6 Worked example

Output of examples/backwater.py

```
250 m3/s on a 0.0008 bed, natural_clean
normal depth    3.54 m
critical depth  1.84 m
reach is mild, so a bridge backs water up upstream
```

```
Piers blocking 18% of the section, Froude 0.35
afflux 0.071 m, so the control sits at 3.61 m
M1 profile reaching 5.14 km
- M1 profile on a mild reach. Control at 3.61 m against normal 3.54 m and critical 1.84 m.
- Backwater reaches 5.14 km to get 99% of the way to normal depth. The curve is asymptotic, so
↪ that figure is a convention and moves a long way with it: quoting an extent without quoting
↪ the criterion is meaningless.
```

```
In flood, 1.2 m over a wooded floodplain:
the channel is 26% of the width
and carries 84% of the flow
contraction scour 1.08 m with the computed split, 1.81 m if it is assumed to be all of it
```

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In the example the channel occupies 26% of the section width and carries 84% of the discharge. The computed fraction gives a contraction scour of 1.08 m in `contraction_scour`; a flow fraction of 1.0 gives 1.81 m.

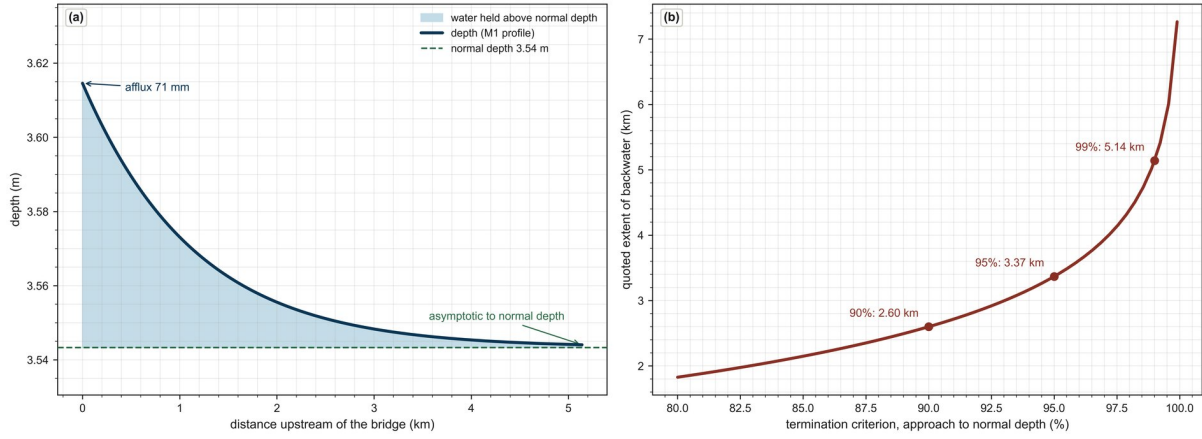


Figure 9: Backwater at the crossing. (a) Depth along the reach upstream of the bridge, against normal depth; the plotted variable is depth, since the bed rises 4 m over the reach while the depth changes by 7 cm. (b) Quoted extent of backwater against the termination criterion.

11 Scour at piers and bridges

Module: `pyCoastal.applications.scour`. *Examples:* `examples/pier_scour.py`, `examples/bridge_scour.py`.

In an estuary the scour drivers are out of phase. The tidal current reverses twice a day, the river current is unidirectional, the wave condition varies with the wind, and the water depth varies with the tide. A single load case combining a design current with a design wave requires an assumption about their co-occurrence. The module evaluates the combined scour at each phase of the tidal cycle and returns the envelope, the governing phase and the component depths.

11.1 Pier scour in combined waves and current

Geometry. A pier is modeled as a stem (`Pier(diameter, shape="circular", length=None)`) on a base (`PierBase(width, length, height, top_level, skew)`). A buried pile cap has no effect until the scour hole reaches its top level. Once exposed, it presents a greater width than the stem, which increases the scour depth and exposes a further part of the base. The two elements are reduced to one effective width by weighting each over the depth of flow it occupies (Equation 29):

$$D_e = \frac{D_{\text{base}} h_{\text{base}} + D_{\text{stem}} (h - h_{\text{base}})}{h}, \quad (29)$$

where h_{base} is the height of the base above the *scoured* bed. Each element carries the HEC-18 shape factor K_1 (`SHAPE_FACTOR`) and the alignment factor (Equation 30)

$$K_2 = \left(\cos \theta + \frac{L}{a} \sin \theta \right)^{0.65}, \quad L/a \leq 12. \quad (30)$$

A base aligned with the ebb is skewed to the flood; the flood skew is taken as the supplement of the ebb skew.

Combined waves and current. Sumer and Fredsoe (2001) (Equation 31):

$$\frac{S}{D} = 1.3 \{1 - \exp[-A(KC - B)]\}, \quad A = 0.03 + \frac{3}{4}U_{cw}^{2.6}, \quad B = 6 \exp(-4.7U_{cw}), \quad (31)$$

with $KC = U_m T/D$ on the *near-bed* orbital velocity and $U_{cw} = U_c/(U_c + U_m)$. At $U_{cw} = 0$ it is exactly the waves-only relation (Equation 24); at $U_{cw} = 1$ the threshold vanishes and $S/D \rightarrow 1.3$ as KC grows.

The steady-current floor. KC is defined on the wave orbital velocity. A strong current with small waves therefore gives a high U_{cw} and a low KC simultaneously, and Equation 31 returns a scour ratio approaching zero. The steady-current case is the $KC \rightarrow \infty$ limit of an oscillatory flow, and a current above the threshold of motion generates a horseshoe vortex independently of the wave condition. The implementation therefore limits the ratio from below at the steady-current value when the approach current alone is live-bed. This limit is an addition to the published relation and is reported as `current_governs` (`scour_ratio_combined(KC, Ucw, current_live_bed=...)`; `None` disables it).

Depth limitation and time scale. `depth_limitation(depth, diameter)` applies $\tanh(h/D)$ (Breusers et al. 1977). The factor is 0.995 at $h/D = 3$ and 0.76 at $h/D = 1$. The skin-friction Shields parameter under the current uses $u_* = U\kappa/\ln(11h/k_s)$ with $k_s = 2.5d_{50}$, and the time scale (Sumer, Christiansen and Fredsoe 1992) is given by Equation 32,

$$T^* = \frac{1}{2000} \frac{h}{D} \theta^{-2.2}, \quad T = T^* \frac{D^2}{\sqrt{g(s-1)d_{50}^3}}, \quad S(t) = S_{eq} [1 - e^{-t/T}]. \quad (32)$$

For a pier of order 2 m width, T is of order 20 hours, against a tidal half cycle of about 6 hours.

Feedback. `equilibrium_scour(pier, base, depth, current, Um, period, ...)` solves the coupling between scour depth and effective diameter by bisection and returns the *deepest* self-consistent root. A scour hole does not refill, and the shallow root corresponds to a transient state.

Protection. `riprap_size(velocity)` follows HEC-23 guideline 12, $d_{50} = 0.692(KV)^2/[2g(s-1)]$ with $K = 1.5$ for round-nosed and 1.7 for rectangular piers, and `scour_protection(pier, base, velocity, scour)` sizes a falling apron reaching two obstacle widths from the face, at least three stones thick, with a launch allowance. An apron relocates the scour hole to its perimeter; it does not remove it.

The tidal cycle. `EstuaryConditions(mean_depth, tidal_amplitude, tidal_current, river_current, Hs, Tp, tidal_period=M2, current_phase=90, bed="medium_sand", wave_follows_tide=False)` takes positive as ebb. The river current adds to the tidal current on the ebb and opposes it on the flood, so the peak combined current occurs on the ebb. `current_phase` sets the lead of the current over the elevation (90 degrees for a standing tide, 0 for a progressive one). `design_pier_scour(pier, conditions, base, samples=73, protection_factor=1.0)` evaluates every phase and returns a `PierScourDesign` with the equilibrium scour (for protection and the long-term foundation check), the `tidal_limited` scour one half cycle can cut, `base_exposed`, `undermined`, the `envelope`, and the `protection`.

```
from pyCoastal.applications.scour import (
    Pier, PierBase, EstuaryConditions, design_pier_scour,
)
```

```
pier = Pier(diameter=2.5, shape="circular")
```

```

base = PierBase(width=7.0, length=12.0, height=2.5, top_level=-1.5, skew=20.0)
estuary = EstuaryConditions(mean_depth=9.0, tidal_amplitude=2.2,
                             tidal_current=1.1, river_current=0.4,
                             Hs=1.2, Tp=5.5, bed="medium_sand")
design = design_pier_scour(pier, estuary, base)
design.equilibrium, design.tidal_limited, design.undermined, design.protection

```

11.1.1 Worked example: pier scour

Output of examples/pier_scour.py

Pier 2.5 m on a 7 x 12 m base, top -1.5 m to the bed
Peak ebb 1.50 m/s, peak flood 0.70 m/s

```

Governing phase      145 deg (ebb)
  current             0.78 m/s (peak is 1.50)
  depth              7.20 m
  effective diameter  5.15 m (stem is 2.50)
  equilibrium scour   5.93 m
  one tidal half      1.76 m (time scale 17.7 h)
  base                UNDERMINED

```

- Governing phase 145 deg, on the ebb: 0.78 m/s current and 0.47 m/s near-bed orbital velocity
↳ in 7.20 m of water.
- $U_{cw} = 0.62$, so the waves and current both matter. $KC = 0.5$ against a threshold of 0.32.
- Equilibrium scour 5.93 m, 1.30 times the effective diameter of 5.15 m.
- Shallow water takes 12% off the unlimited depth: $h/D = 1.4$.
- The hole exposes the base completely (2.5 m) and reaches its underside. The foundation is
↳ undermined, not merely exposed, and the bearing check has to be redone on the scoured section.
- Flow misalignment multiplies the base width by 1.32 at 20 deg of skew. A base set square to
↳ one direction of a reversing tide is skewed to the other.
- Scour time scale 17.7 hours against a 6.2 hour tidal half cycle, so one half cycle reaches
↳ 1.76 m, 30% of equilibrium. The hole does not refill on the reverse flow, so successive tides
↳ work it deeper towards the equilibrium. Size protection on the equilibrium either way.
- Sumer and Fredsoe's relations carry a standard deviation near 0.7 diameters in steady
↳ current. Treat the number as a mean, not a bound, and design protection with margin on it.

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Two results follow from the phase sweep. **The governing phase differs from the phase of peak current.** The effective diameter is weighted over the depth of flow, so an exposed base contributes more at low water. The governing phase in this case occurs after peak ebb, at a current of 0.78 m/s against a peak of 1.50 m/s. **The response to burial depth is discontinuous.** With the base buried deeper than the scour depth of the bare stem, the scour is that of the stem. With the base 1 m shallower, the feedback of Equation 29 raises the equilibrium scour to approximately twice that depth.

11.2 Total bridge scour

HEC-18 divides total scour at a crossing into contraction scour, local scour at the pier and abutment scour. On a contracted crossing the pier component is frequently the smallest of the three. The module implements all three and evaluates each over the tidal cycle.

The opening. BridgeOpening(approach_width, opening_width, pier_blockage=0, abutment_length=25, abutment_shape="spill_through", abutment_skew=90, slope=5e-4, flow_fraction=1.0, abutment_depth_fraction=0.4) gives the contraction ratio W_1/W_2 with the piers subtracted from the gross opening. flow_fraction is Q_2/Q_1 , best

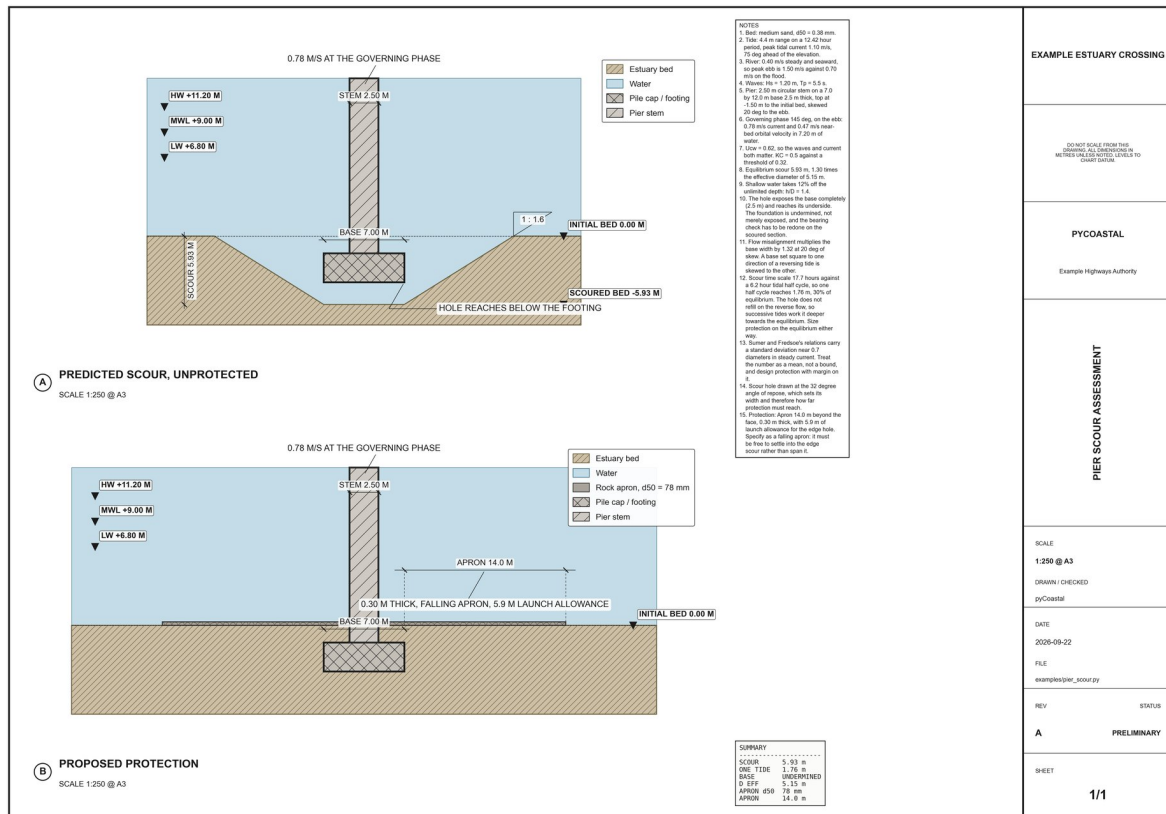


Figure 10: Pier scour assessment sheet. View A is the predicted scour without protection. View B is the proposed apron on an intact bed. The two states are drawn separately.

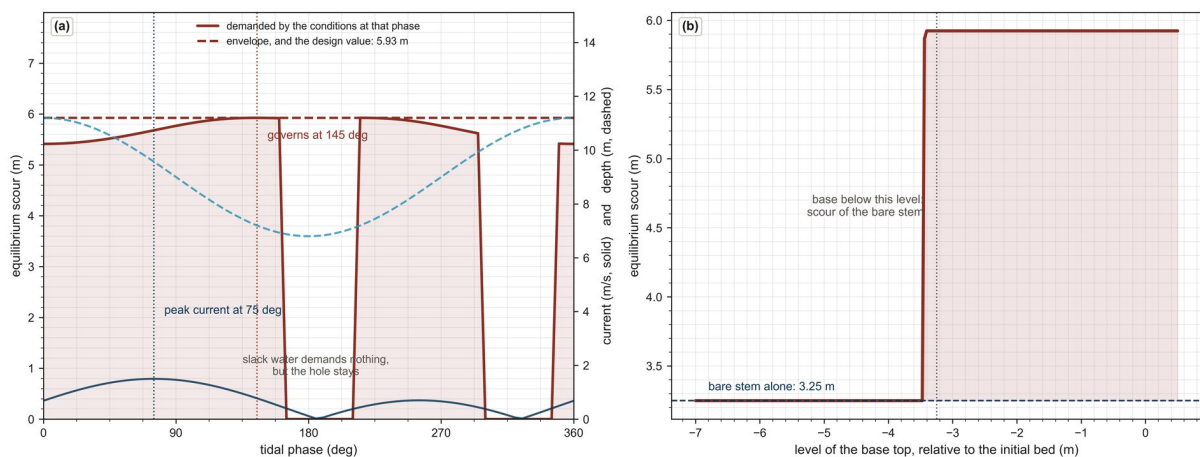


Figure 11: Pier scour. (a) Equilibrium scour, current and depth through the tidal cycle. (b) Equilibrium scour against the level of the top of the base.

computed with `river.flow_distribution` (Section 10). `abutment_depth_fraction` sets Froehlich's y_a , the depth where the embankment actually stands; feeding it a mid-channel depth in a deep estuary returns a hole deeper than the water.

Contraction scour. Two modes, switched by the critical velocity $V_c = 6.19 y^{1/6} D_{50}^{1/3}$ (HEC-18 eq. 6.1). Live bed, Laursen (1960) (Equation 33):

$$\frac{y_2}{y_1} = \left(\frac{Q_2}{Q_1} \right)^{6/7} \left(\frac{W_1}{W_2} \right)^{k_1}, \quad (33)$$

with $k_1 = 0.59, 0.64$, or 0.69 as V_*/w is below 0.5 , between 0.5 and 2 , or above 2 (`transport_exponent`). Clear water (HEC-18 eq. 6.4) (Equation 34):

$$y_2 = \left[\frac{0.025 Q_2^2}{D_m^{2/3} W_2^2} \right]^{3/7}. \quad (34)$$

The live-bed relation is a sediment balance: faster water carries more but delivers more, and the two cancel, leaving the width ratio. The two relations were fitted separately and do not meet at the threshold; `contraction_scour(..., regime="live" | "clear")` computes either on demand so the step can be seen.

Abutment scour. Froehlich (1989) for $L'/y < 25$ and HIRE for longer abutments (Equation 35):

$$\frac{y_s}{y_a} = 2.27 K_1 K_2 \left(\frac{L'}{y_a} \right)^{0.43} Fr^{0.61} + 1, \quad \frac{y_s}{y_a} = 4 Fr^{0.33} \frac{K_1}{0.55} K_2, \quad (35)$$

with K_1 from `ABUTMENT_SHAPE` (1.0 vertical, 0.82 wing wall, 0.55 spill through). The +1 in Froehlich is a factor of safety HEC-18 added, not physics, and it is often more than half the answer; it is reported as `safety_margin`. These are the fourth-edition HEC-18 relations, the edition the pier scour also comes from.

Sequential evaluation. `bridge_scour_state` computes the components in the order the flow encounters them. Contraction scour is computed from the approach flow. Local scour at the pier is then computed from the flow *in the contracted opening*, which has a higher velocity and a greater depth. Abutment scour is computed from the approach flow. The components are summed only where they coexist: contraction plus pier scour at a pier, contraction plus abutment scour at an abutment. `design_bridge_scour(opening, conditions, pier, base)` envelopes each component over the cycle and returns a `BridgeScourDesign` with `contraction`, `pier_local`, `abutment`, `total_at_pier`, `total_at_abutment`, `component(name)` and notes. A total exceeding the water depth is reported in the notes and indicates application of the relations outside their fitted range.

Continuing from the pier above, with the same `estuary`, `pier` and `base`:

```
from pyCoastal.applications.scour import BridgeOpening, design_bridge_scour

opening = BridgeOpening(approach_width=140, opening_width=80, pier_blockage=5,
                        abutment_length=22, abutment_shape="spill_through")
design = design_bridge_scour(opening, estuary, pier, base)
design.total_at_pier, design.total_at_abutment, design.component("pier")
```

11.2.1 Worked example: bridge scour

Output of examples/bridge_scour.py

Waterway 140 m, opening 80 m gross less 5 m of piers
Contraction 1.87 to 1

Governing phase 35 deg (ebb)

approach 1.24 m/s in 10.80 m
through the opening 1.51 m/s in 16.62 m

contraction 6.03 m
pier local 5.93 m
abutment 8.37 m

at a pier 10.72 m
at abutment 14.09 m <- governs

- Opening 80 m gross less 5.0 m of piers in a 140 m waterway: a 1.87 to 1 contraction.
- Governing phase 35 deg on the ebb: 1.24 m/s approaching in 10.80 m, becoming 1.51 m/s in
↪ 16.62 m through the opening.
- Live-bed contraction scour: the approach at 1.24 m/s is 1.9 times the 0.67 m/s threshold, so
↪ the opening is fed from upstream and scours only until it can pass what arrives. Note that
↪ this mode does not depend on how fast the water goes: Laursen's relation is a sediment
↪ balance, and above the threshold it turns on the width ratio alone.
- Components: contraction 6.03 m, pier local 5.93 m, abutment 8.37 m. Each is the worst over
↪ the whole cycle; they do not all peak at the same phase.
- Total 10.72 m at a pier and 14.09 m at an abutment. The abutment governs. These are not added
↪ together: they are depths at two different places, and combining them would invent a hole
↪ that exists nowhere.
- Abutment scour by Froehlich, of which 4.32 m (52%) is the +1 flow depth HEC-18 adds as a
↪ factor of safety rather than as physics. Worth knowing before it is treated as a measurement.
- Local scour is computed on the flow in the contracted opening, not on the approach, since the
↪ contraction has already accelerated and deepened it. Taking approach values here is the usual
↪ way a pier gets understated.
- A total of 14.09 m exceeds the 10.80 m of water at the governing phase. That is a signal the
↪ relations have been pushed past where they were fitted, not a foundation level. Check the
↪ abutment depth and the contraction ratio, and get a physical model or a morphodynamic run
↪ before committing to it.
- Long-term degradation or aggradation of the reach is not included and is a separate study.
↪ HEC-18 adds it to these; on a river with a moving bed it can exceed all three.

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11.3 Limitations

A pile cap at bed level can act as a collar and reduce the scour depth. This effect is not credited in the computed depth. Cohesive beds are not supported. Long-term degradation of the reach and channel migration are outside the scope of the module; HEC-18 adds both to the totals computed here.

12 Navigation channels

Module: pyCoastal.applications.channel. *Example:* examples/navigation_channel.py.
Browser: Channel.

The module sizes the depth and width of an approach channel. Both are built as a sum of allowances, each reported separately. The method is the concept-design procedure of PIANC (2014), Report 121. The width and maneuvering-lane tables are module-level dictionaries and

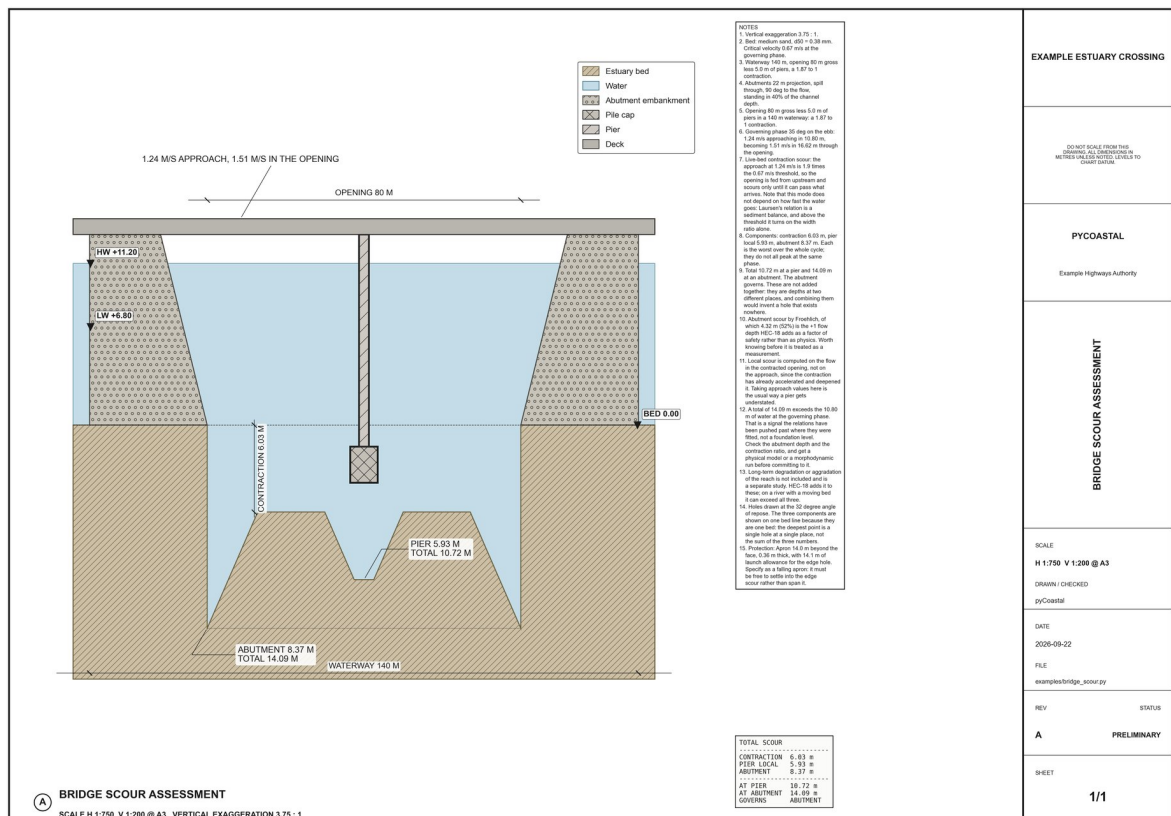


Figure 12: Bridge scour assessment sheet. The three components are drawn on a single bed line, which gives the deepest point at each location.

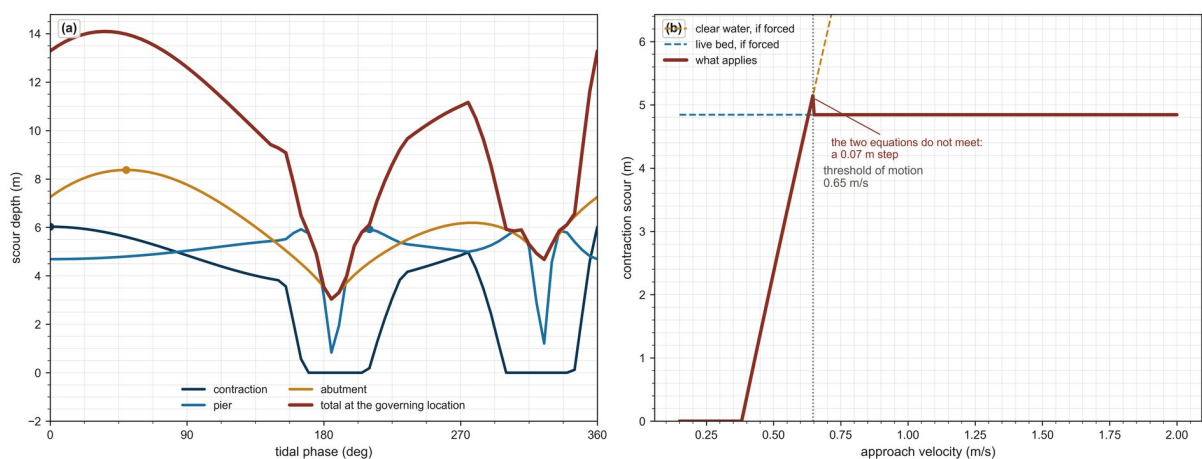


Figure 13: Bridge scour. (a) The three components through the tidal cycle; each reaches its maximum at a different phase and is enveloped separately. (b) Contraction scour against approach velocity, across the live-bed and clear-water modes.

can be edited to match the governing edition of the guideline or local requirements. The procedure is concept design; a final channel layout is confirmed by simulation.

12.1 The vessel

`Vessel(name, length, beam, draught, block_coefficient=0.75, length_pp=None)` is the design ship. L_{pp} defaults to $0.96L_{oa}$. `displaced_volume` is $C_b L_{pp} B T$ and `displacement` is in tonnes at 1025 kg/m^3 . The same class is used by the berthing module.

12.2 The depth chain

Start at the design water level, subtract the static draught, then in turn the squat as the ship moves, its vertical response to waves, the net clearance over the bed, and the tolerances the dredger and the survey cannot beat. What is left is the dredge level.

Squat. ICORELS (1980), as given by PIANC (Equation 36):

$$S = C_s \frac{\nabla}{L_{pp}^2} \frac{F_{nh}^2}{\sqrt{1 - F_{nh}^2}}, \quad F_{nh} = \frac{V}{\sqrt{gh}}, \quad (36)$$

with $C_s = 2.4$ for an open or wide channel. It blows up as $F_{nh} \rightarrow 1$ and is normally kept to $F_{nh} < 0.7$; the function flags speeds where it is unusable. Barrass (1979) is the back-of-envelope check, $S_{\max} = C_b V_k^2 / 100$ in open water and $/50$ in a confined channel, with the speed in knots (`squat_icorels`, `squat_barrass`).

Wave response. `wave_response_allowance(Hs, factor=0.5, period=None, vessel=None)` takes a fraction of H_s as keel motion: about 0.3 for a long vessel in short head seas up to about 0.7 for a short vessel in long beam seas. With any swell this is the single largest judgement in the depth chain. Given the period and vessel, it also reports the ratio of wavelength to ship length, which governs whether the ship contours or bridges the wave.

Clearance stack. `underkeel_clearance(vessel, squat, wave_allowance, net_clearance=0.6, water_level_allowance=0.0, dredging_tolerance=0.3, survey_tolerance=0.2, siltation_allowance=0.2, density_allowance=0.0)` returns each allowance, the gross total below the keel, and the required depth. The dredging and survey tolerances are part of the stack and are included by default.

The squat depends on the water depth and the water depth depends on the squat. `design_channel` resolves this dependency by fixed-point iteration.

12.3 The width chain

`channel_width(vessel, manoeuvrability="moderate", section="outer", two_way=False, speed_class="moderate", bank="sloping_channel_edges", conditions=None)` builds the PIANC width: a basic maneuvering lane (MANOEUVRING_LANE: 1.3, 1.5, 1.8 beams for good, moderate, poor maneuverability), plus additional widths from WIDTH_COMPONENTS for vessel speed, crosswind, crosscurrent, longitudinal current, waves, aids to navigation, bottom surface, depth of waterway, and cargo hazard (with separate outer and inner-channel columns), plus a bank clearance each side (BANK_CLEARANCE), plus a passing distance for two-way traffic (PASSING_DISTANCE). **Unspecified components are assigned the least restrictive class and reported as assumed** in the `assumptions` entry of the result.

12.4 Side slopes, volumes, and turning basins

`dredged_side_slope(bed, factor=2.0)` gives $\cot \beta = f / \tan \phi'$; the factor of 2 covers both the partial factor on a submerged granular slope and the overcut and slumping of dredging, landing near 1:2.6 in gravel and 1:3.8 in silt. `turning_basin_diameter(vessel, assisted=True, current=False)` gives $1.5 L_{oa}$ with tugs, 2.0 unassisted, plus half a length where a current runs through.

`design_channel(vessel, speed, design_water_level, Hs=0, Tp=None, wave_factor=0.5, net_clearance=0.6, ..., bed="medium_sand", existing_bed=None, **width_kwargs)` returns a `ChannelDesign` with the `dredge_level`, `required_depth`, the clearance stack, squat and wave details, `width`, `top_width`, `side_slope`, `dredge_volume(length)`, `warnings`, and `summary()`.

```
from pyCoastal.applications.channel import Vessel, design_channel

vessel = Vessel(name="Post-Panamax container ship", length=336, beam=48.2,
                draught=14.5, block_coefficient=0.68)
channel = design_channel(vessel, speed=8.0, design_water_level=1.20,
                        Hs=1.8, Tp=9.0, existing_bed=-11.5, two_way=True,
                        conditions={"crosswind": "moderate", "waves": "moderate"})
channel.dredge_level, channel.width, channel.dredge_volume(1000)
```

12.5 Worked example

Output of examples/navigation_channel.py

```
Design vessel      Post-Panamax container ship
                  336 x 48.2 x 14.5 m, Cb = 0.68, 157,130 t
Speed             8.0 knots (Fnh = 0.31)
```

Depth chain (m below the design water level)

static draught	14.50
squat	0.37
wave response	0.90
density	0.00
net clearance	0.80
water level uncertainty	0.30
dredging tolerance	0.30
survey tolerance	0.20
siltation	0.20
required depth	17.57

```
Design water level +1.20 m CD
Dredge level      -16.37 m CD
```

Channel width 415 m (8.6 beams, 2-way)

basic manoeuvring lane	72.3
crosswind (moderate)	14.5
crosscurrent (low)	9.6
waves (moderate)	24.1
aids_to_navigation (good)	9.6
bottom_surface (smooth_and_soft)	4.8
depth_of_waterway (shallow)	9.6
bank clearance, each side	24.1
passing distance	77.1

```
Existing bed      -11.50 m CD
Dredge depth      4.87 m
Top width         463 m at 1:5 side slopes
Volume           2,135 thousand m3 per km
```

Turning basin 672 m (2.0 x Loa, assisted, current running)

Squat, two ways

6.0 kn	ICORELS 0.20 m	Barrass 0.24 m	Fnh 0.24
8.0 kn	ICORELS 0.37 m	Barrass 0.44 m	Fnh 0.31
10.0 kn	ICORELS 0.59 m	Barrass 0.68 m	Fnh 0.39
12.0 kn	ICORELS 0.89 m	Barrass 0.98 m	Fnh 0.47

What moves the dredge level

wave factor 0.3 rather than 0.5	-0.35 m	-162 k m3/km
wave factor 0.7 rather than 0.5	+0.35 m	+164 k m3/km
speed 12 kn rather than 8 kn	+0.49 m	+229 k m3/km
net clearance 1.5 m rather than 0.8 m	+0.69 m	+320 k m3/km
no siltation allowance	-0.20 m	-90 k m3/km

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↪ media/navigation_channel.dxf

Wrote media/channel_depth_chain.png

The sensitivity table at the end of the output quantifies the effect of each input. The wave response factor, the fraction of the wave height taken as vertical vessel motion, moves the dredge level by 0.35 m over the range 0.3 to 0.7. The combined dredging and survey tolerances are 0.5 m.

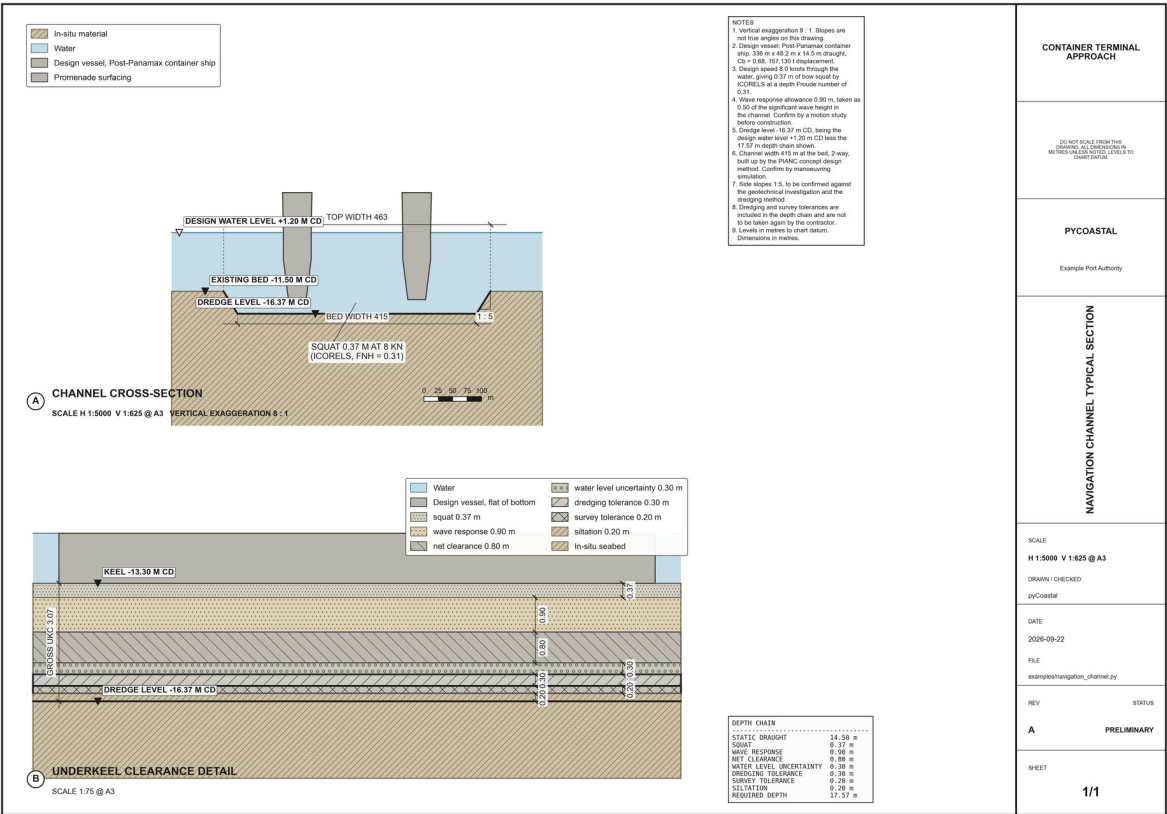


Figure 14: Navigation channel drawing sheet: the channel section at a stated vertical exaggeration, and the underkeel clearance at true scale.

13 Berthing energy and fenders

Module: pyCoastal.applications.berthing. Example: examples/berth_fenders.py.

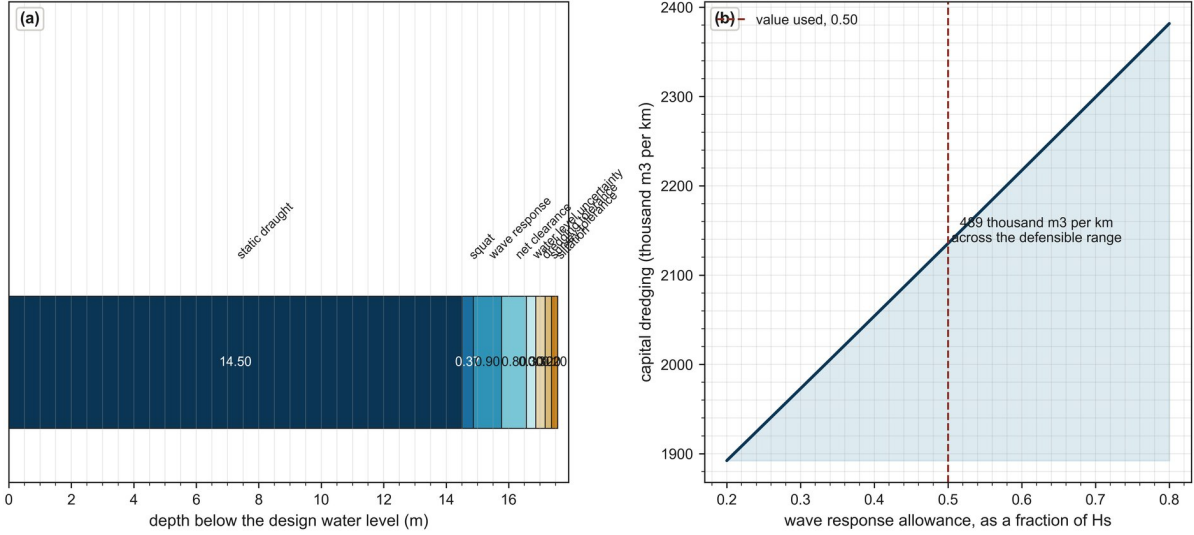


Figure 15: Navigation channel depth. (a) The depth chain, as allowances below the design water level. (b) Capital dredging volume against the wave response factor.

A berthing ship delivers kinetic energy to the fender system. The module checks three limit states. Insufficient fender energy capacity allows contact between the ship and the quay. Insufficient panel area transfers a contact pressure above the limit for the hull. Excessive fender spacing allows hull contact with the structure between fenders. All three are evaluated and reported.

13.1 The energy chain

PIANC (2002, Working Group 33) writes the normal berthing energy as the kinetic energy times four factors (Equation 37):

$$E_N = \frac{1}{2}MV^2 C_m C_e C_s C_c. \quad (37)$$

- **Added mass** (Vasco Costa 1964), $C_m = 1 + 2T/B$, the water the ship drags sideways. It is the only factor that makes the energy larger. Below about 1.1 draughts of water depth the form starts to understate it, and `added_mass_factor(vessel, depth)` warns.
- **Eccentricity**, $C_e = (K^2 + R^2 \cos^2 \gamma) / (K^2 + R^2)$ with the radius of gyration $K = (0.19C_b + 0.11)L_{pp}$, R the distance from the contact point to the center of mass along the berth (default $L_{pp}/4$, the quarter point), and γ the angle between the velocity and that line (default 90 degrees). The rotation carries away energy the fender never sees.
- **Softness** C_s : 1.0 for a soft fender, 0.9 where it is stiff enough that the hull deflects with it.
- **Configuration** C_c (BERTH_CONFIGURATION): 0.8 solid quay, 0.9 semi-solid, 1.0 open piled.

The **abnormal factor**, applied afterwards (ABNORMAL_FACTOR: 1.25 for a large tanker up to 2.0 for ro-ro, ferries and tugs, 1.5 for container ships), covers abnormal berthing conditions such as a parted tug line or a propulsion failure. It is applied after the four factors and is reported separately in the result.

Velocity enters as a square and is the dominant term in the energy. BERTHING_VELOCITY holds the Brolsma et al. (1977) style table (Table 8); `berthing_velocity(vessel, condition)` picks the size class from the displacement.

Table 8: Indicative design berthing velocities (m/s),
BERTHING_VELOCITY.

Condition	small	medium	large
easy_sheltered	0.20	0.15	0.10
difficult_sheltered	0.30	0.20	0.15
easy_exposed	0.40	0.30	0.20
good_exposed	0.50	0.40	0.25
difficult_exposed	0.80	0.60	0.40

13.2 Fender, panel, and spacing

Fender family. Within one family and rubber grade, performance scales with size: energy with the volume of rubber, reaction with the section (Equation 38),

$$E = E_0 \left(\frac{H}{H_0} \right)^3, \quad R = R_0 \left(\frac{H}{H_0} \right)^2. \quad (38)$$

A fender sized for twice the energy returns a reaction larger by a factor of approximately 1.6. `FenderFamily` holds a reference size; the default `CONE_FENDER` is a mid-grade cone fender of 1.0 m, 500 kNm and 1000 kN, with standard heights from 0.3 to 2.5 m. `select_fender(energy, family)` returns the smallest standard size that absorbs the energy and the fraction of its rated capacity used. The routine performs fender scaling. Manufacturer-specific product selection, against tested energy and reaction curves at the design temperature, angle and velocity, is outside its scope.

Hull pressure. `hull_pressure(reaction, panel_width, panel_height, vessel_class)` checks the panel against `HULL_PRESSURE_LIMIT` (150 kN/m² for large tankers, 200 for small tankers and bulk carriers, 400 for container ships and general cargo, 700 for tugs).

Spacing. `fender_spacing(vessel, spacing, projection, bow_radius=None, smallest_vessel=None, clearance=0.15)` applies two checks. Geometric: with the fenders compressed, the standoff must exceed the sagitta of the hull across the gap (Equation 39),

$$s \leq 2\sqrt{R_b^2 - (R_b - p + c)^2}. \quad (39)$$

Practical: the smallest ship using the berth must reach two fenders at once, about $0.15L_{pp}$ of that ship. Without a bow radius the contact point is taken on the parallel midbody, which is straight, and the geometric check is inactive; the length rule then governs. The length rule is evaluated on the *smallest* vessel using the berth, supplied as `smallest_vessel`.

`design_berth(vessel, velocity=None, condition="easy_sheltered", vessel_class="container", configuration="open_piled", ..., smallest_vessel=None, panel_aspect=(1.2, 2.0), family=CONE_FENDER)` runs the whole chain and returns a `BerthDesign` with `normal`, `abnormal`, `energy`, `fender`, `panel`, `pressure`, `spacing`, `adequate`, and `notes`.

```
from pyCoastal.applications.berthing import design_berth
from pyCoastal.applications.channel import Vessel
```

```
ship = Vessel(name="Post-Panamax container", length=366, beam=48.2,
              draught=15.2, block_coefficient=0.68)
feeder = Vessel(name="Feeder", length=140, beam=22.0, draught=8.5,
```

```

        block_coefficient=0.68)
design = design_berth(ship, velocity=0.15, vessel_class="container",
                    configuration="open_piled", depth=17.0,
                    smallest_vessel=feeder)
design.energy, design.fender["height"], design.pressure["pressure"], design.adequate

```

13.3 Worked example

Output of examples/berth_fenders.py

Post-Panamax container: 179 thousand tonnes
Approaching at 0.15 m/s

```

kinetic energy          2018 kNm
x Cm 1.631             added water
x Ce 0.478             rotation about the contact
x Cs 1.000             fender softness
x Cc 1.000             berth configuration
normal berthing        1573 kNm
x 1.50                 abnormal allowance
design energy           2360 kNm

fender                 2.00 m, 4000 kNm rated (59% used)
reaction               4000 kN
hull pressure          325 kN/m2 against 400 allowed
spacing                16.1 m against 20.2 m allowed
ADEQUATE               True

```

- Post-Panamax container: 366 m by 48.2 m, 15.2 m draught, 179 thousand tonnes displacement.
- Berthing at 0.15 m/s, as specified. Velocity is squared in the energy, so it carries more of the answer than anything else here.
- Factors: Cm 1.63 adds the water the ship drags, Ce 0.48 takes away what the ship rotates out, Cs 1.00 and Cc 1.00 take away a little more. Net, the fender sees 78% of the 2018 kNm the ship arrived with.
- Eccentricity is taken at the quarter point, the usual design assumption. A ship that lands flat on its midships rotates about nothing and hands the fender all of it: Ce becomes 1 and the energy rises by 2.09 times.
- Abnormal factor 1.50 for a container, taking the design energy from 1573 to 2360 kNm. This is the allowance for the berthing that goes wrong, and it is the largest single term in this calculation.
- Cone fender, mid grade, 2.00 m: 4000 kNm rated against 2360 kNm required, so it works at 59% of capacity and pushes back 4000 kN.
- Hull pressure 325 kN/m2 over a 2.8 by 4.4 m panel, within the 400 kN/m2 allowed for this class.
- Fenders at 16.1 m centres, inside the 20.2 m limit set by vessel length for Feeder. Contact is on the parallel midbody, which is straight, so only the vessel length rule binds; check the end fenders separately with a bow radius.
- Fender performance here is a scaling model within one family, not a catalogue. Size the fender you are asking for from this, then select against the manufacturer's tested curves at the design temperature, angle and velocity, which is what they guarantee.

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Three of the four PIANC factors reduce the energy and the abnormal factor increases it. The design energy of 2360 kNm is of the same order as the kinetic energy of 2018 kNm at contact. The intermediate values are reported at each step of the chain.

14 Beach nourishment

Module: pyCoastal.applications.nourishment. *Examples:* examples/nourishment_design.py, examples/nourishment_profile.py. *Browser:* Nourishment.

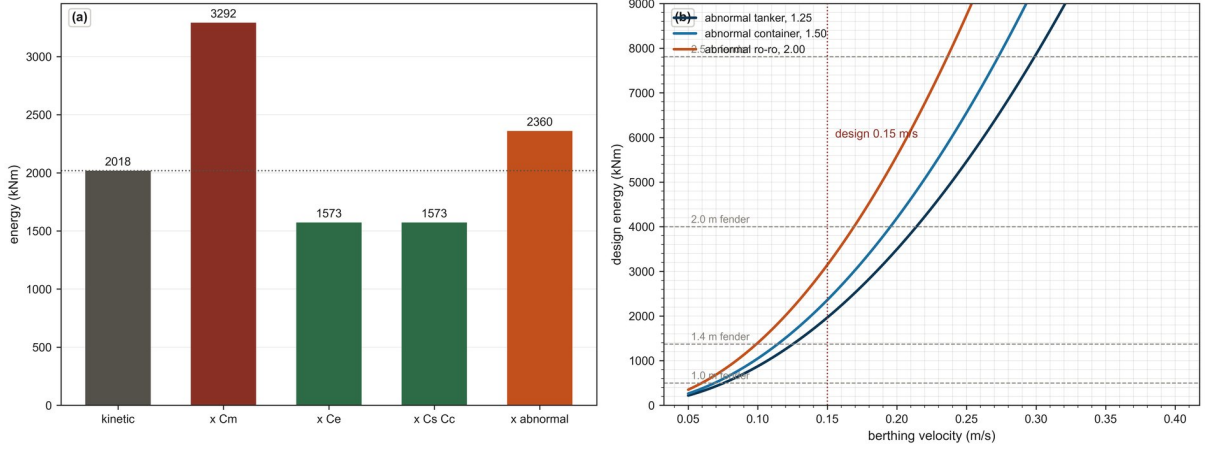


Figure 16: Berthing energy. (a) The PIANC energy chain, from kinetic energy to design energy. (b) Design energy against berthing velocity.

A beach fill is designed in two directions. Alongshore, the fill is a perturbation to a straight shoreline; it spreads and the placed volume leaves the project area over time. Cross-shore, the equilibrium profile is set by the grain size of the fill, which determines the dry beach width obtained from a given placed volume.

14.1 The planform: how long a fill lasts

Under small wave angles the one-line model (Equation 4) reduces to a diffusion equation. With the CERC transport in the solid-volume SI form this model needs (Equation 40),

$$Q = K_{\text{cerc}} H_b^{5/2} \sin 2\alpha, \quad K_{\text{cerc}} = \frac{K \sqrt{g/\gamma_b}}{16(s-1)}, \quad (40)$$

(`cerc_coefficient(K=0.39, s=2.65, gamma_b=0.78)`; the solver divides by $1-p$ itself, so a bulk-volume coefficient would double-count porosity), linearization gives Equation 41,

$$\frac{\partial y}{\partial t} = \varepsilon \frac{\partial^2 y}{\partial x^2}, \quad \varepsilon = \frac{2K_{\text{cerc}} H_b^{5/2}}{(1-p)(D+B)}, \quad (41)$$

(`longshore_diffusivity`). For an initially rectangular fill of width W and length $L = 2a$, Pelnard-Considere (1956) gives Equation 42,

$$y(x, t) = \frac{W}{2} \left[\operatorname{erf} \left(\frac{a-x'}{2\sqrt{\varepsilon t}} \right) + \operatorname{erf} \left(\frac{a+x'}{2\sqrt{\varepsilon t}} \right) \right], \quad (42)$$

with x' measured from the fill center (`pelnard_considere`). The center width $W \operatorname{erf}(a/2\sqrt{\varepsilon t})$ halves when the argument reaches $\operatorname{erf}^{-1}(1/2) = 0.4769$. This defines the spreading half-life returned by `sections.spreading_half_life`, which sets the time scale of the planform evolution.

`NourishmentDesign(length=1000, berm_width=30, taper=100, center=None, D=8, B=2, porosity=0.4)` is the fill: full-width length, shoreline advance at placement, linear tapers at each end, closure depth D , and berm height B . `WaveClimate(Hb=1.0, T=8.0, alpha0=0.0, Kcerc=KCERC_DEFAULT)` drives it; a zero angle means the fill diffuses symmetrically without migrating. `simulate_nourishment(design, climate, duration, domain_length=None,`

`dx=10.0, n_outputs=25, bc="fixed_ends", cfl=0.9)` evolves the tapered planform with `tools.shoreline` on a domain five times the fill footprint and returns a `NourishmentResult` with `planforms`, `volume_retained`, `retained_fraction`, `berm_width`, and `design_life(threshold=0.5)`. `renourishment_schedule(design, climate, horizon, threshold=0.5)` returns the interval, the number of renourishments, the placement times, and the total volume over a planning horizon. The analytical solution verifies the solver in the test suite.

Output of `examples/nourishment_design.py`

```
Placed volume      : 960,000 m3
Active profile height: 10.0 m
Diffusivity       : 0.0175 m2/s (551,100 m2/yr)
```

year	retained %	berm width (m)
0.0	100.0	30.0
4.0	54.0	16.6
8.0	41.5	12.3
12.0	34.9	10.2
16.0	30.7	8.9
20.0	27.8	8.0

```
Design life (50% retained): 5.0 yr
Renourishment interval    : 4.9 yr
Renourishments in 30 yr  : 6
Total sand over 30 yr     : 6,720,000 m3
```

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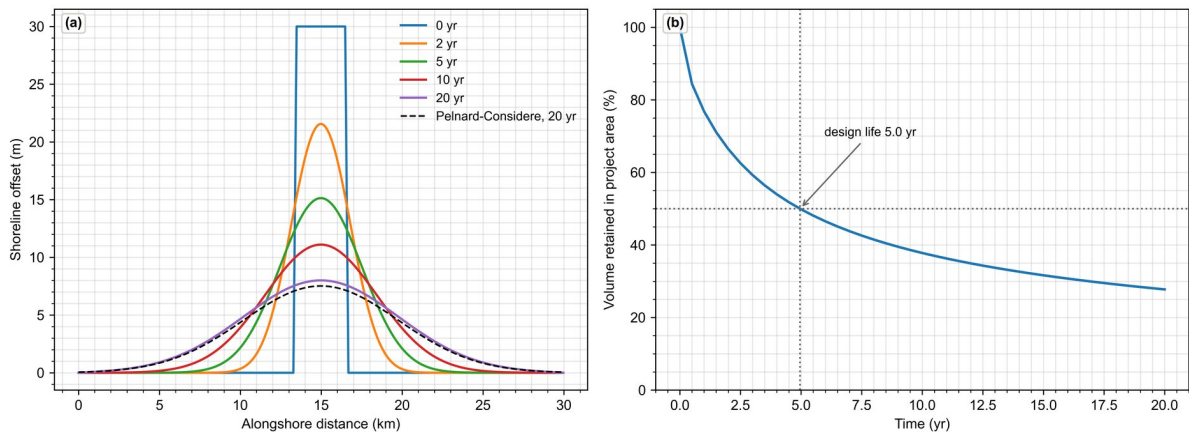


Figure 17: Nourishment planform. (a) Shoreline offset alongshore at the output times, against the Pelnard-Considere solution. (b) Volume retained in the project area against time, with the renourishment threshold.

14.2 The profile: borrow material and shoreline advance

Dean's equilibrium profile is $h = Ay^{2/3}$ (`equilibrium_profile`), with the scale parameter tied to the fall velocity (Kriebel, Kraus and Larson 1991) (Equation 43),

$$A = 0.067 w^{0.44} \quad (w \text{ in cm/s}), \quad (43)$$

so a sediment cannot have one settling velocity for scour and another for its profile (`dean_scale`; fine sand at 0.19 mm gives 0.094 and medium sand at 0.38 mm gives 0.141, against Dean's tabulated 0.10 and 0.14).

A fill with scale A_f placed on a native beach with scale A_n pushes the profile seaward at each depth by $\Delta(h) = a + (h/A_f)^{3/2} - (h/A_n)^{3/2}$, where a is the shoreline advance. The active profile is bounded by the closure contour, so the integration runs over *depth* from the waterline to closure and gives Equation 44,

$$V = Ba + ah_L + 0.4h_L^{5/2} (A_f^{-3/2} - A_n^{-3/2}), \quad (44)$$

where h_L is the closure depth, or the depth at which the profiles intersect for a fill coarser than the native sand (`fill_volume_for_advance`). For matched sand the expression reduces to $V = (B + h_*)a$. Three cases follow. A coarser fill **intersects** the native profile and yields a larger advance per unit volume. A finer fill is **non-intersecting**. Below a **critical volume** (`critical_volume`) a finer fill yields no dry beach: the placed volume is contained in the flattening of the submerged profile. `shoreline_advance(A_native, A_fill, volume, berm_height, closure_depth)` solves for the advance by bisection over the volume, which is monotonic in the advance, and returns the case. `profile_overfill_factor(native, borrow, ...)` returns the ratio of borrow to native volume for the same advance. This quantity is distinct from James's (1975) textural overfill ratio. `grain_compatibility(native, borrow)` compares the phi means and sortings ($\delta = (\phi_b - \phi_n)/\sigma_n$) and gives a verdict; `phi_size` and `size_from_phi` convert grain sizes.

Output of examples/nourishment_profile.py

Native beach: Medium sand, d50 = 0.38 mm, A = 0.141
Berm 2.0 m, closure 6.0 m, fill 250 m3 per metre

borrow	A	delta	profile	dry beach	critical
Very fine sand	0.051	+3.78	submerged	0.0 m	2362 m3/m
Fine sand	0.094	+1.82	submerged	0.0 m	566 m3/m
Medium sand	0.141	+0.00	matched	31.2 m	0 m3/m
Coarse sand	0.183	-1.78	intersecting	54.5 m	0 m3/m
Fine gravel	0.307	-7.24	intersecting	66.2 m	0 m3/m

What that costs, per metre of beach, for 40 m of dry sand

Very fine sand	2682 m3/m	(8.38 times the native requirement)
Fine sand	886 m3/m	(2.77 times the native requirement)
Medium sand	320 m3/m	(1.00 times the native requirement)
Coarse sand	164 m3/m	(0.51 times the native requirement)
Fine gravel	131 m3/m	(0.41 times the native requirement)

Compatibility

Very fine sand	much finer than native, poorly suited
Fine sand	much finer than native, poorly suited
Medium sand	close to native, suitable
Coarse sand	coarser than native, well suited
Fine gravel	coarser than native, well suited

Planform, 1500 m of fill in Hb = 1.2 m:

spreading half-life	0.57 yr
placed volume	719 thousand m3 in place

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Wrote media/nourishment_borrow.png

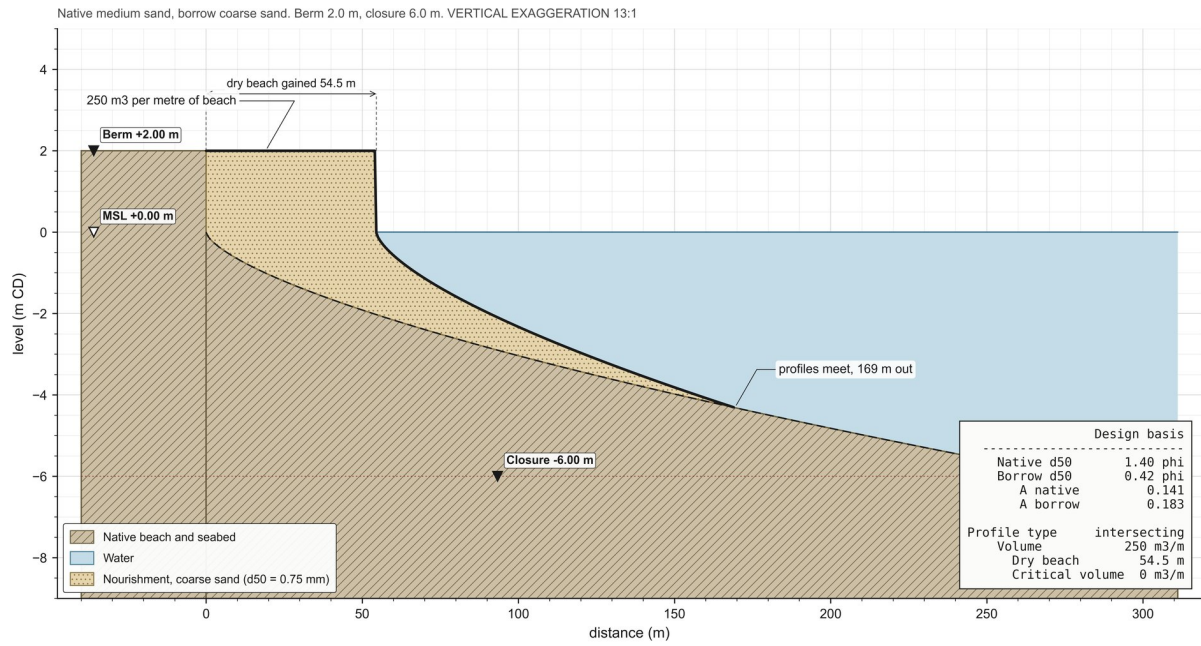


Figure 18: Profiles for borrow sources from very fine sand to fine gravel on a medium sand beach.

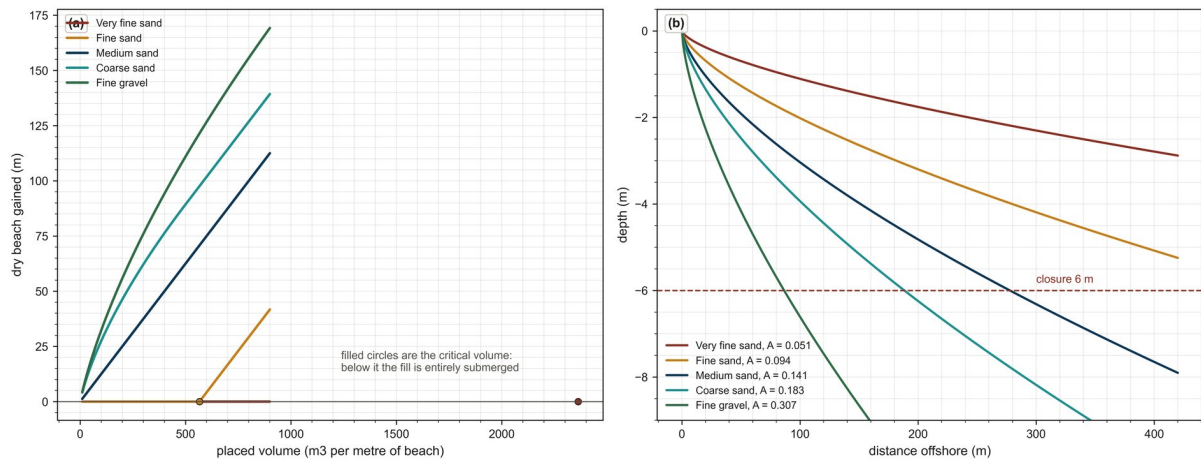


Figure 19: Borrow material. (a) Dry beach gained against placed volume, by borrow source. (b) Equilibrium profiles of the native beach and of each borrow source.

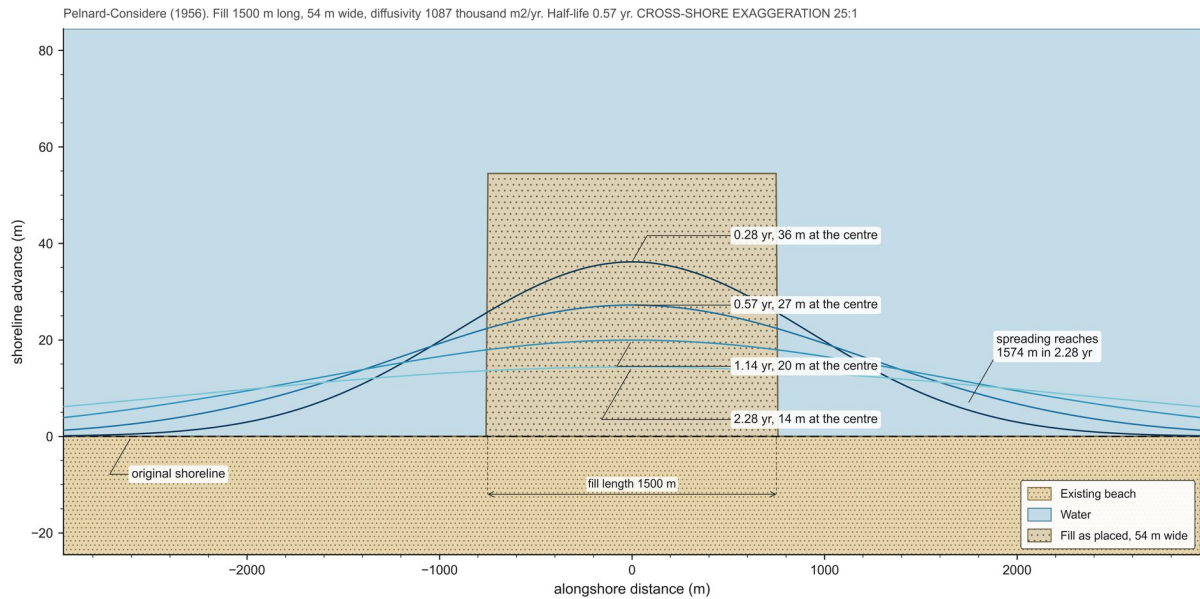


Figure 20: Planform spreading alongshore from the analytical solution, at times taken from the fill's own half-life. The cross-shore axis is stretched to fill the sheet and the factor is stated on the drawing.

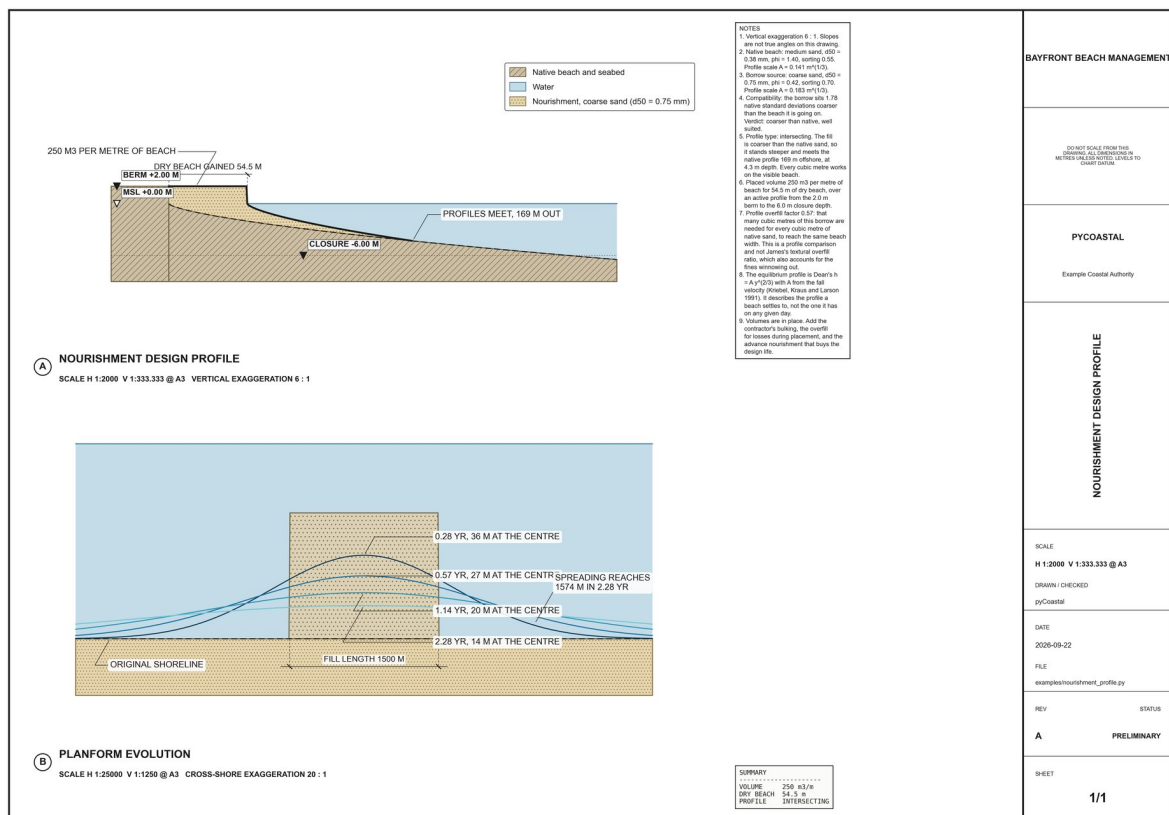


Figure 21: Nourishment drawing sheet with the design profile and the planform on one sheet, each at its own standard scale.

14.3 Limits

The analytical planform is for a rectangular fill with constant diffusivity; the numerical solver handles tapers but still uses one wave climate. The profile method assumes both profiles reach equilibrium and ignores winnowing of fines, which the compatibility check addresses separately.

15 Storm surge and coastal flooding

Module: `pyCoastal.applications.surge`. *Example:* `examples/storm_surge_flooding.py`.

The module builds a total water level from its components and then finds how far a storm floods: along a cross-shore profile, or over a terrain grid with hydraulic connectivity, so that low ground behind a continuous barrier is not flooded by a level that cannot physically reach it.

15.1 The water level budget

The still water level is the sum of the astronomical tide (supplied, not predicted), any sea-level rise, and three setups:

- **Barometric setup**, the inverted-barometer response, $\eta_p = (p_a - p_c)/(\rho g)$, close to 1 cm per hPa (`barometric_setup`). It is the equilibrium response and an upper bound for a fast-moving storm.
- **Wind setup**, from integrating $d\eta/dx = nkW^2 \cos \theta / (gd)$ over the fetch, with $k = 3.3 \times 10^{-6}$ and a return-flow factor n of 1.15 to 1.30. Over a constant depth (Equation 45),

$$\eta_w = \frac{nkW^2 F \cos \theta}{gd}, \quad (45)$$

(`wind_setup`). The setup varies as $1/d$ and is therefore controlled by the shallowest part of the shelf. `wind_setup_profile(depths, dx, ...)` integrates across a measured shelf profile and carries the accumulated setup into the local depth $d + \eta$, which reduces the further increment.

- **Wave setup** in the surf zone, from the breaking height (`surf_zone_setup(breaking_height, gamma=0.8)`).

Runup is added on top for the total water level, from Stockdon et al. (2006) (Equation 46):

$$R_{2\%} = 1.1 \left(0.35\beta \sqrt{H_0 L_0} + \frac{\sqrt{H_0 L_0 (0.563\beta^2 + 0.004)}}{2} \right), \quad (46)$$

with the setup and swash components returned separately (`stockdon_runup`). Runup is a swash oscillation with a period of the order of the wave period. Flood extent is therefore computed from the still water level, and the shoreline wave hazard from the total water level.

`StormConditions(wind_speed=40, central_pressure=95000, tide=0, Hm0=6, Tp=12, fetch=100e3, shelf_depth=20, wind_angle=0, sea_level_rise=0)` is the storm; `total_water_level(storm, beach_slope=0.05, breaker_index=0.78, shelf_depths=None, dx=None, return_flow=1.15)` returns every component, the still water level, and the total water level.

15.2 Flood mapping

A plain elevation threshold floods every low cell, including basins sealed off from the sea by higher ground. `bathtub_flood(terrain, level, seed=None, connectivity=4)` walks outward from the coast instead, so only ground the water can reach is flooded; `seed` marks where the sea enters (default: any below-level cell on the domain edge) and `connectivity` is 4 or 8. `isolated_low_ground` returns the cells below the level that the sea cannot reach, `flooded_area(flooded, dx)` and `flood_volume(terrain, flooded, level, dx)` quantify the map, and `flood_depths(z, level)` gives depths regardless of connectivity. On a profile, `inundation_limit(x, z, level)` walks landward and stops at the first ground above the level.

```
import numpy as np
from pyCoastal.applications.surge import (
    StormConditions, total_water_level, bathtub_flood, isolated_low_ground,
)

storm = StormConditions(wind_speed=45, central_pressure=94000, tide=0.6,
                        Hm0=7.0, Tp=13.0, fetch=120e3)
shelf = np.linspace(60.0, 8.0, 200)
levels = total_water_level(storm, shelf_depths=shelf, dx=120e3 / 199)

sea = np.zeros_like(terrain, dtype=bool); sea[0, :] = True
flooded = bathtub_flood(terrain, levels["still_water_level"], seed=sea)
stranded = isolated_low_ground(terrain, levels["still_water_level"], seed=sea)
```

15.3 Worked example

Output of examples/storm_surge_flooding.py

```
Water level budget (m above datum)
      tide      0.60
    sea level rise  0.00
  barometric setup  0.73
      wind setup   3.28
      wave setup   1.75
still water level  6.36
      runup R2%    2.32
total water level  8.68

Still water level 6.36 m
threshold-only flooded area : 19.17 km2
connected flooded area      : 18.82 km2
overstated by the threshold : 0.35 km2 (2% of it)
water volume on land        : 113.81 Mm3

with the inlet open   : 18.82 km2 flooded
with the inlet sealed : 5.45 km2 flooded
a single breach in the barrier is what floods the bay

inundation limit, intact barrier: 1418 m
inundation limit, at the inlet  : 4990 m
```

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Connectivity matters: on the synthetic barrier island, a single breach in the barrier is the difference between 5.45 km² and 18.82 km² flooded.

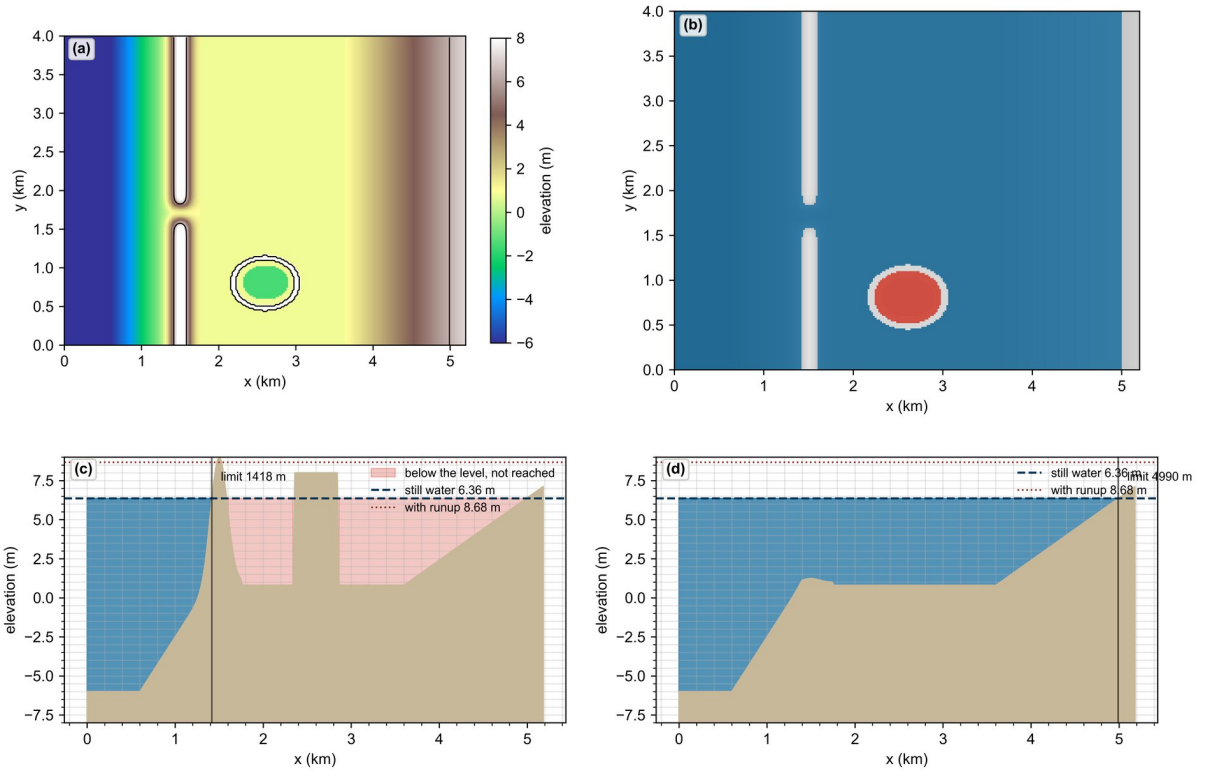


Figure 22: Storm surge flooding on a barrier-island terrain. (a) Terrain with the still water level contour. (b) Connected flooding and the low ground the sea cannot reach. (c) and (d) Cross-shore transects through the intact barrier and through the inlet.

Part IV

Drawings and deliverables

16 Drawings, sections, and sheets

Modules: `pyCoastal.drafting`, `pyCoastal.applications.sections`, `pyCoastal.plotting`. Needs the `plots` extra.

A coastal cross-section is produced as a scaled drawing: geometry in meters, line weights carrying meaning, hatched materials, and the governing dimensions annotated on the sheet. `pyCoastal.drafting` provides the drafting primitives on top of `matplotlib`. `applications.sections` converts each design object into a drawing and a quantity take-off.

16.1 Drafting primitives

`Section(title="", subtitle="", figsize=(13, 7), ax=None, caps=False, dim_style="arrow", frame=True, exaggeration=1.0)` is a drawing in real coordinates (x cross-shore, z elevation), locked to equal aspect so a 1:2 slope looks like a 1:2 slope. Text is sized in points and placed with offsets in points, so annotation stays readable when the section is rescaled; only the geometry scales. Its methods are the vocabulary of a drawing office:

- `material(points, kind, label=None)` fills a closed polygon with a material from the library (Table 9);
- `line(points, weight="medium", style="-")` draws construction, ground, or hidden lines with the line weights in `WEIGHTS` (thin 0.5, medium 1.0, heavy 1.8, extra 2.4);
- `water(x0, x1, level, bed=None, label=None)` fills water between a level and the bed, with the level symbol;
- `dim_h(x0, x1, z, text=None, extend_from=None)` and `dim_v(z0, z1, x, text=None, side="right")` draw dimension lines with extension lines and terminators;
- `level(x, z, text=None, side="left", symbol="level")` draws the levelling triangle and elevation callout;
- `slope(apex, cot_alpha, rise, direction="left")` draws the slope triangle, labelled $1 : \cot \alpha$;
- `note(xy, text, offset=(40, 30))` places a leader line and note;
- `detail_bubble(label, title, scale="")` places the circled detail marker;
- `notes_block(lines, title="NOTES", numbered=True)` places numbered specification notes;
- `table(rows, title="Design parameters")` places the parameter block, which records the design inputs on the sheet;
- `scale_bar(length, divisions=4)` and `key(loc="upper left")` add a checkered scale bar and a material key;
- `fit_scale(xlim, zlim, scales=None, paper="")` sets the view to a true, round drawing scale that fits the extents, and `auto_exaggeration(xlim, zlim, cap=200)` scales the second axis to fill the viewport, for cases where a true scale is not legible;
- `finish(xlim=None, zlim=None, grid=True)`, `save(path, dpi=600)`, and `to_dxf(path, scale=1.0)`.

Table 9: The material library in `pyCoastal.drafting`.

Key	Drawn as
<code>armour, secondary,</code> <code>underlayer, toe</code> <code>core</code>	individual stones at the layer's stone size
<code>concrete, reinforced</code>	quarry run, dotted hatch
<code>blinding</code>	mass and reinforced concrete, diagonal and cross hatch, heavy edge
<code>rock_fill, granular, sand</code>	blinding layer
<code>subgrade</code>	fills, circle and dot hatches
<code>pavement</code>	in-situ seabed, dense diagonal hatch
<code>water</code>	promenade surfacing
	water, thin edge

`Material(name, face, edge, hatch=None, weight="medium", stones=0.0)` defines a new one, and `use_crisp_style()` sets global matplotlib settings for legible technical figures.

`Sheet(titleblock, size="A3", dpi=400, margin=8.0, strip=52.0)` is a drawing sheet with a border and a title block strip down the right edge, in any of PAPER (A0 to A4). `viewport(rect)` adds a Section view within the usable area, `set_scale_from(view)` copies a view's fitted scale into the title block, `save(path)` writes the sheet with no trimming (the border is the edge), and `to_dxf(path)` exports every view to one DXF. `TitleBlock` holds `project`, `title`, `organisation`, `client`, `scale`, `date`, `file`, `sheet`, `revision`, `drawn_by`, `checked_by`, `status`, and a standard disclaimer.

`DXF.write_dxf(path, entities, scale=1.0)` writes polylines on named layers to a minimal DXF R12 file that any CAD package opens.

16.2 Sections for each design

Every structure in `applications.sections` has three entry points. All three take a design object as their input:

- `draw_*` puts the geometry and annotation into an open `Section` and returns the extents it needs;
- `*_section` is a standalone figure with plot axes, for a report or slide;
- `*_sheet` is a full drawing sheet with border, title block, and notes, at a true stated scale; extra keyword arguments go to the `TitleBlock`.

Table 10: Drawing functions in `pyCoastal.applications.sections`.

Structure	draw	section	sheet	notes
Seawall	<code>draw_</code> <code>seawall,</code> <code>draw_</code> <code>seawall_</code> <code>toe_detail</code>	<code>seawall_section</code>	<code>seawall_</code> <code>sheet</code>	<code>seawall_</code> <code>notes</code>
Rubble mound	<code>draw_</code> <code>rubble_</code> <code>mound</code>	<code>rubble_mound_</code> <code>section</code>	<code>rubble_</code> <code>mound_sheet</code> (trunk and head)	<code>rubble_</code> <code>mound_notes</code>

Structure	draw	section	sheet	notes
Channel	draw_ channel, draw_ channel_ detail	channel_section	channel_ sheet	channel_ notes
Nourishment profile	draw_ nourishment	nourishment_ section	nourishment_ sheet	nourishment_ notes
Nourishment plan	draw_ nourishment_ plan	nourishment_ plan_section	via nourishment_ sheet(plan_ design=...)	
Pier scour	draw_pier_ scour	pier_scour_ section	pier_scour_ sheet	pier_scour_ notes
Bridge scour	draw_ bridge_ scour	bridge_scour_ section	bridge_ scour_sheet	bridge_ scour_notes

Supporting geometry: `MoundProfile(x_sea, x_land, crest, cot_sea, cot_land)` describes a trapezoidal mound, with `offset(thickness)` giving the exact perpendicular offset of a layer (the crest edges move by $t(\sqrt{1 + \cot^2 \alpha} - \cot \alpha)$), `band`, and `area`; `mound_layer_volumes(profiles, bed)` takes off the layer quantities. The underlayer follows the Rock Manual rule of a tenth of the armor mass, so its D_{n50} falls by $10^{1/3} \approx 2.15$. `vessel_outline` draws a midship section, `scour_hole_profile` a hole whose faces stand at the angle of repose (`REPOSE_ANGLE = 32` degrees), `bridge_bed_profile` the scoured bed across the whole waterway as one line, and `plan_margin` and `spreading_half_life` set the extents and times of a planform.

16.2.1 Scale and exaggeration

Sections are drawn at true scale by default and fitted to a round standard scale, which the title block records. Where a true scale is not legible, as for a channel 600 m wide and 20 m deep or a beach profile several hundred meters long and a few meters deep, a vertical exaggeration is applied and its factor is printed next to the scale. The corresponding sheets carry a second view at true scale: the channel keel with the underkeel allowances, and the seawall toe.

16.3 The drawing sheets

The worked examples of Part III issue these sheets; they are collected here as the deliverables of the design chain.

The examples also write DXF files of the geometry (`media/seawall_section.dxf`, `media/breakwater_section.dxf`, `media/navigation_channel.dxf`), with each material on its own layer.

16.4 Plot helpers

`pyCoastal.plotting` holds the color maps and helpers shared by the examples: `water_colormap()`, a diverging map for the instantaneous surface built to look like water; `agitation_colormap()`, a sequential map for wave height or disturbance coefficient; `surface_norm(snapshots, percentile=99.5)`, a symmetric color scale centered on still water; `land_overlay()`

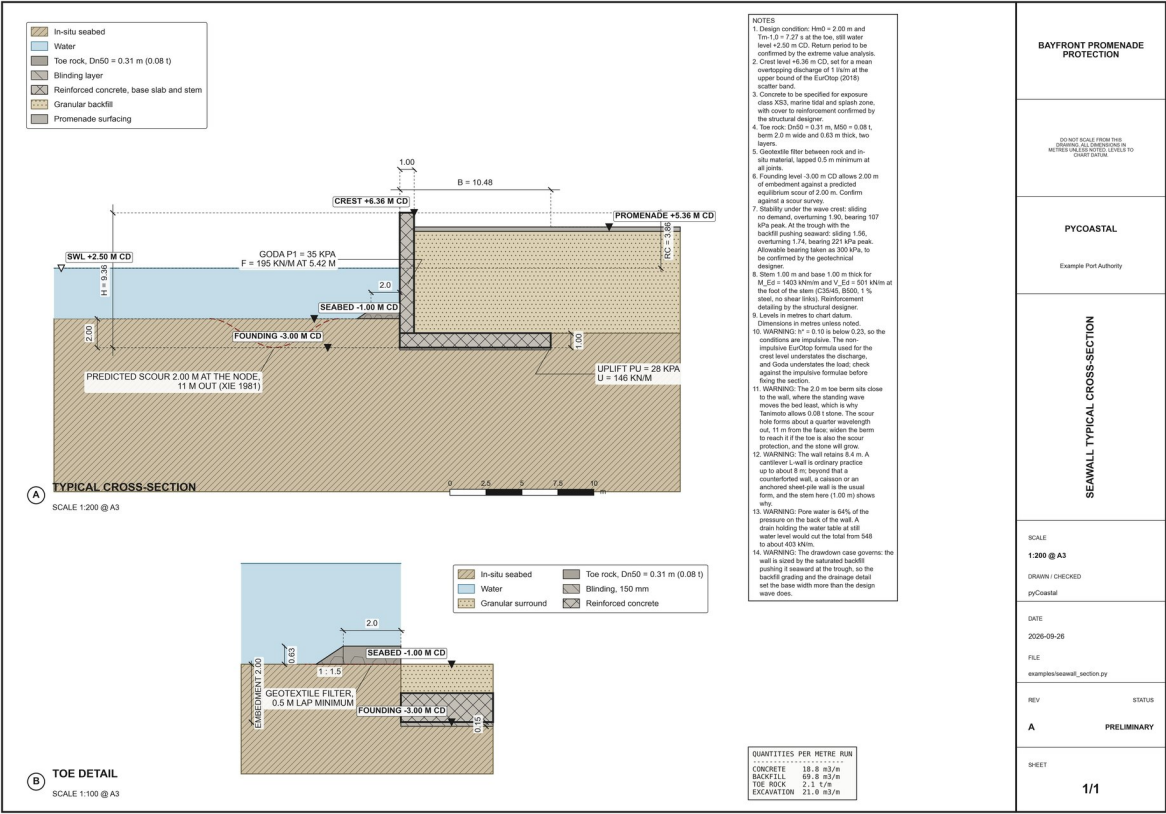


Figure 23: Seawall typical cross-section with toe detail (seawall_sheet).

land), a float array that is 1 on structures and NaN on water; and LAND_COLOR. For quantitative fields use agitation_colormap or a standard scientific map, and keep water_colormap for reading the wave pattern at a glance.

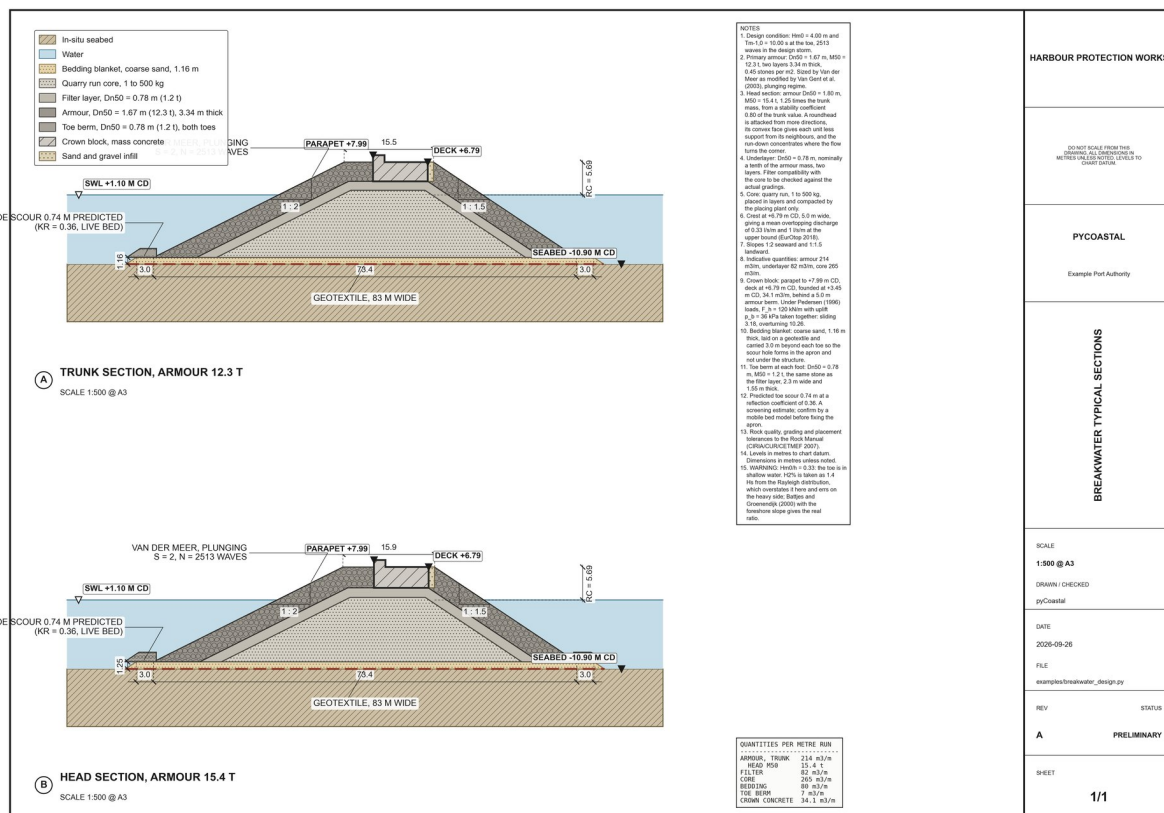


Figure 24: Breakwater trunk and head sections (rubble_mound_sheet).

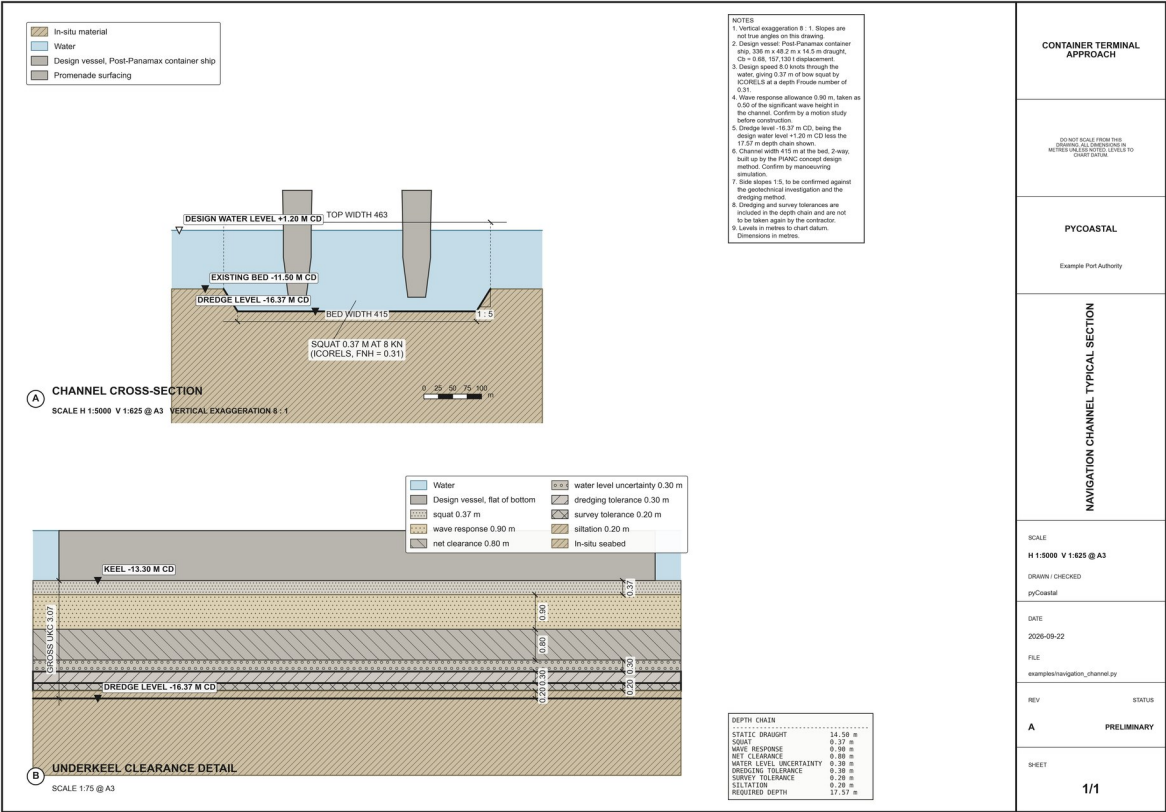


Figure 25: Navigation channel with true-scale keel detail (channel_sheet).

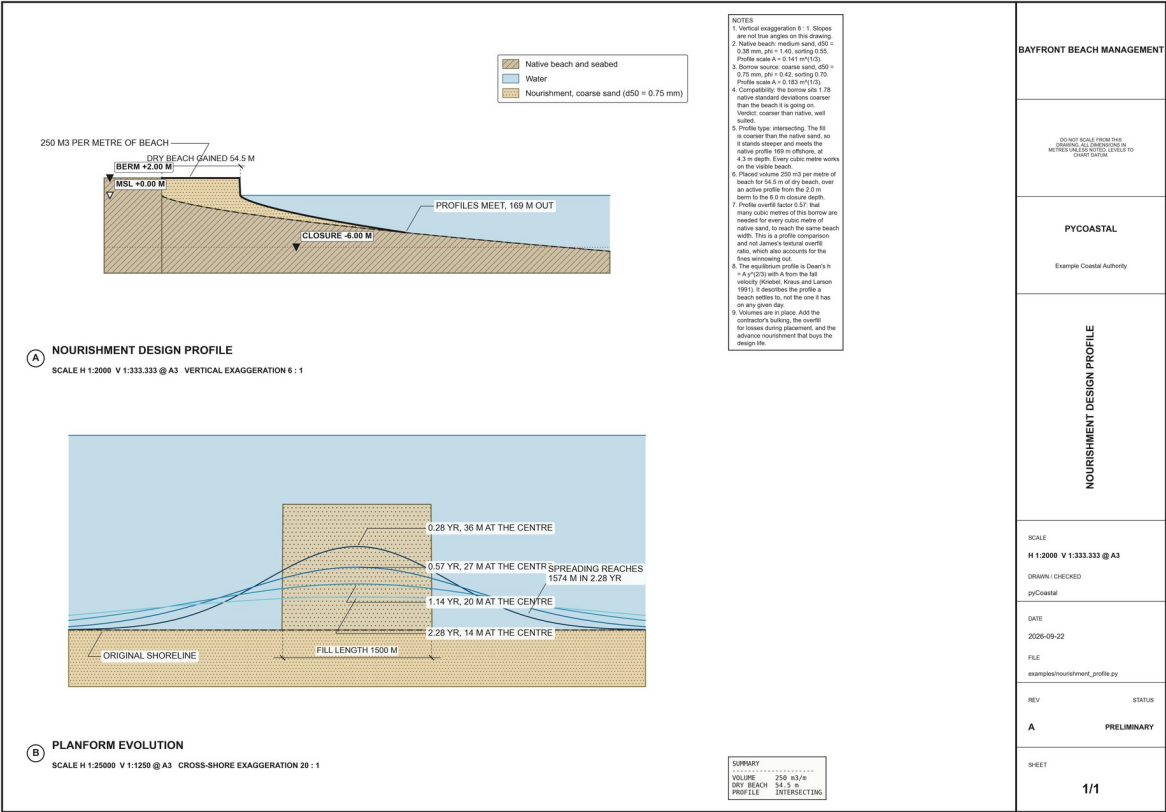
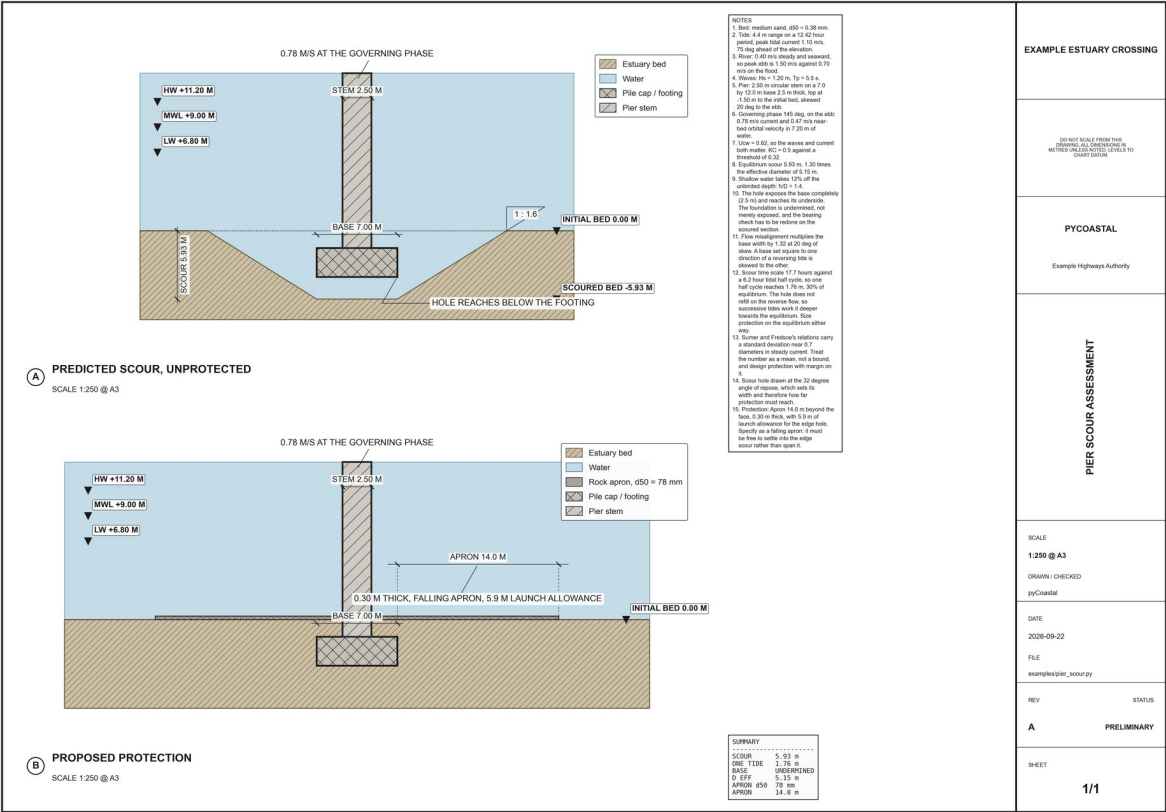


Figure 26: Nourishment design profile and planform (nourishment_sheet).



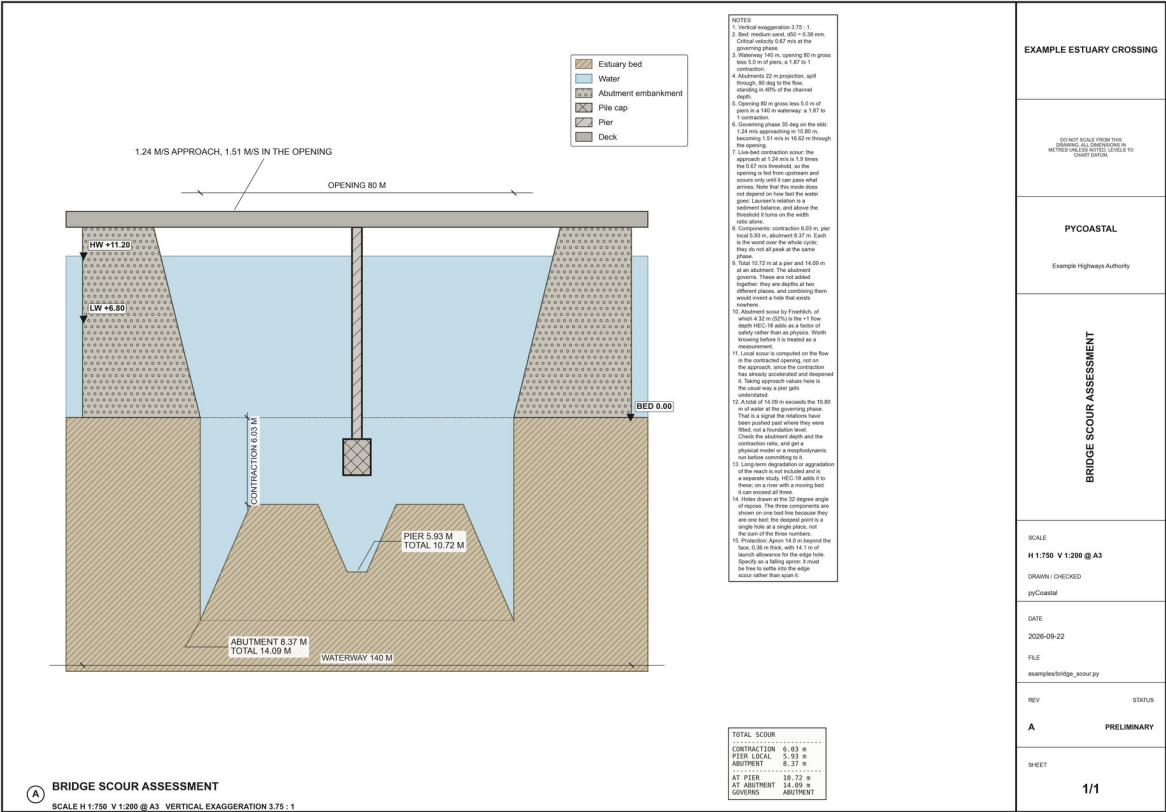


Figure 28: Bridge scour, all three components on one bed line (bridge_scor_sheet).

Part V

PyCoaTools

17 The browser app

Directory: `webapp/`.

PyCoaTools is a static web page that runs the pyCoastal design modules in the browser. It renders the result as a drawing sheet, writes the design report, and displays the supporting literature for the relations used, taken from PyCoaPedia.

17.1 How it works

A browser has no Python, so `webapp/engine.js` carries a JavaScript transliteration of the design modules, function for function: where the Python bisects, the JavaScript bisects; where the Python iterates on the base width, so does the JavaScript. Units follow the Python. The two implementations are verified against each other:

- `make_vectors.py` dumps reference cases straight out of pyCoastal into `vectors.json`, including the branch cases: the surging armor branch, an impulsive wall, a depth-limited Goda wave, drag- and inertia-dominated piles, and a resultant outside the middle third;
- `verify_engine.py` runs `engine.js` in QuickJS against those cases and fails on any disagreement beyond floating-point noise;
- the page re-runs the same comparison on load and shows the result as a chip in the masthead, so the user sees it too.

`app.js` defines the modules, their inputs, the run functions, the report, and the checks; `draw.js` draws each module's sheet as SVG; and `index.html` is the application frame: a fixed masthead, an input panel, the drawing, the design report, and the Theory and sources panel, each scrolling on its own. Above 1560 px the report sits beside the drawing in its own pane. The page uses the fonts the desktop already has (Segoe UI and Consolas) and fetches no web font, so the first frame is correct.

17.2 Modules and inputs

The app has six modules. Each input is a slider with a range, a select, or a toggle, and the defaults follow the worked examples of Part III.

Seawall

Gravity L-wall: crest from EurOtop, base from Goda and the backfill, stem as an RC cantilever.
(module key `seawall`)

Input	Label	Unit	Control	Range or options	Default
Hm0	Wave height Hm0	m	slider	0.5 to 8, step 0.1	2.0
Tp	Peak period Tp	s	slider	4 to 18, step 0.1	8.0
swl	Still water level	m CD	slider	-1 to 6, step 0.1	2.5

Input	Label	Unit	Control	Range or options	Default
bed	Seabed level	m CD	slider	-20 to 2, step 0.1	-1.0
use	Tolerable overtopping	-	select	pedestrians_unaware, pedestrians_aware, harbour_quay_equipment, trained_staff, vehicles_low_speed	trained_staff
beta	Wave angle	deg	slider	0 to 60, step 5	0
scour	Scour coefficient	-	slider	0 to 0.6, step 0.05	0.4
backfill	Backfill material	-	select	silt, very_fine_sand, fine_sand, medium_sand, coarse_sand, fine_gravel, coarse_gravel, rock_fill, soft_clay, stiff_clay	medium_sand
table	Water table below surface	m	slider	0 to 16, step 0.5	0
surcharge	Surcharge on the promenade	kPa	slider	0 to 60, step 5	10
bearing	Allowable bearing	kPa	slider	100 to 800, step 25	300

Breakwater

Rock by Van der Meer (Van Gent 2003 form), concrete units by their own formulae, crest by EurOtop, crown wall by Pedersen. (module key **breakwater**)

Input	Label	Unit	Control	Range or options	Default
Hm0	Wave height Hm0	m	slider	0.5 to 9, step 0.1	4.0
Tp	Peak period Tp	s	slider	4 to 18, step 0.1	11.0
depth	Depth at toe	m	slider	2 to 40, step 0.5	12.0
cot	Seaward slope 1 :	-	slider	1.5 to 4, step 0.25	2.0
storm	Storm duration	h	slider	1 to 48, step 1	6
damage	Damage level S	-	slider	2 to 17, step 1	2

Input	Label	Unit	Control	Range or options	Default
use	Tolerable overtopping	-	select	pedestrians_unaware, pedestrians_aware, harbour_quay_equipement, trained_staff, vehicles_low_speed	trained_staff
armour	Armour type	-	select	rock_two_layer_permeable, rock_two_layer_impermeable, cubes_two_layer_random, tetrapod, accropode, xbloc	rock_two_layer_permeable
bed	Seabed material	-	select	silt, very_fine_sand, fine_sand, medium_sand, coarse_sand, fine_gravel, coarse_gravel, rock_fill, soft_clay, stiff_clay	medium_sand
section	Section	-	select	trunk, head	trunk

Channel

PIANC depth chain and width build-up. (module key `channel`)

Input	Label	Unit	Control	Range or options	Default
length	Vessel length	m	slider	80 to 400, step 5	336
beam	Beam	m	slider	12 to 62, step 0.5	48.2
draught	Draught	m	slider	3 to 20, step 0.1	14.5
cb	Block coefficient	-	slider	0.5 to 0.9, step 0.01	0.68
speed	Speed	kn	slider	3 to 14, step 0.5	8
wl	Design water level	m CD	slider	-1 to 4, step 0.1	1.2
Hs	Wave height in channel	m	slider	0 to 4, step 0.1	1.8
factor	Wave response factor	-	slider	0.2 to 0.8, step 0.05	0.5
net	Net clearance	m	slider	0.3 to 2, step 0.1	0.8
bed	Existing bed	m CD	slider	-25 to -5, step 0.5	-11.5
twoway	Two-way traffic	-	toggle	on / off	true

Input	Label	Unit	Control	Range or options	Default
material	Bed material	-	select	silt, very_fine_sand, fine_sand, medium_sand, coarse_sand, fine_gravel, coarse_gravel, rock_fill, soft_clay, stiff_clay	medium_sand

Monopile

Morison loads on a stream function wave, swept through the cycle; scour under waves and current. (module key `monopile`)

Input	Label	Unit	Control	Range or options	Default
diameter	Pile diameter	m	slider	0.5 to 12, step 0.1	8.0
H	Wave height H	m	slider	1 to 20, step 0.5	12
T	Period T	s	slider	4 to 20, step 0.5	13
depth	Water depth	m	slider	5 to 60, step 1	30
current	Tidal current	m/s	slider	0 to 2.5, step 0.1	1.0
rough	Marine growth	-	toggle	on / off	true
bed	Seabed material	-	select	silt, very_fine_sand, fine_sand, medium_sand, coarse_sand, fine_gravel, coarse_gravel, rock_fill, soft_clay, stiff_clay	medium_sand

Nourishment

What a borrow source is worth, on Dean's equilibrium profile. (module key `nourishment`)

Input	Label	Unit	Control	Range or options	Default
native	Native beach sand	-	select	silt, very_fine_sand, fine_sand, medium_sand, coarse_sand, fine_gravel, coarse_gravel	medium_sand
borrow	Borrow source	-	select	silt, very_fine_sand, fine_sand, medium_sand, coarse_sand, fine_gravel, coarse_gravel	coarse_sand
volume	Placed volume	m ³ /m	slider	20 to 1200, step 10	250
berm	Berm height	m	slider	0.5 to 5, step 0.1	2.0

Input	Label	Unit	Control	Range or options	Default
closure	Depth of closure	m	slider	2 to 14, step 0.5	6.0
view	View	-	select	profile, planform	profile
length	Fill length (plan-form)	m	slider	200 to 6000, step 50	1500
Hb	Breaking wave height (plan-form)	m	slider	0.3 to 3.0, step 0.1	1.2

Return values

Generalised Pareto tail above a storm threshold. (module key **extremes**)

Input	Label	Unit	Control	Range or options	Default
threshold	Threshold	m	slider	0.5 to 8, step 0.1	2.5
scale	Scale	m	slider	0.1 to 3, step 0.05	0.55
shape	Shape	-	slider	-0.4 to 0.4, step 0.01	-0.08
rate	Peaks per year	-	slider	0.5 to 30, step 0.5	6
record	Record length	yr	slider	5 to 100, step 1	40
period	Design return period	yr	slider	5 to 500, step 5	100

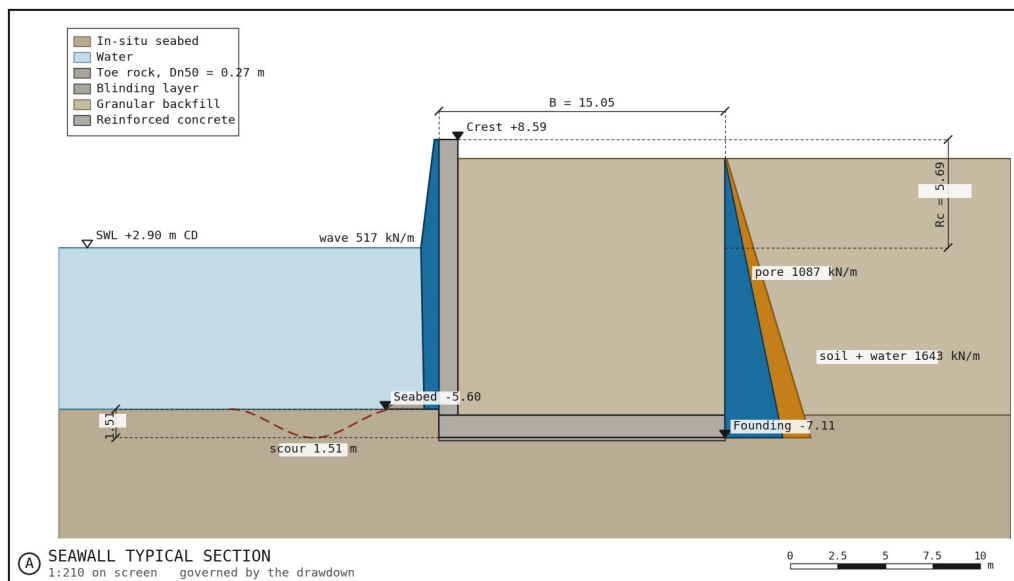


Figure 29: Seawall module sheet as rendered by **draw.js**.

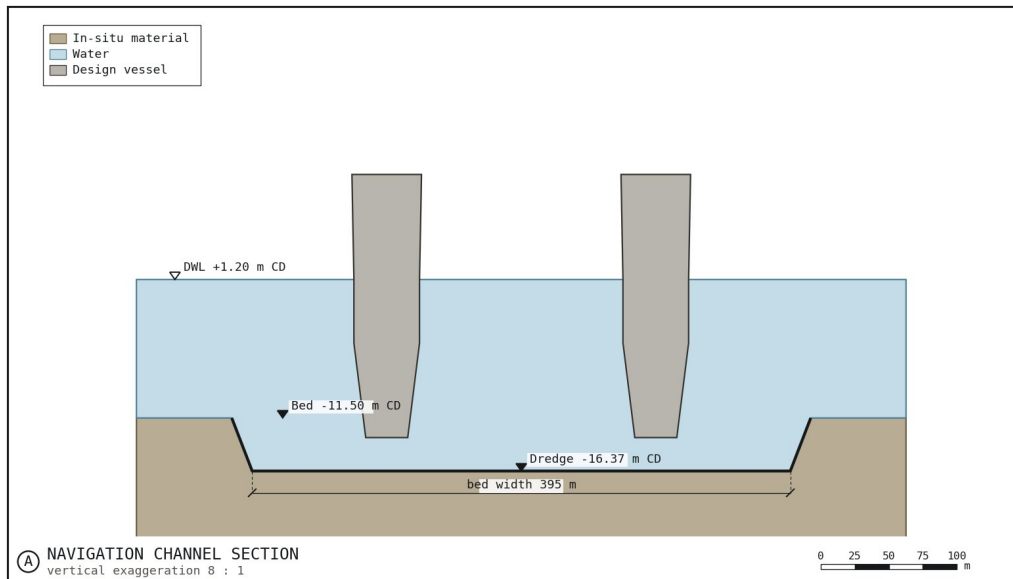


Figure 30: Channel module sheet.

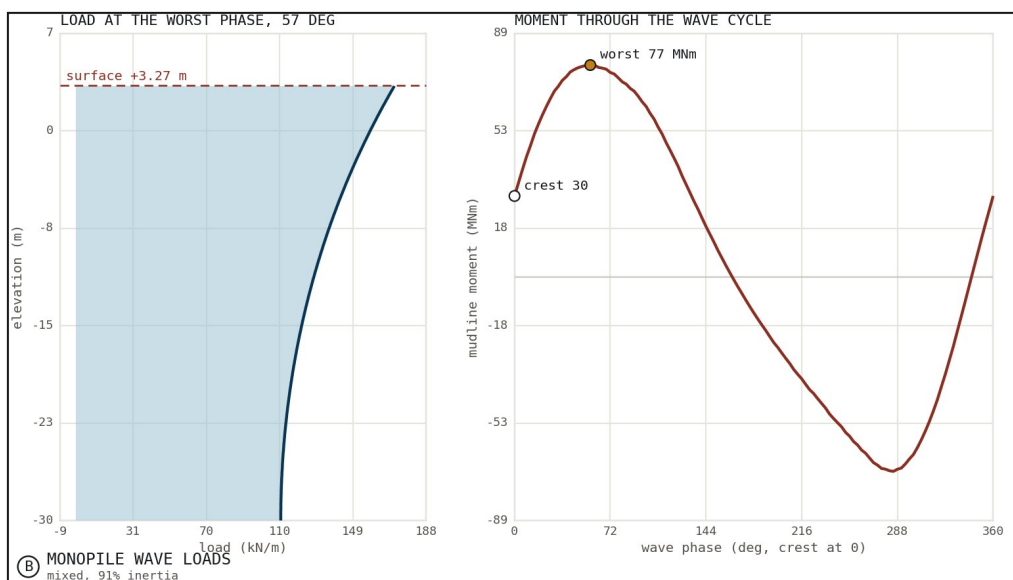


Figure 31: Monopile module sheet.

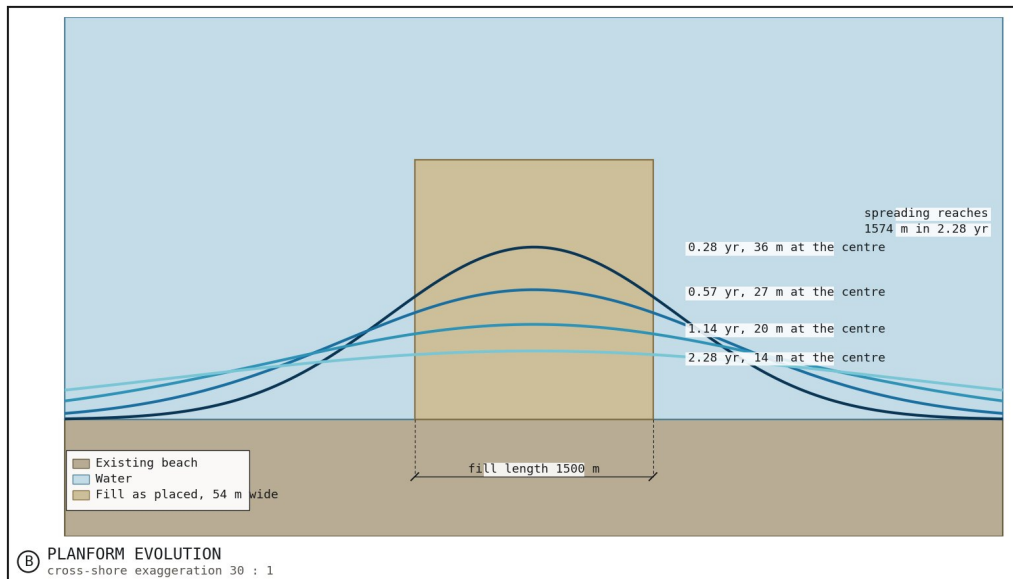


Figure 32: Nourishment module, planform view.

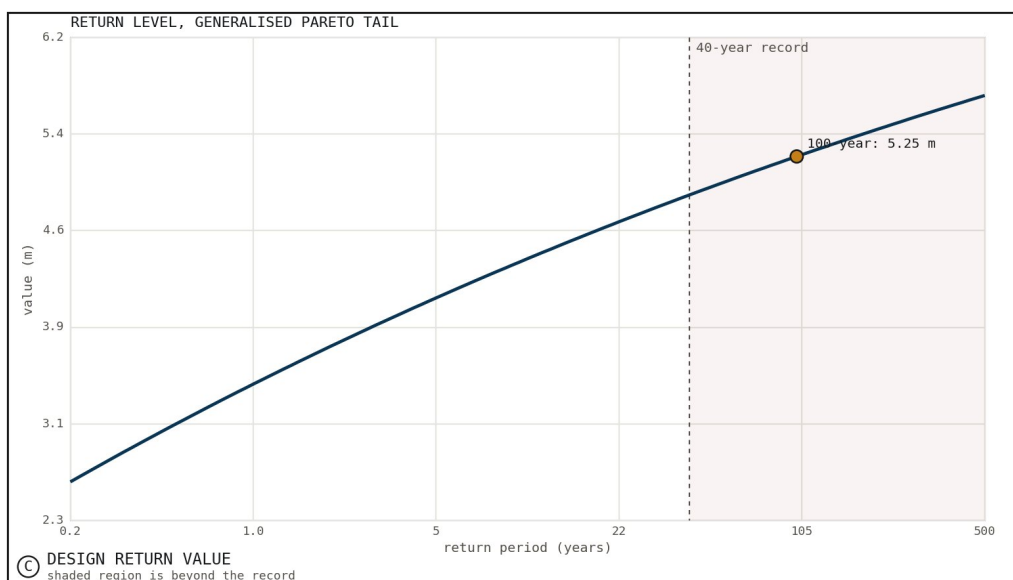


Figure 33: Return values module.

17.3 Theory and sources

The Theory and sources panel reads `knowledge.json`, the extract of PyCoaPedia that `pedia/build_pedia.py` writes. The source database is about 150 MB of SQLite, which no browser will open; the app needs the topic tree, the curated synthesis for each topic, the atomic claims with their regime bounds, the equations, and enough paper metadata to cite and link, which comes to under 2 MB. Each design module is mapped to the PyCoaPedia topics that bear on it. The mapping is inclusive: a design relation draws on several topics. The panel shows each topic's synthesis, claims and equations, each with its DOI. Coverage follows the screening: PyCoaPedia records the literature that has been screened. The PyCoaPedia button in the masthead opens the full explorer, described with the rest of the knowledge base in [Section 18](#).

17.4 Verification and review tools

- `verify_app.py` loads the page's own scripts into QuickJS behind a stub DOM, so the module definitions, run functions, report and check builders, and in-page verification all execute at least once before publishing. It catches the things that break a static page: a typo in a function name, an input key that no run function reads, a check that throws on the defaults.
- `render_preview.py` runs `draw.js` in QuickJS behind a DOM stub that serializes to SVG, then rasterizes the result. This verifies the drawing output. Execution alone does not: a sheet that draws its background after the geometry executes without error and renders blank. Output goes to `media/webapp/`.
- `rasterise.py` renders the app's SVG with matplotlib. The SVG is restricted to paths of M, L, H, V and Z plus rects, circles and text, which matplotlib reproduces. It verifies the presence, position and draw order of each element.

```
pip install quickjs svglib
python webapp/make_vectors.py --out webapp/vectors.json
python webapp/verify_engine.py
python webapp/verify_app.py
python webapp/render_preview.py --out media/webapp
```

To use the app locally, serve the folder (`python -m http.server` inside `webapp/`) and open `index.html`; the page loads `knowledge.json` and `vectors.json` with `fetch`, which requires an HTTP origin.

Part VI

PyCoaPedia and agentic retrieval

18 PyCoaPedia: the knowledge base

PyCoaPedia is the knowledge base distributed with pyCoastal. It holds the peer-reviewed coastal and ocean engineering literature as structured data: a tree of topics, a curated synthesis per topic, atomic claims with the regime in which each was established, equations with their variables and regime, and structured extractions of each paper. Every record is traceable to a DOI. The validity range of a relation is therefore available by query.

18.1 What is in it

The build shipped with this version of pyCoastal was generated on 2026-09-22. It holds 156 topics in 26 top-level branches, 1,779 claims linked by 1,359 claim-to-claim relationships, 163 equations, and 1,655 papers, 1,655 of them with a structured extraction, drawn from 4,926 papers screened.

The knowledge is organized around four kinds of record.

Topics form a tree, from 26 top-level branches (waves, structures, scour, sediment transport, storm surge, ports, numerical modeling, and so on) down to specific subjects such as `structures.overtopping` or `waves.transformation.breaking.runup`. The dotted identifier is the path.

Synthesis is written per topic under fixed headings: what is well established, the governing physics, the dimensionless parameters, the major equations, typical methods, numerical models, experimental datasets, validated ranges, recent advances, disagreements, limitations, open questions, and the seminal papers. It is the place to start on a topic.

Claims are atomic findings. Each carries the regime it holds in, a confidence class (a direct finding, a literature review statement, an inferred relationship, or a proposed hypothesis), the kind of evidence (field, experimental, numerical, analytical, mixed, or review), the location in the paper, and the paper itself. Claims are linked to each other where one supports, contradicts, extends, validates or compares against another. Disagreement in the literature is therefore queryable.

Papers carry their bibliographic record and abstract, the topics they are classified under, and a structured extraction: the research question, the principal results, the quantitative findings, the parameter ranges, the engineering implications, the limitations and the applicability conditions. Parameters, numerical models, datasets and field sites used by each paper are recorded as well.

18.2 How it is built

PyCoaPedia is distilled from a larger research database that an ingestion pipeline keeps growing: literature discovery through Crossref, OpenAlex and Semantic Scholar, relevance screening, classification into the topic tree, and extraction of claims, equations and paper summaries. That database also holds discovery logs, raw source records and full-text bookkeeping, and is not shipped. `pedia/build_pedia.py` takes a snapshot of it and writes three things from one pass, so they cannot disagree (Figure 34):

- `pyCoastal/pedia/pycoapedia.sqlite`, the distilled database with a full-text index, shipped inside the Python package;
- `pedia/topics/*.md`, one Markdown page per topic, with an index (`pedia/README.md`), a module map (`pedia/modules.md`) and the schema (`pedia/SCHEMA.md`);
- `webapp/knowledge.json`, the extract the PyCoaTools app and the PyCoaPedia explorer load.

```
python pedia/build_pedia.py --db ../2026_CoastalWiki/data/coastalwiki.db
```

Rerunning it after the pipeline has added papers refreshes all three, and the tests in `tests/test_pedia.py` check the result.

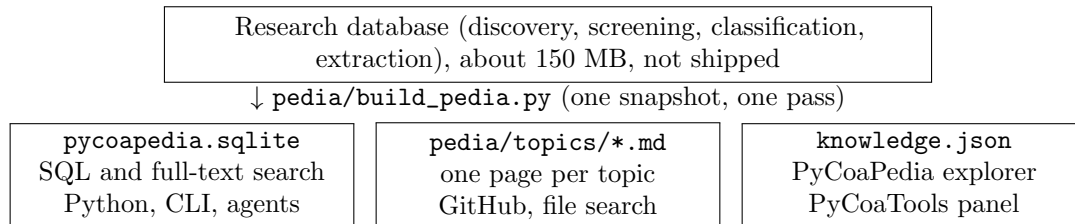


Figure 34: How PyCoaPedia is built and where each output is read.

18.3 Navigating the knowledge

The same knowledge can be reached five ways, and they suit different readers. The workflow from a design question to the supporting evidence is: select the pyCoastal module in use, read the synthesis of its topics, in particular the validated ranges and the disagreements, select the claims whose regime matches the case, and follow each claim to its paper and the quantitative findings recorded there.

18.3.1 In the browser: the PyCoaPedia explorer

`webapp/pedia.html` is a standalone page that loads the same extract as PyCoaTools. The topic tree is on the left with the number of claims on each branch; a search box searches topics, synthesis, claims and equations at once; and chips filter everything to the topics behind one pyCoastal design module. A topic page shows its breadcrumb and subtopics, the synthesis under its headings, the equations (typeset when KaTeX is reachable), and the claims, filterable by kind of evidence, each linking to its paper by DOI. Topic and claim views have their own URLs, so a finding can be shared as a link. From PyCoaTools, the PyCoaPedia button in the masthead and the Theory and sources panel beside each design lead to the same pages.

18.3.2 From Python

`pyCoastal.pedia` is a small read-only interface to the database, with no dependencies beyond the standard library:

```
from pyCoastal import pedia
```

```

pedia.stats()                                # build date and counts
pedia.tree("scour")                          # a branch of the topic tree
hits = pedia.search("overtopping shallow foreshore", limit=10)
t = pedia.topic("structures.overtopping")    # synthesis, claims, equations
t["synthesis"]["validated_ranges"]
pedia.claims("structures.overtopping", evidence="experimental")
pedia.related_claims(410)                   # supports, contradicts, ...
pedia.for_module("seawall")                 # topics behind a design module
pedia.paper(doi="10.1016/j.coastaleng.2016.07.002")

```

`search` runs on an FTS5 index with Porter stemming, so every word must appear but “overtopping” also finds “overtopped”; `kind=` restricts it to topics, synthesis, claims, equations or papers, `topic=` to one branch, and `raw=True` accepts the full FTS5 syntax (`OR`, `NEAR`, prefixes). For anything else, `pedia.connect()` returns a read-only connection to the database, and the environment variable `PYCOAPEDIA_DB` points the module at another build.

18.3.3 From the shell

```
python -m pyCoastal.pedia stats
python -m pyCoastal.pedia search "wave overtopping" --kind claim --limit 10
python -m pyCoastal.pedia tree structures
python -m pyCoastal.pedia topic structures.overtopping --claims
python -m pyCoastal.pedia module seawall
python -m pyCoastal.pedia paper --doi 10.1016/j.coastaleng.2016.07.002
python -m pyCoastal.pedia sql "SELECT id, label FROM topics WHERE level = 0"
```

Every command accepts `--json` for machine-readable output.

18.3.4 In SQL

The database is an ordinary SQLite file, so any client or language can query it. `pedia/SCHEMA.md` documents every table and its controlled vocabularies, with worked queries; two of them show the idea:

```
-- the validated ranges behind a pyCoastal design module
SELECT m.topic_id, s.statement
FROM module_topics m JOIN synthesis s ON s.topic_id = m.topic_id
WHERE m.module = 'seawall' AND s.section = 'validated_ranges';

-- where the literature disagrees
SELECT a.text, b.text, r.rationale
FROM claim_relationships r
JOIN claims a ON a.id = r.source_claim_id
JOIN claims b ON b.id = r.target_claim_id
WHERE r.relationship = 'contradicts';
```

18.3.5 As Markdown

`pedia/README.md` draws the whole topic tree with links, `pedia/modules.md` lists the topics behind each design module, and each page in `pedia/topics/` holds one topic: breadcrumb, subtopics, synthesis, equations in LaTeX, claims with their regime and citation, and the most relevant papers. The pages read on GitHub and suit tools that search files.

18.4 The design modules and their topics

Each design module of pyCoastal, and of PyCoaTools, is mapped to the topics whose literature applies to it (Table 17). The mapping is inclusive: a design relation draws on several topics. A crest-level calculation maps to both the overtopping topic and the runup topic.

Table 17: Design modules and their PyCoaPedia topics.

Module	Topics
Navigation channel (channel)	Navigation and vessel response (<code>ports.navigation</code>); Harbor sedimentation and dredging (<code>ports.sedimentation</code>); Mooring hydrodynamics (<code>ports.mooring</code>); Codes and design guidance (<code>design.guidance</code>)

Module	Topics
Design conditions (extremes)	Extreme wave climates (<code>waves.generation.extremes</code>); Structural and system reliability (<code>uncertainty.reliability</code>); Aleatory and epistemic uncertainty (<code>uncertainty.sources</code>); Probabilistic hazard analysis (<code>risk.probabilistic</code>); Extreme-event observations (<code>field.extreme</code>)
Beach nourishment (nourishment)	Beach nourishment (<code>beaches.nourishment</code>); Longshore currents (<code>nearshore.longshore</code>); Planform evolution (<code>morphodynamics.planform</code>); Shoreline change (<code>beaches.change</code>)
Pile wave loads (piles)	Wave loads (<code>wave_structure.loads</code>); Impact and impulsive loads (<code>wave_structure.loads.impact</code>); Local scour at structures (<code>scour.structures</code>); Pipeline and cable scour (<code>scour.pipelines</code>)
Harbour agitation (port)	Harbor agitation and resonance (<code>ports.agitation</code>); Diffraction (<code>waves.transformation.diffraction</code>); Reflection (<code>waves.transformation.reflection</code>)
Seawall design (seawall)	Seawalls and revetments (<code>structures.seawalls</code>); Overtopping (<code>structures.overtopping</code>); Structural failure and damage (<code>structures.failure</code>); Local scour at structures (<code>scour.structures</code>); Scour protection (<code>scour.protection</code>); Design conditions and criteria (<code>design.criteria</code>)
Breakwater design (structures)	Breakwaters (<code>structures.breakwaters</code>); Overtopping (<code>structures.overtopping</code>); Transmission and reflection (<code>wave_structure.transmission</code>); Porous structures (<code>wave_structure.porous</code>); Performance-based design (<code>design.performance</code>)
Storm surge and flooding (surge)	Surge forcing (<code>storm_surge.forcing</code>); Wind stress and drag (<code>storm_surge.forcing.wind</code>); Atmospheric pressure forcing (<code>storm_surge.forcing.pressure</code>); Wave setup contribution to storm surge (<code>storm_surge.coupling.wave_setup</code>); Compound flooding (<code>storm_surge.compound</code>); Wave runup (<code>waves.transformation.breaking.runup</code>); Inundation modeling (<code>inundation.models</code>); Flood-map uncertainty (<code>inundation.uncertainty</code>)

18.5 Coverage and its limits

Coverage follows the screening process. PyCoaPedia records the literature that has been screened; the paper count of each topic gives the extent of that screening. A low claim count indicates limited screening of the topic. Each claim carries the regime in which it was established, which defines its range of application.

19 The repository for AI agents

pyCoastal is written for two kinds of reader. Engineers read this manual. AI agents, working on a design or answering a question on a user's behalf, search the repository, and they work best from plain text, structured data, and a clear map of where things are. The repository is laid out so that every piece of knowledge in it has a machine-readable form next to the human one (Table 18).

Table 18: Human and machine-readable forms of the same knowledge.

Knowledge	For people	For agents
Where everything is	this manual, <code>README.md</code>	<code>AGENTS.md</code> , <code>llms.txt</code>
Theory and usage	<code>pyCoastal manual.pdf</code>	<code>docs/manual/chapters/*.md</code>
API	the source	<code>docs/reference/*.md</code> , one file per module
Worked examples	Part III	<code>examples/*.py</code> , <code>docs/examples.md</code>
Literature	PyCoaPedia explorer	<code>pycoapedia.sqlite</code> , <code>pedia/topics/*.md</code> , <code>python -m pyCoastal.pedia --json</code>

19.1 Entry points

AGENTS.md at the repository root is the map an agent should read first: what each directory holds, the conventions (units, levels, axis order), how to run the tests and the examples, how to query PyCoaPedia, and the rules the code follows (every relation names its source, nothing silently extrapolates, warnings are part of the result). **llms.txt** gives the same map in the compact format some tools look for.

The manual chapters in `docs/manual/chapters/` are Markdown with LaTeX mathematics, one file per chapter, which is exactly the source of this PDF; an agent can search them without parsing a PDF.

The API reference in `docs/reference/` is generated from the docstrings by the manual build, one Markdown file per module with every public class, function and constant, its signature, and its full docstring, so it cannot drift from the code.

The example index `docs/examples.md` lists every script in `examples/` with what it demonstrates and how to run it.

19.2 Querying the literature

For literature questions, query the database. The Markdown pages carry the same content in a form suited to file search. The sequence is search, open, cite:

```
python -m pyCoastal.pedia --json search "toe scour vertical wall" --limit 20
python -m pyCoastal.pedia --json topic scour.structures
python -m pyCoastal.pedia --json paper 2229
```

Every claim returned carries its regime and its source paper, with the DOI, so an answer built from PyCoaPedia can state where each statement holds and where it comes from. **pedia/SCHEMA.md** lists the tables, the controlled vocabularies, and worked SQL queries for anything the command line does not cover.

19.3 Keeping it current

`python docs/manual/build_manual.py` regenerates the PDF, the API reference and the example index; `python pedia/build_pedia.py --db <source>` refreshes the knowledge base in all its forms. Both are deterministic, so their output belongs in version control and a diff shows exactly what changed.

Part VII

Numerical framework

20 Grids, operators, and time integration

`pyCoastal.numerics` provides grid definitions, finite-difference operators, explicit time integrators and boundary-condition classes. Each component is separate and can be replaced independently.

20.1 Grids

20.1.1 UniformGrid

Most numerical examples use a uniform Cartesian mesh, the simplest setting in which to illustrate discretization ideas. Two storage layouts are common: cell-centered (unknowns at cell centers) and node-centered (unknowns at grid intersections). `pyCoastal` adopts the cell-centered layout, which keeps stencils symmetric and boundary handling simple on rectangular domains.

```
from pyCoastal.numerics.grid import UniformGrid

grid = UniformGrid(shape=(200, 100), spacing=(2.0, 2.0), origin=(0.0, 0.0))
grid.Xc          # cell-center coordinates, a list [X, Y] built with indexing="ij"
grid.Xf          # face coordinates
grid.spacing     # (dx, dy)
grid.n_cells, grid.cell_volume
grid.boundary_indices["west"] # flat indices of the west boundary cells
```

`UniformGrid(shape, spacing, origin=None)` works in one and two dimensions. A three-dimensional shape raises `NotImplementedError` on construction: the boundary indices are built in the constructor and are defined for 1D and 2D only. Cell centers are at $x_i = x_0 + (i + \frac{1}{2})\Delta x$. In 2D the flat index of cell (i, j) is $in_y + j$, so that `x` is axis 0 and `y` is axis 1 in every array. The grid precomputes, for each side ("west", "east", "south", "north"), the flat indices of the boundary cells in `boundary_indices` and a copy in `sponge_indices`, which the boundary conditions of Section 20.4 use. `neumann_indices(side)` returns the pair of boundary and first-interior indices needed to impose a gradient.

20.1.2 Mesh1D, Mesh2D, and Domain

`pyCoastal.numerics.domain` provides simple structured meshes built from a case file. `Mesh1D(x0, x1, nx)` stores cell centers `x` and spacing `dx`; `Mesh2D(x0, x1, nx, y0, y1, ny)` stores 2D center arrays `x`, `y` and spacings `dx`, `dy`. `Domain(cfg)` reads a `domain:` block with `dimension:` 1 or 2 and instantiates the right mesh; `Domain.info()` prints a summary. `Mesh2D` builds its arrays with NumPy's default `indexing="xy"`, giving shape (ny, nx) , which is the transpose of the `UniformGrid` convention. `UniformGrid` and `Mesh2D` use different axis conventions and should not be combined without explicit dimension handling.

Indexing and memory layout. Arrays follow NumPy C order, so the last index is contiguous. Spacings Δx , Δy are constant.

20.2 Finite differences

Finite differences approximate derivatives by algebraic expressions in neighboring grid values. On a uniform grid a first derivative at a cell center is approximated by a centered difference,

which samples one point on each side and is second-order accurate. Higher derivatives and one-sided formulas follow the same construction. The discretization converts the differential equations into a system of algebraic equations on a structured grid.

20.2.1 Centered operators

Centered operators combine values symmetrically around the point of interest, which keeps the approximation free of directional bias and gives higher accuracy for smooth solutions. For first derivatives and the Laplacian (Equations 47, 48):

$$\left. \frac{\partial f}{\partial x} \right|_{i,j} \approx \frac{f_{i+1,j} - f_{i-1,j}}{2\Delta x}, \quad \left. \frac{\partial f}{\partial y} \right|_{i,j} \approx \frac{f_{i,j+1} - f_{i,j-1}}{2\Delta y}, \quad (47)$$

$$\nabla^2 f|_{i,j} \approx \frac{f_{i+1,j} - 2f_{i,j} + f_{i-1,j}}{\Delta x^2} + \frac{f_{i,j+1} - 2f_{i,j} + f_{i,j-1}}{\Delta y^2}. \quad (48)$$

20.2.2 Upwind differences for advection

For an advection term with velocity u the solution at a point is determined by what arrives from upstream, so the stencil is taken from the upstream side (Equation 49):

$$\left. \frac{\partial f}{\partial x} \right|_i \approx \begin{cases} \frac{f_i - f_{i-1}}{\Delta x}, & u > 0, \\ \frac{f_{i+1} - f_i}{\Delta x}, & u < 0. \end{cases} \quad (49)$$

The one-sided stencil introduces numerical diffusion but is stable for hyperbolic problems and prevents non-physical oscillations.

20.2.3 The operator library

`pyCoastal.numerics.operators` implements these stencils on a `UniformGrid` (Table 19). The operators use periodic (roll) indexing, so boundary values must be imposed afterwards by the boundary conditions.

Table 19: Operators in `pyCoastal.numerics.operators`.

Function	Operation
<code>laplacian(field, grid)</code>	5-point Laplacian, Equation 48
<code>gradient(field, grid)</code>	centered gradient, returns (fx, fy)
<code>grad_x, grad_y</code>	centered first derivatives
<code>upwind_x(field, u, grid), upwind_y(field, v, grid)</code>	first-order upwind derivatives
<code>divergence(ux, uy, grid)</code>	centered $\nabla \cdot \mathbf{u}$
<code>curl_z(u, v, grid)</code>	$\partial v / \partial x - \partial u / \partial y$
<code>biharmonic(field, grid)</code>	$\nabla^4 f$ as the Laplacian of the Laplacian
<code>mixed_xy(field, grid)</code>	$\partial^2 f / \partial x \partial y$
<code>advect(u, v, field, grid, scheme="upwind")</code>	$u \partial_x f + v \partial_y f$
<code>smooth3(field, grid)</code>	5-point smoothing filter, weights summing to one

`pyCoastal.numerics.scheme` adds axis-generic versions, `central_difference(phi, spacing, axis)` and `upwind(phi, vel, spacing, axis)`.

20.3 Time integration

Once space is discretized, the semi-discrete variable $u(t)$ evolves under a right-hand side that gathers fluxes and sources, $du/dt = \text{RHS}(u, t)$. Explicit schemes compute the new state from information at the current level only, which makes them simple to apply and to check. The choice of integrator controls stability, accuracy, and how fast information moves through the grid.

Forward Euler (first order) treats the rate as constant over the step (Equation 50):

$$u^{n+1} = u^n + \Delta t \text{RHS}(u^n). \quad (50)$$

Heun / RK2 (second order) averages the slopes at the start and the end of the step (Equation 51):

$$k_1 = \text{RHS}(u^n, t), \quad k_2 = \text{RHS}(u^n + \Delta t k_1, t + \Delta t), \quad u^{n+1} = u^n + \frac{\Delta t}{2}(k_1 + k_2). \quad (51)$$

Classical RK4 (fourth order) is Equation 52:

$$u^{n+1} = u^n + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4). \quad (52)$$

SSP RK3 (third order, strong stability preserving) blends each stage with the earlier state, which keeps monotonicity for advection-dominated problems under a CFL limit (Equation 53):

$$\begin{aligned} u^{(1)} &= u^n + \Delta t \text{RHS}(u^n), \\ u^{(2)} &= \frac{3}{4}u^n + \frac{1}{4}[u^{(1)} + \Delta t \text{RHS}(u^{(1)})], \\ u^{n+1} &= \frac{1}{3}u^n + \frac{2}{3}[u^{(2)} + \Delta t \text{RHS}(u^{(2)})]. \end{aligned} \quad (53)$$

SSP RK3 is the default choice for the advection-dominated cases in Section 22.

Adams-Bashforth 2 (second order, multistep) reuses the previous right-hand side, and needs one starting step from another method (Equation 54):

$$u^{n+1} = u^n + \frac{\Delta t}{2}(3 \text{RHS}(u^n) - \text{RHS}(u^{n-1})). \quad (54)$$

These live in `pyCoastal.numerics.time_intg` as array functions, `euler_step(u, t, dt, rhs)`, `rk2_step`, `rk4_step`, `rk3_ssp_step`, and `ab2_step(u, u_prev, t, dt, rhs)`, where `rhs(u, t)` returns an array of the same shape as `u`.

For multi-field states, `pyCoastal.numerics.scheme` provides integrator classes that act on a dictionary of arrays: `EulerIntegrator(dt)` and `SSPRK2Integrator(dt)` (Shu-Osher second-order TVD), both with `step(state, rhs, t, **kwargs)` returning `(new_state, new_time)`, where `rhs(state, t, **kwargs)` returns a dictionary of tendencies.

20.3.1 The Solver driver

`pyCoastal.numerics.solver.Solver(grid, physics, bc, integrator=None)` sketches the loop: `Solver.run(t0, t_end, callback=None)` applies the boundary conditions, takes one step through the integrator, and calls `callback(state, t)` after each. It expects a `physics` object with `initialize_state(grid)`, `rhs(state, t, grid=..., bc=...)` and a `dt`, and a `bc` object with `apply(state, t)`.

Two adaptations are required before the class runs with the components of this package. `BoundaryManager` exposes `apply_all(fields, grid, t)` and has to be wrapped to provide `apply(state, t)`. `physics.navier_stokes.rhs` takes the keyword `bc_mgr`, while the solver passes `bc`. The examples in `examples/` implement their own time loops, as do the design applications that march in time: the port solver of Section 6 and the one-line model of Section 3.4.

20.3.2 Stability

The CFL condition states that information carried by advection cannot move farther than one cell per step. For an advective speed c (Equation 55):

$$\text{CFL} = \max \left(\frac{c \Delta t}{\Delta x}, \frac{c \Delta t}{\Delta y} \right) < 1. \quad (55)$$

Diffusion imposes its own, often more severe, limit (Equation 56):

$$\Delta t \leq \frac{1}{2\nu} \left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} \right)^{-1}. \quad (56)$$

The time step must satisfy both. The design applications apply the same rule internally: the port solver steps at a Courant number of 0.35 by default, and the one-line shoreline model at 0.9 of its diffusive limit (`tools.shoreline.suggest_dt`).

20.4 Boundary conditions

Boundary conditions close the discrete system. They fix a value, prescribe a gradient, let waves or flow enter or leave, or enforce a wall, and they control how signals reflect and how energy enters or exits the domain.

20.4.1 Ghost-cell formulas

For a cell-centered scheme the classical formulas at $x = 0$ are:

- **Dirichlet** (value), $q|_{\Gamma} = q_D(x, t)$: $q_0 = 2q_b - q_1$ with $q_b = q_D$.
- **Neumann** (gradient), $\partial q / \partial n|_{\Gamma} = g_N$: $q_0 = q_1 - \Delta x g_N$.
- **Solid wall**: $u_n = 0$, mirroring the normal component antisymmetrically; free slip mirrors the tangential velocity symmetrically, no slip antisymmetrically.
- **Sponge layer** (wave absorption): a damping zone relaxes the solution toward a reference state (Equation 57),

$$\frac{\partial q}{\partial t} = -\sigma(x) [q - q_{\text{ref}}], \quad \sigma(s) = \sigma_{\text{max}} \sin^2 \left(\frac{\pi s}{2} \right), \quad s = \frac{x - x_0}{L_{\text{sp}}} \in [0, 1]. \quad (57)$$

A sponge of $L_{\text{sp}} \approx 1$ to 3 wavelengths with $\sigma_{\text{max}} \sim \kappa c / L_{\text{sp}}$, $\kappa \approx 3$ to 6, absorbs well; monitor reflections with gauges inside the domain.

20.4.2 The boundary classes

`pyCoastal.numerics.boundary` implements boundary conditions that act directly on the boundary cells, using the flat indices precomputed by the grid. Every class takes a side name and a list of field keys and implements `apply(fields, grid, t)`:

- `DirichletBC(location, var_names, value)`, where `value` is a number or a function of time;
- `NeumannBC(location, var_names, gradient=0.0)`, which copies the interior value plus `gradient * dx` onto the boundary;
- `WallBC(location, var_names)`, which sets the named fields (normally the normal velocity or flux) to zero;
- `SpongeBC(location, var_names, damping)`, which multiplies the boundary strip by a constant factor or by `damping(t, k)` evaluated per flat index `k`, so a spatial ramp can be built from the index;
- `BoundaryManager(bcs=())` holds a list of conditions; `add(bc)` registers one and `apply_all(fields, grid, t)` applies them all, in order, at time `t`.

```
import numpy as np
from pyCoastal.numerics.grid import UniformGrid
from pyCoastal.numerics.boundary import (
    DirichletBC, NeumannBC, WallBC, SpongeBC, BoundaryManager,
)

grid = UniformGrid(shape=(200, 200), spacing=(2.0, 2.0))
fields = {"eta": np.zeros(grid.shape), "u": np.zeros(grid.shape),
          "v": np.zeros(grid.shape)}

bcs = BoundaryManager()
bcs.add(DirichletBC("south", ["eta"], lambda t: 0.5 * np.sin(2 * np.pi * t / 3.0)))
bcs.add(DirichletBC("west", ["u"], value=0.2))           # inflow
bcs.add(NeumannBC("east", ["u", "eta", "v"]))           # zero-gradient outflow
bcs.add(WallBC("north", ["v"]))                         # no normal flow
bcs.add(SpongeBC("north", ["eta"], damping=0.9))        # damp the strip
bcs.apply_all(fields, grid, t=0.0)
```

The complete version of this demonstration is `examples/numerics/2D_irr_turb.py`.

21 Governing equations and closures

`pyCoastal.physics` provides the governing-equation components: flux functions and source terms for assembly into a right-hand side, a pressure-projection step, Poisson solvers, and eddy-viscosity closures.

21.1 Shallow water equations

The shallow water equations apply to a fluid layer whose horizontal scales are large compared with its depth. The variables are the total depth and the depth-averaged velocity; the pressure is hydrostatic. The equations represent wave propagation, flooding and draining, and large-scale currents (Equations 58, 59, 60):

$$\frac{\partial h}{\partial t} + \frac{\partial(hu)}{\partial x} + \frac{\partial(hv)}{\partial y} = 0, \quad (58)$$

$$\frac{\partial(hu)}{\partial t} + \frac{\partial}{\partial x} (hu^2 + \frac{1}{2}gh^2) + \frac{\partial(huv)}{\partial y} = S_x, \quad (59)$$

$$\frac{\partial(hv)}{\partial t} + \frac{\partial(huv)}{\partial x} + \frac{\partial}{\partial y} (hv^2 + \frac{1}{2}gh^2) = S_y. \quad (60)$$

Here h is the total depth, u and v the depth-averaged velocities, and S_x , S_y collect bed slope, bottom friction, and Coriolis. When adding sources, preserve the lake-at-rest balance.

`ShallowWater2D(g=9.81)` provides `fluxes(h, hu, hv)`, which returns the three flux pairs (F_h, G_h) , (F_{hu}, G_{hu}) , (F_{hv}, G_{hv}) , and `source_bed_slope(h, zb, grid=None)`, which returns $S_x = -gh \partial z_b / \partial x$ and $S_y = -gh \partial z_b / \partial y$. Pass the grid so the derivatives use the real spacing; without it the spacing is taken as one.

21.2 Incompressible Navier-Stokes

For incompressible viscous flow a popular explicit strategy is Chorin's projection (fractional step) method: predict the velocity with advection and diffusion, solve a Poisson equation for the pressure that removes the divergence, then correct (Equations 61, 62, 63):

$$\tilde{\mathbf{u}} = \mathbf{u}^n + \Delta t (N(\mathbf{u}^n) + \nu \nabla^2 \mathbf{u}^n), \quad (61)$$

$$\nabla^2 p^{n+1} = \frac{1}{\Delta t} \nabla \cdot \tilde{\mathbf{u}}, \quad (62)$$

$$\mathbf{u}^{n+1} = \tilde{\mathbf{u}} - \Delta t \nabla p^{n+1}, \quad (63)$$

with $\mathbf{u} = (u, v)$, p the kinematic pressure, and ν the viscosity. `pyCoastal.physics.navier_stokes` provides `initialize_state(grid)`, which returns zero $\mathbf{u}$, $\mathbf{v}$, $\mathbf{p}$ fields, and `rhs(state, t, grid, bc_mgr, v=0.001)`, which returns the convection and diffusion tendencies for $\mathbf{u}$ and $\mathbf{v}$ (the predictor of Equation 61), with the velocity boundary conditions enforced on the result.

21.3 Poisson solver

A Poisson solver computes a field whose Laplacian matches a given source. It appears in the pressure projection, in potential flow, and in diffusion steady states (Equation 64):

$$\nabla^2 \phi = b \text{ in } \Omega, \quad \phi|_{\Gamma_D} = \phi_D, \quad \frac{\partial \phi}{\partial n} \Big|_{\Gamma_N} = g_N, \quad (64)$$

with the compatibility condition $\int_{\Omega} b \, d\Omega + \int_{\Gamma_N} g_N \, d\Gamma = 0$ for a pure Neumann problem. The five-point Laplacian of Equation 48 gives a sparse linear system. The Jacobi iteration, with $h = \Delta x = \Delta y$, is Equation 65,

$$\phi_{i,j}^{(k+1)} = \frac{1}{4} \left(\phi_{i+1,j}^{(k)} + \phi_{i-1,j}^{(k)} + \phi_{i,j+1}^{(k)} + \phi_{i,j-1}^{(k)} - h^2 b_{i,j} \right), \quad (65)$$

stopping when $\|b - A\phi^{(k)}\|_2 / \|b\|_2 < \varepsilon$.

`pyCoastal.physics.poisson` offers:

- `build_laplacian(nx, ny, dx, dy)`, the sparse CSR operator with zero Dirichlet boundaries (needs SciPy);
- `solve_direct(rhs, dx, dy, bc_mask=None, bc_values=None)`, a sparse direct solve (the right-hand side is shaped (ny, nx) here, not (nx, ny));
- `solve_jacobi(rhs, dx, dy, bc_mask=None, bc_values=None, tol=1e-6, maxiter=5000)`;
- `solve_poisson(rhs, dx, dy, ..., method="auto")`, which picks the direct solver when SciPy is available and Jacobi otherwise;
- `apply_dirichlet_bc(phi, bc_mask, bc_values)`, which imposes Dirichlet values in place.

21.4 Turbulence closures

When the grid cannot resolve all scales, eddy-viscosity closures emulate the transport by unresolved motions. The closures here are educational scaffolds: production RANS or LES needs careful calibration and near-wall treatment.

Smagorinsky (LES). The eddy viscosity is proportional to the local strain rate and the filter width (Equation 66):

$$\nu_t = (C_s \Delta)^2 |S|, \quad |S| = \sqrt{2S_{ij}S_{ij}}, \quad S_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right), \quad (66)$$

with $C_s \approx 0.1$ to 0.2 . `SmagorinskyModel(Cs=0.17, filter_width=1.0)` computes `eddy_viscosity(u, v, grid)`.

k - ε (RANS). Two transport equations for the turbulent kinetic energy k and its dissipation ε , with $\nu_t = C_\mu k^2 / \varepsilon$ (Equations 67, 68):

$$\partial_t k + U_j \partial_{x_j} k = P - \varepsilon + \partial_{x_j} \left[\left(\nu + \frac{\nu_t}{\sigma_k} \right) \partial_{x_j} k \right], \quad (67)$$

$$\partial_t \varepsilon + U_j \partial_{x_j} \varepsilon = C_{\varepsilon 1} \frac{\varepsilon}{k} P - C_{\varepsilon 2} \frac{\varepsilon^2}{k} + \partial_{x_j} \left[\left(\nu + \frac{\nu_t}{\sigma_\varepsilon} \right) \partial_{x_j} \varepsilon \right], \quad (68)$$

with $P = 2\nu_t S_{ij} S_{ij}$. `KEpsilonModel(Cmu=0.09, sigma_k=1.0, sigma_e=1.3, C1=1.44, C2=1.92)` returns the eddy viscosity from `k` and `eps`; the transport equations belong to the solver.

k - ω (RANS). Pairs k with the specific dissipation $\omega = \varepsilon / k$, which behaves better near walls, with $\nu_t = k / \omega$ (Equations 69, 70):

$$\partial_t k + U_j \partial_{x_j} k = P - \beta^* k \omega + \partial_{x_j} \left[(\nu + \sigma_k^* \nu_t) \partial_{x_j} k \right], \quad (69)$$

$$\partial_t \omega + U_j \partial_{x_j} \omega = \alpha \frac{\omega}{k} P - \beta \omega^2 + \partial_{x_j} \left[(\nu + \sigma_\omega \nu_t) \partial_{x_j} \omega \right]. \quad (70)$$

`KOmegaModel(sigma_k=2.0, sigma_omega=2.0, beta_star=0.09, beta=0.075, gamma=0.52)` returns `k / omega` from `eddy_viscosity(k, omega)`.

22 Simulation examples

Each numerical example combines one governing equation, one discretization and one diagnostic. The scripts read their parameters from YAML files in `examples/configs/`, so a case is repeated or swept by editing the case file. The fields are animated with matplotlib. The scripts run from the repository root.

```
python examples/numerics/water_drop.py
python examples/waves2D.py
python examples/wave2D_irregular.py
python examples/current.py
python examples/pollutant.py
python examples/numerics/viscous_fluid.py
python examples/numerics/2D_irr_turb.py
python examples/equilibrium_shoreline.py
```

22.1 Case files

The YAML keys used by the examples are collected in Table 20. Grid keys give cell counts and either spacings or domain lengths; physics keys the parameters of the equation; solver keys the time step (directly or through a CFL number) and the run length; output keys the gauge locations as grid indices.

Table 20: YAML keys used by the numerical examples.

Config	grid	physics	solver / forcing	output
water_drop. yaml	Nx, Ny, Lx, Ly	wave_speed, sigma	CFL, t_final	
waves2D. yaml	nx, ny, dx, dy	gravity, depth	amplitude, period; cfl, t_ final	obs_points
waves2D_ irregular. yaml	nx, ny, dx, dy	gravity, depth	type (pm or jonswap), gamma, Hs, Tp; dt, duration	gauge
current. yaml	nx, ny, dx, dy	speed	cfl, t_final	gauge
pollutant. yaml	Lx, Ly, Nx, Ny	diffusivity, U0, R	t_final, c0_sigma	
viscous_ fluid.yaml	Nx, Ny, Lx, Ly	viscosity, background_ speed, vortex_sigma	dt, t_final	
equilibrium_ shoreline. yaml	Lshore, Gb, Lb1, Lb2, Yi		n_points, sag_side_factor	

22.2 Water drop: the 2D wave equation

Objective. Verify the accuracy of the Laplacian and the stability of the time stepping on a simple hyperbolic system (Equation 71),

$$\frac{\partial^2 \eta}{\partial t^2} = c^2 \nabla^2 \eta. \quad (71)$$

A Gaussian bump of width `sigma` is released at the center of the domain with zero Dirichlet boundaries, and the circular front should travel at c . Refining the grid and the time step should converge the solution at the expected order.



Figure 35: Circular wave propagation from a central disturbance (`examples/numerics/water_drop.py`).

22.3 Regular waves: `waves2D.py`

A depth-averaged wave field in uniform depth, forced by a sinusoid of amplitude `amplitude` and period `period` at the $y = 0$ boundary and periodic in x . The script solves linear advection along y column by column with an upwind stencil and forward Euler, animates η , and records the time series at the observation points. The wave speed comes from the linear dispersion relation for the configured depth and period.

22.4 Irregular waves: a numerical flume

Objective. Generate a target spectrum and manage reflections. The script synthesizes a surface elevation record from a Pierson-Moskowitz or JONSWAP spectrum (Section 3.1), imposes it as a Dirichlet condition at the inflow boundary, and records a gauge inside the domain, so the input and interior spectra (H_s , T_p) can be compared. A sponge at the outflow absorbs the outgoing waves.

22.5 Current and pollutant: advection and diffusion

Objective. Observe upwind smearing and diffusive spreading. `current.py` advects a passive field with a uniform current, $u_t + c u_x = 0$, with a Dirichlet inlet on the west and a Neumann out-

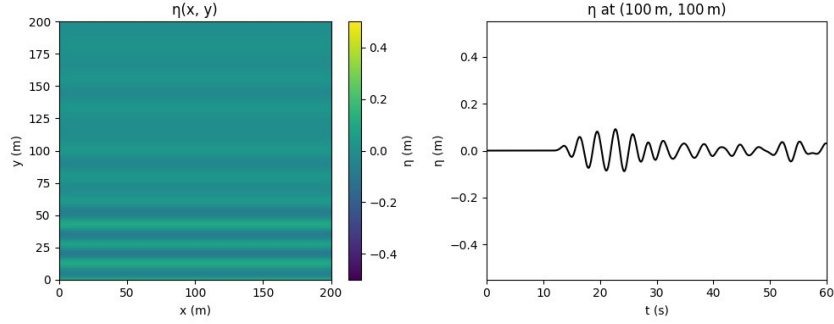


Figure 36: Irregular wave field at the end of the run (`examples/wave2D_irregular.py`).

let on the east, and records a gauge. `pollutant.py` solves the full advection-diffusion equation for a Gaussian patch in a vortex flow in a shallow pond (Equation 72),

$$\frac{\partial C}{\partial t} + u \frac{\partial C}{\partial x} + v \frac{\partial C}{\partial y} = D \nabla^2 C, \quad (72)$$

with upwind advection, centered diffusion (`operators.gradient` and `operators.laplacian`), and zero-flux boundaries. For pure diffusion the variance of the patch grows as $2Dt$, which the run can be checked against.

22.6 Viscous fluid: velocity decay

Objective. Check the diffusive time scale and operator accuracy. `viscous_fluid.py` evolves two counter-rotating vortices on a periodic domain with central differences and forward Euler. The diffusion part can be checked against the decaying mode (Equation 73)

$$u(x, y, t) = U_0 \sin(k_x x) \sin(k_y y) \exp[-\nu(k_x^2 + k_y^2)t]. \quad (73)$$

22.7 Boundary-condition test: 2D_irr_turb.py

A pure boundary-condition exercise on a 400 m by 400 m grid: a sinusoidal Dirichlet wave maker on the south side, a spatially ramped sponge on the north, a Dirichlet inflow velocity and a zero-gradient surface on the west, zero-gradient outflow on the east, and walls for the normal velocity. No physics is advanced; the script applies the `BoundaryManager` every step and records a gauge, which isolates what the boundary classes do.

22.8 Equilibrium shoreline behind detached breakwaters

`equilibrium_shoreline.py` builds a schematic equilibrium planform behind a pair of detached breakwaters from three circular arcs (a left lateral arc, a central salient, and a right lateral arc), following the Hsu and Evans style of parabolic bay construction. The geometry comes from `equilibrium_shoreline.yaml` (shoreline length, gap between breakwaters, their lengths, and offshore distance) and the construction uses `pyCoastal.tools.morphodynamics` (Section 3.5), which also reports the retreat and the accretion percentage.

22.9 Diagnostics, verification, and validation

Numerical experiments should be accompanied by simple, repeatable checks:

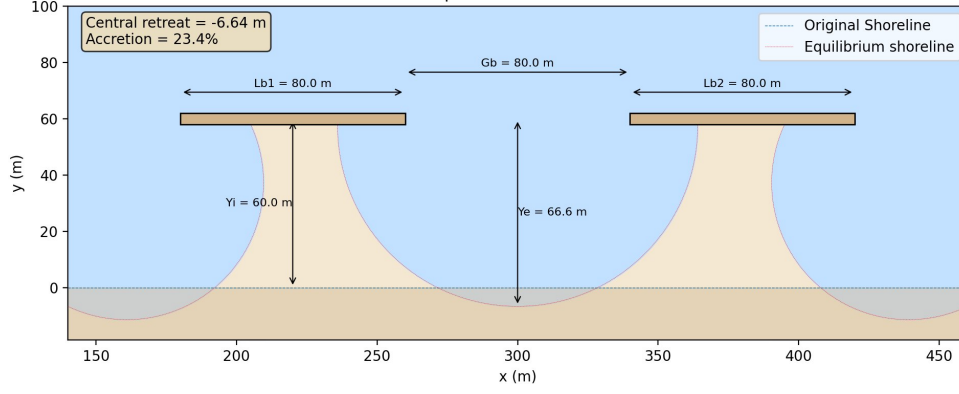


Figure 37: Equilibrium shoreline behind two detached breakwaters (`examples/equilibrium_shoreline.py`).

- **CFL monitor.** Compute and report the Courant number each step, and adapt Δt if needed.
- **Divergence.** After a projection step, $\|\nabla \cdot \mathbf{u}\|_2$ should sit near the Poisson solver tolerance.
- **Energy and variance.** Monitor $\int \eta^2$ or the kinetic energy to spot drift or spurious reflections.
- **Convergence.** Refine Δx , Δy , and Δt and verify the expected order.
- **Spectral targets.** Compare H_s and T_p at interior gauges with the input, and estimate reflection by separating incident and reflected waves.

Save arrays as NumPy files and figures with matplotlib; export animations as `.gif` or `.mp4`. Store the run metadata (git hash, YAML, random seeds) beside the outputs so a figure can be regenerated exactly.

22.10 Extending the framework

To add physics or numerics:

1. implement a right-hand side `rhs(u, t)` and pass it to an integrator;
2. provide boundary handlers consistent with the stencils;
3. add YAML keys with sensible defaults;
4. add a small example and a docstring that states the equations, the variables, the numerics, and the stability limits.

22.11 Limitations of the numerical framework

- Uniform Cartesian grids only; no curvilinear or unstructured meshes.
- Explicit time stepping, which is restrictive for stiff sources such as the ε or ω transport.
- The turbulence closures are educational scaffolds, not production RANS or LES.
- No wetting and drying or moving shorelines in the 2D solvers. (The design applications handle shoreline change with the one-line model and flooding with connectivity-aware mapping; see Section 14 and Section 15.)

Part VIII

Quality assurance

23 The test suite

pyCoastal carries a test suite covering every module. It runs with

```
python -m pytest tests/
```

and in continuous integration on Ubuntu and Windows under Python 3.10, 3.11, and 3.13 on every push to `main` and every pull request.

23.1 What the tests check

Most tests compare a result against an independent reference. The categories recur across modules:

- **Published values and limits.** The Van der Meer plunging and surging branches against the published forms; the Sumer and Fredsoe relation collapsing to the waves-only form at $U_{cw} = 0$ and to $S/D = 1.3$ in pure current; the toe scour recovering Xie at $K_r = 1$; Dean's profile scale against his table.
- **Analytical solutions.** The one-line solver against Pelnard-Considere; the port solver against the semi-infinite breakwater shadow; the fill volume and critical volume for matched sand; normal depth against Manning's equation.
- **Physical consistency.** Monotonic responses (more overtopping at lower crest, deeper scour at higher velocity), conservation (volume in the nourishment model), and conventions (axis order, units, sign of levels).
- **Failure modes.** Unsupported input raises: cohesive beds in the scour relations, normal depth on an adverse slope, non-positive freeboard in the overtopping relations. Input outside a documented validity range returns a result with a warning.
- **Drawings.** Sheets reject unknown paper sizes and report their fitted scale, and DXF export round-trips one line per edge with every layer queued.

23.2 Tests by file

The suite holds 877 test functions in 25 files (parametrized tests expand to more cases when run).

Table 21: Test files and what they cover.

File	Tests	Scope
<code>test_applications_berthing.py</code>	49	Berthing energy and fender selection.
<code>test_applications_bridge_scour.py</code>	47	Contraction and abutment scour: the other two thirds of HEC-18.
<code>test_applications_channel.py</code>	46	Navigation channel design: vessel, squat, clearance, width and depth.
<code>test_applications_extremes.py</code>	51	Extreme value analysis: L-moments, fits, return values and diagnostics.
<code>test_applications_nourishment.py</code>	25	Beach nourishment simulator.

File	Tests	Scope
<code>test_applications_nourishment_profile.py</code>	30	Cross-shore nourishment: equilibrium profiles and borrow compatibility.
<code>test_applications_piles.py</code>	52	Morison wave loads on piles: kinematics, coefficients, loads and scour.
<code>test_applications_plan_view.py</code>	34	The nourishment planform: the analytical map and the scales it is drawn at.
<code>test_applications_port.py</code>	28	Port layout simulator.
<code>test_applications_port_layouts.py</code>	24	Built-in port layouts and oblique wave incidence.
<code>test_applications_river.py</code>	47	Open channel hydraulics: normal and critical depth, backwater, afflux.
<code>test_applications_scour.py</code>	76	Scour at a pier in combined waves, tide and river current.
<code>test_applications_seawall.py</code>	62	Vertical seawall design: Goda pressures, stability, scour and toe.
<code>test_applications_sections.py</code>	34	Section drawings: geometry, layer offsets, quantities and sheets.
<code>test_applications_sediment.py</code>	36	Sediment and soil: material properties, mobility, and earth pressure.
<code>test_applications_structures.py</code>	78	Armour sizing and wave overtopping.
<code>test_applications_surge.py</code>	48	Storm surge components and flood mapping.
<code>test_drafting.py</code>	23	Drawing primitives: geometry, scale, and the DXF export.
<code>test_numerics_grid.py</code>	8	Grid geometry and boundary-index bookkeeping.
<code>test_numerics_operators.py</code>	17	Finite-difference operators, checked against analytic derivatives.
<code>test_pedia.py</code>	10	PyCoePedia: the shipped knowledge base and its reader.
<code>test_physics_conventions.py</code>	8	The whole package uses one array layout: x is axis 0, y is axis 1.
<code>test_tools_sediment.py</code>	13	Sediment transport formulae: closed-form checks and physical scaling.
<code>test_tools_structural.py</code>	16	Armour sizing, run-up and wave force formulae.
<code>test_tools_wave.py</code>	15	Wave kinematics and spectra, checked against analytic limits.

23.3 The browser engine

The JavaScript engine of PyCoeTools is checked against the Python by `webapp/verify_engine.py` on the reference cases of `webapp/make_vectors.py`, and the page repeats the comparison on load (Section 17). `webapp/verify_app.py` executes every module of the page once. Both verifiers are run manually before the app is published; they are not part of the continuous integration workflow.

Appendices

A List of symbols

Symbol	Meaning	Unit
x, y, z	horizontal coordinates and elevation (positive up)	m
$t, \Delta t$	time and time step	s
$\Delta x, \Delta y$	grid spacings	m
$u, v, \mathbf{u}$	velocity components and vector	m/s
h, d	water depth	m
η	free-surface elevation, setup	m
g	gravitational acceleration, 9.81	m/s ²
ρ, ρ_s	water and sediment (or rock) density	kg/m ³
Δ	relative buoyant density $\rho_s/\rho - 1$	-
ν, ν_t	kinematic and eddy viscosity	m ² /s
k, ε, ω	turbulent kinetic energy, dissipation, specific dissipation	m ² /s ² , m ² /s ³ , 1/s
C_s	Smagorinsky constant	-
$\sigma(x), L_{sp}$	sponge damping rate and thickness	1/s, m
H, H_s, H_{m0}, H_b	wave height, significant height, spectral height, breaking height	m
$T, T_p, T_{m-1,0}$	period, peak period, spectral period	s
L, L_0, k, ω	wavelength, deep-water wavelength, wavenumber, angular frequency	m, m, 1/m, 1/s
c, c_g	phase and group celerity	m/s
ξ	surf similarity (Iribarren) number	-
U	Ursell number	-
$S(f), m_n$	variance density spectrum, spectral moment	m ² /Hz
α, β	structure slope angle; wave obliquity or beach slope	rad or deg
D_{n50}, M_{50}	nominal stone diameter and median mass	m, kg
P, S, N	notional permeability, damage level, number of waves	-
K_D	Hudson stability coefficient	-
R_c	crest freeboard	m
q	mean overtopping discharge	l/s/m
$\gamma_f, \gamma_\beta, \gamma_b, \gamma_v$	EurOtop roughness, obliquity, berm, and wall factors	-
p_1, p_3, p_4, p_u	Goda pressures and uplift	kPa
K_r, K_d	reflection and disturbance coefficients	-
C_d, C_m	Morison drag and inertia coefficients	-

Symbol	Meaning	Unit
KC	Keulegan-Carpenter number	-
U_c, U_m, U_{cw}	current, near-bed orbital velocity, current share	m/s, m/s, -
S, S_{eq}	scour depth, equilibrium scour depth	m
D_e	effective pier diameter	m
θ, θ_{cr}	Shields parameter and its critical value	-
d_{50}, D_*, w_s	median grain size, dimensionless grain size, fall velocity	m, -, m/s
ϕ', K_a, K_0	friction angle, active and at-rest pressure coefficients	deg, -
n, K, S_f	Manning coefficient, conveyance, friction slope	s/m ^{1/3} , m ³ /s, -
Fr	Froude number	-
F_{nh}	depth Froude number of a ship	-
C_b, L_{pp}, B, T	block coefficient, length between perpendiculars, beam, draught	-, m, m, m
C_m, C_e, C_s, C_c	PIANC berthing factors	-
A	Dean profile scale	m ^{1/3}
ε	longshore diffusivity (one-line model)	m ² /s
Q, K_{cerc}	longshore transport and CERC coefficient	m ³ /s
u, σ, ξ, λ	threshold, scale, shape, and rate of an extreme value fit	-

B Engineering references

The design relations implemented in pyCoastal, as cited in the module docstrings.

- Barrass, C. B. (1979). A unified approach to squat calculations for ships. *PIANC Bulletin*, 32.
- Battjes, J. A., and Groenendijk, H. W. (2000). Wave height distributions on shallow foreshores. *Coastal Engineering*, 40(3), 161-182.
- Breusers, H. N. C., Nicollet, G., and Shen, H. W. (1977). Local scour around cylindrical piers. *Journal of Hydraulic Research*, 15(3), 211-252.
- Brolsma, J. U., Hirs, J. A., and Langeveld, J. M. (1977). On fender design and berthing velocities. *PIANC 24th Congress*.
- BS 6349-4 (2014). *Maritime works. Code of practice for design of fendering and mooring systems*. British Standards Institution.
- CEN (2004). *EN 1992-1-1, Eurocode 2: Design of concrete structures. Part 1-1: General rules and rules for buildings*. European Committee for Standardization.
- Chorin, A. J. (1968). Numerical solution of the Navier-Stokes equations. *Mathematics of Computation*, 22(104), 745-762.
- Chow, V. T. (1959). *Open-Channel Hydraulics*. McGraw-Hill.
- CIRIA, CUR, and CETMEF (2007). *The Rock Manual: The Use of Rock in Hydraulic Engineering*, 2nd ed. CIRIA C683.
- Coles, S. (2001). *An Introduction to Statistical Modeling of Extreme Values*. Springer.

- Dean, R. G., and Dalrymple, R. A. (1991). *Water Wave Mechanics for Engineers and Scientists*. World Scientific.
- DNV (2010). *DNV-RP-C205: Environmental Conditions and Environmental Loads*. Det Norske Veritas.
- EurOtop (2018). *Manual on Wave Overtopping of Sea Defences and Related Structures*, 2nd ed. Van der Meer, J. W., Allsop, N. W. H., Bruce, T., De Rouck, J., Kortenhaus, A., Pullen, T., Schüttrumpf, H., Troch, P., and Zanuttigh, B.
- FEMA (2005). *Guidelines and Specifications for Flood Hazard Mapping Partners*. Federal Emergency Management Agency.
- Fenton, J. D. (1988). The numerical solution of steady water wave problems. *Computers and Geosciences*, 14(3), 357-368.
- Ferziger, J. H., and Perić, M. (2002). *Computational Methods for Fluid Dynamics*. Springer.
- Froehlich, D. C. (1989). Local scour at bridge abutments. *Proceedings, ASCE National Hydraulic Conference*, 13-18.
- Goda, Y. (1974). New wave pressure formulae for composite breakwaters. *Proceedings of the 14th International Conference on Coastal Engineering*, 1702-1720.
- Goda, Y. (2010). *Random Seas and Design of Maritime Structures*, 3rd ed. World Scientific.
- Gottlieb, S., Shu, C.-W., and Tadmor, E. (2001). Strong stability-preserving high-order time discretization methods. *SIAM Review*, 43(1), 89-112.
- Hasselmann, K., et al. (1973). Measurements of wind-wave growth and swell decay during the Joint North Sea Wave Project (JONSWAP). *Deutsche Hydrographische Zeitschrift*, A8(12).
- Henderson, F. M. (1966). *Open Channel Flow*. Macmillan.
- Hosking, J. R. M., and Wallis, J. R. (1997). *Regional Frequency Analysis: An Approach Based on L-Moments*. Cambridge University Press.
- Hudson, R. Y. (1959). Laboratory investigation of rubble-mound breakwaters. *Journal of the Waterways and Harbors Division*, 85(3), 93-121.
- Hydrologic Engineering Center (2016). *HEC-RAS River Analysis System, Hydraulic Reference Manual, Version 5.0*. US Army Corps of Engineers.
- James, W. R. (1975). *Techniques in evaluating suitability of borrow material for beach nourishment*. Technical Memorandum 60, US Army Coastal Engineering Research Center.
- Kriebel, D. L., Kraus, N. C., and Larson, M. (1991). Engineering methods for predicting beach profile response. *Proceedings of Coastal Sediments '91*, 557-571.
- Lagasse, P. F., et al. (2009). *Bridge Scour and Stream Instability Countermeasures*, HEC-23, 3rd ed. Federal Highway Administration.
- Launder, B. E., and Spalding, D. B. (1974). The numerical computation of turbulent flows. *Computer Methods in Applied Mechanics and Engineering*, 3(2), 269-289.
- Laursen, E. M. (1960). Scour at bridge crossings. *Journal of the Hydraulics Division*, 86(2), 39-54.
- LeVeque, R. J. (2002). *Finite Volume Methods for Hyperbolic Problems*. Cambridge University Press.
- Morison, J. R., O'Brien, M. P., Johnson, J. W., and Schaaf, S. A. (1950). The force exerted by surface waves on piles. *Petroleum Transactions, AIME*, 189, 149-154.
- Nørgaard, J. Q. H., Andersen, T. L., and Burcharth, H. F. (2013). Wave loads on rubble mound breakwater crown walls in deep and shallow water wave conditions. *Coastal Engineering*, 80, 137-147.
- Pedersen, J. (1996). *Experimental study of wave forces and wave overtopping on breakwater crown walls*. Series Paper 12, Hydraulics and Coastal Engineering Laboratory, Aalborg University.

- Pelnard-Considère, R. (1956). Essai de théorie de l'évolution des formes de rivage en plages de sable et de galets. *4èmes Journées de l'Hydraulique*, Question III, 289-298.
- PIANC (2002). *Guidelines for the Design of Fender Systems*. Report of Working Group 33, Maritime Navigation Commission.
- PIANC (2014). *Harbour Approach Channels: Design Guidelines*. Report 121.
- Pierson, W. J., and Moskowitz, L. (1964). A proposed spectral form for fully developed wind seas. *Journal of Geophysical Research*, 69(24), 5181-5190.
- Pope, S. B. (2000). *Turbulent Flows*. Cambridge University Press.
- Richardson, E. V., and Davis, S. R. (2001). *Evaluating Scour at Bridges*, HEC-18, 4th ed. Federal Highway Administration.
- Seelig, W. N., and Ahrens, J. P. (1981). *Estimation of wave reflection and energy dissipation coefficients for beaches, revetments, and breakwaters*. Technical Paper 81-1, US Army Coastal Engineering Research Center.
- Smagorinsky, J. (1963). General circulation experiments with the primitive equations. *Monthly Weather Review*, 91, 99-164.
- Soulsby, R. L. (1997). *Dynamics of Marine Sands*. Thomas Telford.
- Soulsby, R. L., and Whitehouse, R. J. S. (1997). Threshold of sediment motion in coastal environments. *Proceedings of Pacific Coasts and Ports '97*, 149-154.
- Stockdon, H. F., Holman, R. A., Howd, P. A., and Sallenger, A. H. (2006). Empirical parameterization of setup, swash, and runup. *Coastal Engineering*, 53(7), 573-588.
- Sumer, B. M., and Fredsøe, J. (2001). Scour around pile in combined waves and current. *Journal of Hydraulic Engineering*, 127(5), 403-411.
- Sumer, B. M., and Fredsøe, J. (2002). *The Mechanics of Scour in the Marine Environment*. World Scientific.
- Sumer, B. M., Christiansen, N., and Fredsøe, J. (1992). Time scale of scour around a vertical pile. *Proceedings of the 2nd International Offshore and Polar Engineering Conference*, 308-315.
- Sumer, B. M., Fredsøe, J., and Christiansen, N. (1992). Scour around vertical pile in waves. *Journal of Waterway, Port, Coastal, and Ocean Engineering*, 118(1), 15-31.
- Takahashi, S. (2002). *Design of vertical breakwaters*. Reference Document 34, Port and Airport Research Institute, Japan.
- Takahashi, S., Tanimoto, K., and Shimosako, K. (1994). A proposal of impulsive pressure coefficient for design of composite breakwaters. *Proceedings of the International Conference on Hydro-Technical Engineering for Port and Harbor Construction*, 489-504.
- Tanimoto, K., Yagyu, T., and Goda, Y. (1982). Irregular wave tests for composite breakwater foundations. *Proceedings of the 18th International Conference on Coastal Engineering*, ASCE.
- US Army Corps of Engineers (1984). *Shore Protection Manual*. Coastal Engineering Research Center.
- US Army Corps of Engineers (2002). *Coastal Engineering Manual*, EM 1110-2-1100.
- Van der Meer, J. W. (1988). *Rock slopes and gravel beaches under wave attack*. PhD thesis, Delft University of Technology; Delft Hydraulics Publication 396.
- Van Gent, M. R. A., Smale, A. J., and Kuiper, C. (2003). Stability of rock slopes with shallow foreshores. *Proceedings of Coastal Structures 2003*, ASCE.
- Van Rijn, L. C. (1984). Sediment transport, part I: bed load transport. *Journal of Hydraulic Engineering*, 110(10), 1431-1456.
- Vasco Costa, F. (1964). The berthing ship: the effect of impact on the design of fenders and other structures. *The Dock and Harbour Authority*, 45.
- Wheeler, J. D. (1970). Method for calculating forces produced by irregular waves. *Journal of Petroleum Technology*, 22(3), 359-367.
- Wilcox, D. C. (1998). *Turbulence Modeling for CFD*. DCW Industries.

- Xie, S. L. (1981). *Scouring patterns in front of vertical breakwaters and their influence on the stability of the foundations of the breakwaters*. Delft University of Technology.
- Yarnell, D. L. (1934). *Bridge piers as channel obstructions*. Technical Bulletin 442, US Department of Agriculture.